

THE CHAISE LONGUE THEOREM

Conjecture 1.2 of Degtyarev–Shimada for the Fermat varieties of every odd degree in every even dimension, and the integral Hodge conjecture for those whose degree is prime or prime to $(n+2)!$

Rafael Amichis Luengo

Madrid, Spain · tretoef@gmail.com · 29 September 2026 · version 8

Abstract

Let $X \subset P^{n+1}$ be the complex Fermat variety $z_0^m + \dots + z_{n+1}^m = 0$ of even dimension $n = 2k$ and degree $m \geq 3$, and let $L(X) \subseteq H_n(X; \mathbb{Z})$ be the subgroup generated by the classes of its standard k -dimensional linear subspaces. Degtyarev and Shimada conjectured that $H_n(X; \mathbb{Z})/L(X)$ is torsion free [DS, Conjecture 1.2]. **We prove this conjecture for every odd degree m , in every even dimension.** In algebraic form, we prove with no topology that the group $Z[G]/(\psi_f)$ of [DS] is free abelian for every odd m ; the topology of [DS] identifies its torsion with that of $H_n(X; \mathbb{Z})/L(X)$. This algebraic statement has also been checked by machine: a complete proof in Lean 4 with Mathlib, with no gap and no axiom beyond the three standard ones (§12.8). Combined with the classical description of the rational Hodge classes of X (Shioda–Katsura, Shioda, Ran) and with Aoki's determination of the degrees in which every Hodge class is of «pair type», this gives **the integral Hodge conjecture for X whenever m is an odd prime or every prime factor of m exceeds $n + 2$** : in these degrees every integral Hodge class of middle degree is an integral combination of classes of standard linear subspaces. For every other odd m the standard subspaces span a primitive sublattice of the integral Hodge classes, with a free complement whose rank is the number of Hodge characters of X that are not of pair type (in closed form for $m = 9$). As a further consequence, the linear subspaces that [DS] exhibit on the partial Fermat varieties generate primitive sublattices, a statement that [DS] had proved conditionally on Conjecture 1.2 (these are the $(2s+1)!! \cdot m^{d+1}$ subspaces exhibited in [DS]; we do not claim that they are all the linear subspaces).

By the reduction of [DS] to commutative algebra, the conjecture at m is a dimension count over F_p for each prime p dividing m . For $m = q = p^v$ it reads

$$\dim_{F_p}(D_J : J \in \mathcal{J}) \cdot C = Q_k(q), \quad C = F_p[y_1, \dots, y_{2k+1}]/(y_1^{q-1}, \dots, y_{2k+1}^{q-1}),$$

where J runs over the $(2k+1)!!$ perfect matchings of $\{0, \dots, 2k+1\}$, D_J is the product, over the pairs $\{a, b\}$ of J avoiding 0, of $D(y_a, y_b) = (y_b^{q-1} - y_a^{q-1})/(y_a + y_b)$, and $Q_k(q) = (2k+2)! \cdot [x^{2k+2}] I_0(2x)^{(q-1)/2}$. We prove this count (Theorem B) with equality over every field and for every odd q , and with a free quotient over the integers. The proof peels off one variable at a time: the slices of the ideal are matched with the fibres of a finite point set, the bookkeeping collapses onto partitions under weak dominance, the possible values of one new coordinate always form a single chain, and Vandermonde divided differences are the only glue. For $m = q \cdot r$ with $p \nmid r$ and $r > 1$, the group algebra over $\bar{F}_p$ splits over the colourings of the coordinates by r -th roots of unity; on each factor the ideal becomes a tensor product of blocks, which are instances of Theorem 5.3 (the generalisation of Theorem B to down-sets of partitions) and a new family of **bipartite ideals**, generated by the products $\prod_i (x_i - z_{\sigma(i)})^{q-1}$ over bijections σ . We prove the corresponding lower bound (Theorem C), which is an equality over $\bar{F}_p$ (Corollary 7.8), by the same method, with pairs of partitions under the product of two weak dominance orders. Both arguments use the full family of matchings, and they must: for some proper subfamilies the analogous statement is false (an exact finite computation, Fact 9.2).

We also prove, and keep separate, a different count (Theorem A): for every field of characteristic $\neq 2$, every odd q and every n , $\dim F[x_1, \dots, x_n]/(e_1, e_3, e_5, \dots; x_1^q, \dots, x_n^q) = n! \cdot [y^n] e^y I_0(2y)^{(q-1)/2}$. Theorem A does **not** imply the conjecture (§9).

1. Introduction and main results

1.1 The conjecture

We follow [DS]. Let $n = 2k$ with $k \geq 0$, let $m \geq 3$, and let $X \subset P^{n+1}$ be the Fermat variety $z_0^m + \dots + z_{n+1}^m = 0$. All homology is with coefficients in $\mathbb{Z}$.

- $\mathcal{J}$ is the set of partitions of the index set $\{0, 1, \dots, n+1\}$ into unordered pairs, written $J = [[j_0, k_0], \dots, [j_k, k_k]]$ with $j_i < k_i$ and $j_0 < j_1 < \dots < j_k$; in particular $j_0 = 0$. There are $|\mathcal{J}| = (2k+1)!!$ of them.
- For $J \in \mathcal{J}$ and $\beta = (\beta_0, \dots, \beta_k)$ with $\beta_i^m = -1$, the **standard k -space** $L_{J,\beta} \subset X$ is given by $z_{k_i} = \beta_i z_{j_i}$ ($i = 0, \dots, k$).
- For $\emptyset \neq K \subseteq \mathcal{J}$, $L_K(X) \subseteq H_n(X)$ is the subgroup generated by the classes $[L_{J,\beta}]$ with $J \in K$; $L(X) := L_{\mathcal{J}}(X)$.

Conjecture 1.2 of [DS]. *The group $H_n(X)/L(X)$ is torsion free.*

For $k = 1$ the conjecture is a theorem of Degtyarev [De14]; for m prime to 6 it implies that the Picard group of the Fermat surface is generated by its lines. For $k \geq 2$ it has been confirmed by computer for $(n, m) = (4, m)$ with $3 \leq m \leq 12$, and for $(6,3), (6,4), (6,5), (8,3)$ [DS, §5]. Aljovin, Movasati and Villaflor [AMV] compared the lattice of linear cycles with the lattice of Hodge cycles by computing elementary divisors, for the Fermat surfaces of degree $3 \leq m \leq 14$ and the Fermat fourfolds of degree $3 \leq m \leq 6$ [AMV, Table 1]. The conjecture is restated as open in [De15, Conjecture 4.4]. Before the present work we knew of no proof for any $k \geq 2$ beyond these cells; in particular it had not been decided for any odd degree that is not a prime power when $k \geq 2$ (the odd degrees in the table of [DS, §5] are prime powers). As [DS, §1] point out, for prime m the rational Hodge classes of X are spanned by the classes of the standard spaces, so that the conjecture would upgrade this to the *integral* Hodge classes; §10 carries this out, for prime m and for more degrees.

1.2 The main theorem

Put $N := 2k+2 = n+2$, and for odd $q \geq 3$ let

$$Q_k(q) := N! \cdot [x^N] I_0(2x)^{(q-1)/2}, \quad I_0(2x) = \sum_{b \geq 0} x^{2b} / (b!)^2.$$

For odd m , $Q_k(m)$ is the number $|\Gamma_{\mathcal{J}}|$ of [DS, Definition 1.3] (Lemma 2.2), so that $\text{rank } L(X) = Q_k(m) + 1$ by Theorem 0(c) below.

MAIN THEOREM. *Let $m \geq 3$ be odd and $k \geq 0$. Then $H_{2k}(X; \mathbb{Z})/L(X)$ is torsion free for the Fermat variety X of degree m and dimension $2k$. Equivalently, $L(X)$ is a primitive sublattice of $H_{2k}(X; \mathbb{Z})$, of rank $Q_k(m) + 1$.*

For $k = 0$ the statement is trivial: X is a set of m points and $L(X) = H_0(X)$. From now on $k \geq 1$ whenever topology is involved.

The Main Theorem has an algebraic form, which is proved here with no topology at all (Corollary 7.7). Let $R := \mathbb{Z}[t_1, \dots, t_{2k+1}]/(t_1^m - 1, \dots, t_{2k+1}^m - 1)$ be the integral group ring of $G = (\mathbb{Z}/m)^{2k+1}$ and let $\psi_J \in R$ be the generators of [DS], recalled in §2.

MAIN THEOREM' (algebraic form). *Let $m \geq 3$ be odd and $k \geq 1$. Then $R/(\psi_J : J \in \mathcal{J})$ is a free abelian group of rank $m^{2k+1} - Q_k(m)$. Equivalently, $\dim_{\mathbb{F}_p} \mathbb{F}_p[G]/(\bar{\psi}_J : J \in \mathcal{J}) = m^{2k+1} - Q_k(m)$ for*

every prime p .

By [DS, Theorem 1.1(a)] (Theorem 0(a) below) the torsion of $R/(\psi_J : J \in \mathcal{J})$ is that of $H_{2k}(X; Z)/L(X)$, so Main Theorem' implies the Main Theorem; the rank of $L(X)$ is [DS, Theorem 1.4]. These two statements are the only places where topology enters.

The proof has three parts.

- **§2 (the translation).** Using the results of [DS] collected in Theorem 0, torsion-freeness at m is equivalent to a dimension count over F_p for each prime $p \mid m$ (Proposition 2.1); for $m = q = p^v$ the count is **(S)** $\dim_{F_p}(D_J : J \in \mathcal{J})C = Q_k(q)$ (Proposition 2.5).
- **§5 (prime powers).** Theorem B below proves (S).
- **§6–§7 (degrees that are not prime powers).** For $m = q \cdot r$ with $q = p^v$, $p \nmid r$, $r > 1$, the count at p splits over the colourings of the coordinates by r -th roots of unity into a sum of products of block dimensions (Proposition 6.9). The blocks are instances of Theorem 5.3 and bipartite ideals, whose count is Theorem C (§7).

Sections 5–7 prove lower bounds for the dimensions of the ideals: Theorems 5.3 and 7.6. The matching upper bound is the inequality of Proposition 2.1, and it is algebraic too: it is [DS, Claim 4.3] (Theorem 0(b)), a count of points by the Chinese remainder theorem over a field of characteristic 0 (Appendix B, Step 6). So Main Theorem' is proved without topology; topology enters only through Theorem 0(a) and 0(c), which identify the torsion of $R/(\psi_J)$ with that of $H_{2k}(X; Z)/L(X)$ and give the rank of $L(X)$. Theorems 5.9 and 5.9', §4, §8 and §9 are not on the path of the proof; Corollary 7.8 is a consequence of it. Main Theorem' has been proved again in the proof assistant Lean 4 (§12.8).

The cubic case $q = 3$ has a shorter proof with an explicit basis given by ballot paths (§4, Theorem 4.1).

1.3 The integral Hodge conjecture

For a smooth complex projective variety Y , the integral Hodge conjecture in degree $2j$ asks whether every class in $H^{2j}(Y; Z) \cap H^{j,j}(Y)$ is the class of an algebraic cycle with integer coefficients. It is false in general [AH], [Kol], and it remains open for many varieties for which the rational Hodge conjecture is known; for Fermat varieties it is the question studied in [AMV]. Write $\text{Hdg}(X) \subseteq H_n(X; Z)$ for the group of integral Hodge classes of middle degree (§10.1).

COROLLARY H. *Let $m \geq 3$ be odd, $k \geq 1$, and let X be the Fermat variety of degree m and dimension $2k$.*

- (i) *If m is prime, or if every prime factor of m is at least $2k + 3$, then $\text{Hdg}(X) = L(X)$: every integral Hodge class of middle degree is an integral linear combination of classes of standard k -spaces. In particular the integral Hodge conjecture holds for X in every degree.*
- (ii) *For every odd m , $L(X)$ is a primitive sublattice of $\text{Hdg}(X)$ and $\text{Hdg}(X)/L(X)$ is free of rank $|\mathfrak{B}| - Q_k(m)$, where $\mathfrak{B}$ is the set of Hodge characters of §10.1. It is zero if and only if m is prime or every prime factor of m is at least $2k + 3$.*

For odd m , «every prime factor of m is at least $2k + 3$ » is the same as $\gcd(m, (n+2)!) = 1$, and also the same as the condition $\gcd(m, (n+1)!) = 1$ of [AMV, §1]. The proof is in §10. It combines the Main Theorem with the classical description of the rational Hodge classes of X by characters [SK], [Shi79], [Ran] ((H3) of §10.1), and with the determination, due to Aoki [Aok, Theorem A], of the degrees in which every Hodge character is of pair type; for prime m we give a short proof of this last fact (§10.1, (H4)). The new input is the primitivity of $L(X)$. The classical fact that the rational Hodge classes of pair type are spanned by linear subspaces [Ran, Theorem 4.9] is not used: over Z it follows from the Main Theorem.

What was known. For $k = 1$ the integral Hodge conjecture is the Lefschetz (1,1) theorem, and (i) says that the lines generate the Néron–Severi group, which was known when $\gcd(m, 6) = 1$ [SSvL], [De14], and classically for the cubic surface; for odd m these are exactly the degrees of (i) at $k = 1$. For $(k, m) = (2, 3)$ the integral Hodge conjecture itself follows from Voisin's theorem on cubic fourfolds [Voi, Theorem 2.11]; as [AMV, §1] note, the stronger statement $Hdg(X) = L(X)$ does not follow from it. The cells $(2, 3)$ and $(2, 5)$ were verified by computer in [AMV]. In each prime cell where [DS, §5] confirmed the primitivity of $L(X)$ by computer, namely $(k, m) = (2, 3), (2, 5), (2, 7), (2, 11), (3, 3), (3, 5), (4, 3)$, (i) already followed from that computation and the classical facts by the argument of §10.3; [AMV, §1] make this observation for the cells of their table. The author had also verified the primitivity of $L(X)$ by computer, before the present proof, in the prime cells $(k, m) = (2, 13), (2, 17), (2, 19), (3, 7), (4, 5), (5, 3)$ [Rep]; three of them, $(2, 13), (3, 7)$ and $(5, 3)$, are re-run in §12.1. What is new, for $k \geq 2$, is the uniform statement: every other cell with m prime, and every cell of (i) with m not prime, for instance $(k, m) = (2, 49), (2, 77), (3, 121), (3, 143)$; to our knowledge no cell of (i) with m not prime and $k \geq 2$ had been decided before. Part (ii) is new for $k \geq 2$, to our knowledge; for $k = 1$ it follows from [De14] and the rank $|\mathcal{B}| + 1$ of (H3). Jumagulov [Jum] recently gave a computer-assisted proof of the *rational* Hodge conjecture for the Fermat fourfolds of odd degree at most 199; that statement is rational, and complementary to ours.

1.4 Partial Fermat varieties

Let $0 \leq s \leq d$. Following [DS, §1], a **partial Fermat variety** is a smooth hypersurface $W_s \subset P^{2d+1}$ given by $f_0(z_0, z_1) + f_1(z_2, z_3) + \cdots + f_d(z_{2d}, z_{2d+1}) = 0$, where the f_i are binary forms of degree m , $f_0 = \cdots = f_s = x^m + y^m$, and $f_{s+1}, \dots, f_d$ are generic, pairwise distinct and distinct from $x^m + y^m$. [DS, §1] exhibit $(2s+1)!! \cdot m^{d+1}$ linear d -spaces on it.

COROLLARY W. *Let $m \geq 3$ be odd and $0 \leq s \leq d$.*

- (i) *The classes of the $(2s+1)!! \cdot m^{d+1}$ linear d -spaces of [DS, §1] on W_s generate a primitive sublattice of $H_{2d}(W_s; \mathbb{Z})$.*
- (ii) *On the Fermat variety X of dimension $2d$, let $\mathcal{J}(s, d) \subseteq \mathcal{J}$ be the set of matchings that contain the pairs $\{2i, 2i+1\}$ for $s < i \leq d$. Then $L_{\mathcal{J}(s,d)}(X)$ is primitive in $H_{2d}(X; \mathbb{Z})$, of rank $Q_s(m)(m-1)^{d-s} + 1$.*

[DS, Corollary 1.7] states (i) under the assumption that Conjecture 1.2 holds for the Fermat varieties of dimension $2s$ and degree m , and proves it for $s = 0, 1$; [De15, §4.2–§4.3] reduces (i) and (ii) to the same assumption and proves them in the same cases (in [De15] the index is the dimension $2s$). The Main Theorem removes the assumption. The proof is in §11, where (ii) is proved directly.

1.5 The algebraic statements

Let F be a field and $q \geq 3$ odd. Put $C_F := F[y_1, \dots, y_{2k+1}]/(y_1^{q-1}, \dots, y_{2k+1}^{q-1})$ and define $D(a, b)$ and D_J as in the abstract: D is the polynomial of §2.3, which has integer coefficients.

THEOREM B. *For every field F , every odd $q \geq 3$ and every $k \geq 0$,*

$$\dim_F(D_J : J \in \mathcal{J}) \cdot C_F = Q_k(q).$$

Moreover, with $C_Z := \mathbb{Z}[y_1, \dots, y_{2k+1}]/(y_1^{q-1}, \dots, y_{2k+1}^{q-1})$, the group $C_Z/(D_J : J \in \mathcal{J})C_Z$ is free abelian of rank $(q-1)^{2k+1} - Q_k(q)$, and the dimension of each graded piece of $(D_J : J \in \mathcal{J})C_F$ does not depend on F .

Theorem B is the root case of a more general statement about down-sets of partitions (Theorem 5.3 for $\geq$; Theorems 5.9 and 5.9' for $=$ and for the integral statement). Neither the topology nor [DS] enters its proof;

they enter only in the translation of §2. With Theorem B and Proposition 2.5, the Main Theorem follows for $m = p^v$ (Corollary 5.4).

The degrees that are not prime powers need a second count, in two sets of variables $x_1, \dots, x_\alpha$ and $z_1, \dots, z_\beta$ with the box (x_i^q, z_i^q) , for $q = p^v$: the **balanced** bipartite ideal I_a^{bal} , generated by $\prod_{i=1}^a (x_i - z_{\sigma(i)})^{q-1}$ over all $\sigma \in S_a$ ($\alpha = \beta = a$), and the **phantom** bipartite ideal I_a^{ph} , generated by the same products over all bijections from a of the $a + 1$ variables x_i onto the z_i ($\alpha = a + 1, \beta = a$). **Theorem C** (§7) says that their dimensions are at least the numbers of pairs (ξ, η) of tuples in a set of q elements, of lengths (a, a) and $(a+1, a)$, such that the multiset of η is equal to, respectively contained in, the multiset of ξ . Over $\bar{F}_p$ these are equalities (Corollary 7.8), and the dimensions do depend on the characteristic.

1.6 A different count: Theorem A

Write $E := (e_1, e_3, \dots, e_{2k+1}) \subseteq F_p[x_0, \dots, x_{2k+1}]$ for the ideal of the odd elementary symmetric polynomials. Over an algebraic closure its zero set is the union of the linear spaces $x_a + x_b = 0$ ($\{a, b\} \in J$), over all $J \in \mathcal{J}$.

THEOREM A. *Let F be a field of characteristic $\neq 2$, let $q \geq 1$ be odd and let $n \geq 0$. Then $\dim_F F[x_1, \dots, x_n]/(e_1, e_3, e_5, \dots; x_1^q, \dots, x_n^q) = n! \cdot [y^n] e^y I_0(2y)^{(q-1)/2}$.*

With $F = F_p$, $q = p^v$ and $n = N$, Theorem A says that $A_k(q) := \dim F_p[x_0, \dots, x_{2k+1}]/(E + (x_i^q))$ equals the number $P_k(q)$ of F_q -points of the union of the matching spaces (Corollary 8.1). **This is not the Main Theorem, and it does not imply it.** Theorem A is a *sum* statement at the *odd* box q on the grid *with* 0; the Main Theorem for $m = q$ is equivalent to an *intersection* statement at the *even* box $q - 1$ on the grid *without* 0 (Proposition 2.6), and §9 shows that the two are independent for families of matchings. The numbers differ as well: $P_2(3) = 141$ against $Q_2(3) = 20$.

1.7 The idea of the proofs

Write the ideal of Theorem B as $V \subseteq C_{2k+1}$ and the point set it should count as $Z \subseteq T^{2k+1}$, $T = F_q^\times$. Peel off the variable y_1 : the ideal splits into $q - 1$ slices $W_0 \subseteq \dots \subseteq W_{q-2}$, ideals in the remaining variables, and the point set splits into $q - 1$ layers according to how many values of the first coordinate complete a given tail. If every layer is the point set of an ideal contained in the matching slice, induction on the number of variables gives $\dim V \geq |Z|$. To make this work we enlarge the family: to every down-set Λ of partitions under weak dominance we attach an ideal V_Λ , generated by products of divided differences D and Vandermonde determinants Δ , and a point set Z_Λ , the tuples whose «unpaired part» has shape in Λ . Three facts then carry the induction. The number of completions of a tail depends only on the shape μ of the tail (P1). The $q - 1$ possible new shapes form a chain for weak dominance, so the completions in Λ are an initial segment of the chain, and the layers are again down-sets (P2). And each generator of a layer can be lifted to the right slice by one of three explicit constructions, with no room to spare (P3). The full family of matchings enters only once, at the root: the down-set generated by the partition (1) has as ideal exactly the ideal of all the D_J .

For $m = q \cdot r$ with $q = p^v$, $p \nmid r$ and $r > 1$, $t^m - 1 = \prod_{\zeta \in \mu_r} (t - \zeta)^q$ over $\bar{F}_p$, so the group algebra is a product of local rings, one for each colouring c of the coordinates by r -th roots of unity. On the factor of c , the generator ψ_J vanishes unless every pair of J has colours ζ, ζ^{-1} ; the surviving generators are pure tensors along the colour classes, and the ideal is a tensor product of one ideal per class of colour 1 and one per pair of inverse colours. The class of colour 1 is an instance of Theorem 5.3. A pair of inverse colours becomes, after the change of variables $x = \zeta^{-1}t - 1$, $z = (\zeta t)^{-1} - 1$, a bipartite ideal, and the colour of the index 0, which is determined by the others, decides whether it is balanced or phantom. The count of the points factors over the colourings in the same way. For the bipartite ideals, the tail of a point now has two residues, one for each set of variables, and its shape is a pair of partitions: adding a value to the x -side removes a box from the z -residue, adds a box to the x -residue, or starts a new part, and these q options again form a chain, now in the product of two weak

dominance orders. The proof of Theorem B then goes through with the box q in place of $q - 1$ and the binomial coefficients $C(q - 1, t) \equiv (-1)^t$ in place of the signs of D .

1.8 A note on the history

For several months the project that produced this paper took the count $A_k(q) = P_k(q)$ of Theorem A to be the algebraic form of Conjecture 1.2. Two earlier notes of the author, on the cases $k = 2$ (*The Sofa Theorem* [Sofa]) and $k = 3$ (*The Hammock Theorem* [Ham]), were written under that identification and stated that they established Conjecture 1.2 in those dimensions. On 23 September 2026, re-deriving the reduction from the original [DS], we found that the identification had never been proved, and that it is false for families of matchings (§9). The conclusions of those notes about Conjecture 1.2 in dimensions 4 and 6 are nevertheless true: for odd m they are the cases $k = 2, 3$ of the Main Theorem, proved here by a different route. The previous version of this paper proved the Main Theorem for odd prime powers and stated the colour reduction for the degrees that are not prime powers as an open problem; the reduction and the bipartite count (§6–§7) complete it. The present version (v8) adds the machine-checked proof of Main Theorem' (§12.8), records that the repository [Rep] is public, states the status of the *Double Ladder* theorem more precisely (Remark 10.2), and adds one clause to the proof of Lemma 6.8; no statement and no proof of the previous version changes.

1.9 Notation

q is odd throughout. In §2, §6 and §9, p is an odd prime and $q = p^v$. §5 and §7 are stated for every odd q over a field (in §7, a field in which the binomial coefficients $C(q - 1, t)$ are non-zero), and are applied with $q = p^v$. The word «block» means a column block B_c of a tight pattern in §5 and §7, and a block of a colouring in §6; the letter R , with subscripts, denotes several rings (R of §2, R_c of §6, $R_{A,B}$ of §7), each defined where it is used. Symbols introduced inside a section are local to it (§6–§7 have their own local symbols), and Appendix A has its own symbols.

symbol	meaning	first use
$\mathcal{J}, J, (2k+1)!!$	perfect matchings of $\{0, \dots, 2k+1\}$, written with $j_0 = 0$	§1
N	$2k+2$, the number of homogeneous coordinates	§1
R, ψ_J, Γ_K	the ring, the generators and the point set of [DS]	§
T, h	$F_q^\times$ (or any set of $q-1$ elements with a fixed-point-free involution $u \mapsto -u$); $h = (q-1)/2$ classes $\{u, -u\}$	§2
$Q_k(q)$	$N! \cdot [x^N] I_0(2x)^h$	§1
B, C, C_m	the box rings (y_i^q) and (y_i^{q-1})	§2 §
$D(a, b), D_J, D_P$	divided differences and their products over pairs	§2 §5
$\Delta(B)$	the Vandermonde determinant of an index set B	§5
$\nu(M), \lambda(M)$	the number of negation pairs of a tuple, and the partition of its residue	§: §5
$Par_m^{(h)}, \preceq, S_t$	partitions allowed at m coordinates; weak dominance; sums of the t largest parts	§5
V_Λ, Z_Λ	the ideal and the point set of a down-set Λ (§5.2); of a down-set of pairs of partitions (§7.2)	§5 §7
$W_f(V), F_\Lambda(\mu), \Lambda_i$	slices, completion counts, layers	§5 §5
$gr(I)$	the ideal of top-degree forms of an ideal I	§5
$c, \mathcal{C}_\zeta, \mu_r$	a colouring of the coordinates, its colour classes, the r -th roots of unity	§
$I_a^{bal}, I_a^{ph}, N_{bal}, N_{ph}$	the bipartite ideals and their counts	§6 §6
$BPar_q(\alpha, \beta)$	pairs of partitions, with the product of weak dominance orders	§7
$E, A_k(q), P_k(q)$	odd symmetric ideal; the sum count and the point count of Theorem A	§1 §
m	the degree of the Fermat variety (§1, §2, §6, §10, §11); in §5 and Appendix A, the number of variables	—
$\mathfrak{U}, \mathfrak{B}, \alpha $	characters of the primitive cohomology; Hodge characters; the degree of a character	§1

2. The translation: Conjecture 1.2 as a dimension count

This section is written from the original [DS] (arXiv:1405.4683v3, the latest version; J. Math. Soc. Japan 68 (2016)), and uses its notation.

- $R := \mathbb{Z}[t_1, \dots, t_{n+1}]/(t_1^m - 1, \dots, t_{n+1}^m - 1)$, the group ring of $G \cong (\mathbb{Z}/m)^{n+1}$, which acts on X by multiplying the coordinates by m -th roots of unity. The variable $t_0 := (t_1 \cdots t_{n+1})^{-1}$ never enters below.
- $\varphi(u) := u^{m-1} + \cdots + u + 1$.
- For $J \in \mathcal{J}$: $\tau_J := \prod_{i=0}^k (t_{k_i} - 1)$ and $\psi_J := \tau_J \cdot \prod_{i=1}^k \varphi(t_{j_i} t_{k_i})$. Since $j_0 = 0$, the pair containing 0 contributes only the factor $t_{k_0} - 1$.
- $\Gamma_K \subseteq \mu_m^{n+1}$ [DS, Definition 1.3] is the set of $(a_1, \dots, a_{n+1})$ with $a_i \neq 1$ for every i such that, for some $J \in K$, $a_{j_i} a_{k_i} = 1$ for $i = 1, \dots, k$.

We use four results of [DS], collected here.

THEOREM 0 [DS]. Let $n = 2k \geq 2$, $m \geq 3$, $\emptyset \neq K \subseteq \mathcal{J}$, and $A_K := R/(\psi_J : J \in K)$.

- (a) [DS, Theorem 1.1(a)] $\text{Tors}(H_n(X)/L_K(X)) \cong \text{Tors}(A_K)$.
- (b) [DS, Claim 4.3] If K_p is an algebraically closed field of characteristic $p = 0$ or $p \nmid m$, then $\dim_{K_p}(A_K \otimes K_p) = m^{n+1} - |\Gamma_K|$.
- (c) [DS, Theorem 1.4] $\text{rank } L_K(X) = |\Gamma_K| + 1$.
- (d) [DS, Corollary 1.5] The order of the torsion of $H_n(X)/L_K(X)$ is divisible only by primes dividing m .

Part (b) is algebraic: it is a count of points by the Chinese remainder theorem (Appendix B, Step 6). Parts (c) and (d) follow from (a), (b) and the proof of (a) in a few lines. For the reader who wants to check the only topological input, Appendix B gives a self-contained proof of Theorem 0 modulo two results that we do not re-prove: Pham's description of the homology of the affine Fermat variety [Ph], and the intersection numbers of the standard spaces with the Pham cycle [DS, Theorem 2.2].

2.1 Torsion as one dimension

Proposition 2.1. Let $\emptyset \neq K \subseteq \mathcal{J}$. For every prime p , $\dim_{F_p}(A_K \otimes F_p) \geq m^{n+1} - |\Gamma_K|$, and $H_n(X)/L_K(X)$ is torsion free if and only if equality holds for every prime p dividing m . More precisely, for each prime p , equality holds iff A_K has no p -torsion, iff $H_n(X)/L_K(X)$ has no p -torsion.

Proof. R is a free $\mathbb{Z}$ -module of rank m^{n+1} (basis t^ν , $\nu \in (\mathbb{Z}/m)^{n+1}$), so A_K is a finitely generated abelian group; write $A_K \cong \mathbb{Z}^\rho \oplus \bigoplus_\ell T_\ell$ with T_ℓ a finite ℓ -group. Tensoring with F_p gives $\dim_{F_p}(A_K \otimes F_p) = \rho + c_p$, where c_p is the number of cyclic factors of T_p (since $\mathbb{Z}/\ell^a \otimes F_p$ is F_p for $\ell = p$ and 0 otherwise). By Theorem 0(b) with $p = 0$, $\rho = \dim_{\mathbb{C}}(A_K \otimes \mathbb{C}) = m^{n+1} - |\Gamma_K|$. So the inequality holds, and equality at p means $T_p = 0$. (This part of the proof uses only Theorem 0(b), which is algebraic.) By Theorem 0(a), $H_n(X)/L_K(X)$ is torsion free iff A_K is; by Theorem 0(d), only primes dividing m can occur. ■

From now on, until the end of §5, p is an odd prime, $v \geq 1$ and $m = q = p^v$ (odd m that are not prime powers are treated in §6–§7). Then only the prime p matters, and $A_K \otimes F_p = F_p[G]/(\bar{\psi}_J : J \in K)$, where $F_p[G] := F_p[t_1, \dots, t_{2k+1}]/(t_i^q - 1)$ and $\bar{\psi}_J$ is the image of ψ_J .

2.2 The count

Let T be a finite set of $q - 1$ elements with a fixed-point-free involution $u \mapsto -u$ (for instance $T = F_q^\times$). It has $h := (q-1)/2$ orbits $\{u, -u\}$, which we call **classes**.

Lemma 2.2. For odd q , $|\Gamma_{\mathcal{J}}| = Q_k(q)$. More precisely, $|\Gamma_{\mathcal{J}}|$ is the number of N -tuples in $\mu_q \setminus \{1\}$ whose multiset is closed under inversion, and this is also the number of N -tuples in T whose multiset is closed under the involution.

Proof. Let $a = (a_1, \dots, a_{2k+1}) \in \Gamma_{\mathcal{J}}$ and put $a_0 := (a_1 \cdots a_{2k+1})^{-1}$. The pairs $\{j_i, k_i\}$, $i \geq 1$, cover $\{1, \dots, 2k+1\} \setminus \{k_0\}$ and their products are 1, so $\prod_{i \geq 1} a_i = a_{k_0}$ and $a_0 = a_{k_0}^{-1} \neq 1$. Hence $a \in \Gamma_{\mathcal{J}}$ iff the multiset $\{a_0, \dots, a_{2k+1}\} \subseteq \mu_q \setminus \{1\}$ splits into inverse pairs. Conversely, if an N -tuple in $\mu_q \setminus \{1\}$ is closed under inversion, its product is 1, so it is determined by its last $2k+1$ entries. Since q is odd, inversion has no fixed point on $\mu_q \setminus \{1\}$, which splits into h pairs $\{c, c^{-1}\}$. By the exponential formula (the generating function of a product of labelled structures is the product of the generating functions), each pair contributes the factor $\sum_b x^{2b}/(b!)^2 = I_0(2x)$ (choose b coordinates equal to c and b equal to c^{-1}). This gives $Q_k(q)$, and the same computation applies to T . ■

Lemma 2.2 and its proof hold for every odd m in place of q ; §6, §10 and §11 use it in that form. Likewise Lemma 2.3 below holds in any ring $F[t]/((t-1)^q)$ with $F \supseteq F_p$ a field and $q = p^v$, and Lemma 2.4(i) over any such field F ; this is how §6.4 uses them.

External check. At $k = 1$, $Q_1(q) = 3(q-1)(q-2)$, so $\text{rank } L(X) = 3(m-1)(m-2) + 1$. For $m = 3$ this is 7, the Picard number of the cubic surface, which is generated by its 27 lines. It agrees with the constant-term de-

scription of $\text{rank } L(X)$ in [DS, Remark 4.4].¹

¹The polynomials displayed in [DS, Remark 4.4] are, for odd m , the corresponding values of $|\Gamma_{\mathcal{J}}|$, in accordance with the constant-term description $\text{rank } L(X) = 1 + |\Gamma_{\mathcal{J}}|$ given there; this is how we quote them in §3.

2.3 A coordinate in which the relations become linear

Lemma 2.3 (identities in characteristic p). *In $F_p[t]/(t^q - 1)$ put $y := t - t^{-1}$. Then:*

- (i) $F_p[t]/(t^q - 1) = F_p[y]/(y^q)$, and $t \mapsto t^{-1}$ becomes $y \mapsto -y$;
- (ii) in $F_p[t_j, t_k]/(t_j^q - 1, t_k^q - 1)$, $t_j t_k - 1 = (\text{unit}) \cdot (y_j + y_k)$;
- (iii) $\varphi(u) = (u - 1)^{q-1}$ in $F_p[u]$.

Proof. (i) $t^q - 1 = (t - 1)^q$ in characteristic p , because $q = p^v$; so $t - 1$ is nilpotent, and the ring is local with maximal ideal $(t - 1)$. The elements t and $t + 1 = 2 + (t - 1)$ are units, the second because $2 \in F_p^\times$. Hence $y = t^{-1}(t - 1)(t + 1)$ is a unit times $t - 1$, so $y^j \neq 0$ for $j \leq q - 1$ and $y^q = (\text{unit}) \cdot (t - 1)^q = 0$. The q elements $1, y, \dots, y^{q-1}$ lie in successive powers of the maximal ideal and not in the next, so they are linearly independent; the ring has dimension q , so it is $F_p[y]/(y^q)$. Inversion sends $t - t^{-1}$ to $t^{-1} - t = -y$. (ii) $y_j + y_k = t_j + t_k - t_j^{-1} - t_k^{-1} = (t_j + t_k)(t_j t_k)^{-1}(t_j t_k - 1)$, and $t_j + t_k = 2 + (t_j - 1) + (t_k - 1)$ is a unit. (iii) $(u - 1)\varphi(u) = u^q - 1 = (u - 1)^q$, and $F_p[u]$ is a domain. ■

Write $B := F_p[y_1, \dots, y_{2k+1}]/(y_1^q, \dots, y_{2k+1}^q)$; by (i), $F_p[G] \cong B$. Put

$$D(a, b) := \sum_{u=0}^{q-2} (-1)^u a^u b^{q-2-u} = (b^{q-1} - a^{q-1})/(a + b).$$

The second expression holds in $Z[a, b]$ because $q - 2$ is odd. D is antisymmetric, $D(b, a) = -D(a, b)$, and at $q = 3$ it is $D(a, b) = b - a$.

Lemma 2.4 (identities in characteristic p). (i) $(a + b)^{q-1} \cdot b = -ab \cdot D(a, b)$ in $F_p[a, b]/(b^q)$. (ii) For every $J \in \mathcal{J}$, $\bar{\psi}_J = (\text{unit}) \cdot Y \cdot D_J$ in B , where $Y := y_1 y_2 \cdots y_{2k+1}$ and $D_J := \prod_{i=1}^k D(y_{j_i}, y_{k_i})$ is the product over the k pairs of J that do not contain 0.

Proof. (i) The sum $(a + b) \cdot \sum_{s=0}^{q-1} (-1)^s a^s b^{q-1-s}$ telescopes to $(-1)^{q-1} a^q + b^q = a^q + b^q$, which is $(a + b)^q$ in characteristic p . So $(a + b)^{q-1} = \sum_{s=0}^{q-1} (-1)^s a^s b^{q-1-s}$ in the domain $F_p[a, b]$. Multiply by b and drop b^q : $\sum_{s=1}^{q-1} (-1)^s a^s b^{q-s} = -ab \cdot \sum_{u=0}^{q-2} (-1)^u a^u b^{q-2-u}$. (ii) By Lemma 2.3, up to a unit, $\bar{\psi}_J = \prod_{i=0}^k y_{k_i} \cdot \prod_{i=1}^k (y_{j_i} + y_{k_i})^{q-1}$. By (i), $y_{k_i}(y_{j_i} + y_{k_i})^{q-1} = -y_{j_i} y_{k_i} D(y_{j_i}, y_{k_i})$ for each $i \geq 1$. The indices $\{k_0\} \cup \bigcup_{i \geq 1} \{j_i, k_i\}$ are exactly $\{1, \dots, 2k+1\}$, so the monomial factors multiply to Y . ■

2.4 The sum form (S)

Let $C := F_p[y_1, \dots, y_{2k+1}]/(y_1^{q-1}, \dots, y_{2k+1}^{q-1})$, of dimension $(q-1)^{2k+1}$.

Proposition 2.5 (the translation). Let $q = p^v$, $k \geq 1$ and $\emptyset \neq K \subseteq \mathcal{J}$. Then $\dim_{F_p} F_p[G]/(\bar{\psi}_J : J \in K) = q^{2k+1} - \dim_{F_p} (D_J : J \in K)C$, and

- $\dim_{F_p} (D_J : J \in K)C \leq |\Gamma_K|$ always;
- $H_{2k}(X)/L_K(X)$ is torsion free iff $\dim_{F_p} (D_J : J \in K)C = |\Gamma_K|$.

In particular ($K = \mathcal{J}$), the Main Theorem at (k, q) is equivalent to

(S) $\dim_{F_p} (D_J : J \in \mathcal{J})C = Q_k(q)$.

Proof. Multiplication by Y on B kills exactly the monomials having some exponent $\geq q - 1$, so its kernel is $(y_1^{q-1}, \dots, y_{2k+1}^{q-1})B$, and it induces an injective linear map $C \rightarrow B, f \mapsto Yf$. By Lemma 2.4(ii), the ideal $(\bar{\psi}_J : J \in K)B = (Y D_J : J \in K)B$ is the image of the ideal $(D_J : J \in K)C$ under this map, so the two have the same dimension. The two claims follow from Proposition 2.1 and Lemma 2.2. ■

Remarks 2.5'. (1) The generators D_J all have degree $k(q-2)$. (2) **What the topology is needed for.** The inequality $\dim(D_J)C \leq Q_k(q)$ can also be proved with no topology: it is part of Theorem 5.9, and it also follows from Propositions 2.6 and 9.1. The inequality is also the case $p \mid m$ of Proposition 2.1, whose proof uses only Theorem 0(b), a Chinese-remainder count. [DS] is needed only for two things: that equality in (S) is *equiva-*

lent to torsion-freeness, and the rank of $L(X)$. (3) [DS, §5] formulate a computational criterion of the same kind in the ring $\bar{R} := Z[t_1, \dots, t_{n+1}]/(\varphi(t_1), \dots, \varphi(t_{n+1}))$, through their Lemma 4.5 ($\psi_j = \pm \lambda \rho_j$, $\lambda = \Pi(t_i - 1)$). We do not use it.

2.5 An equivalent intersection form

Proposition 2.6. *Let $I'_J := (y_{j_i} + y_{k_i} : i = 1, \dots, k)$ and $b_k(r) := \dim F_p[y_1, \dots, y_{2k+1}] / \bigcap_{J \in \mathcal{J}} (I'_J + (y_1^r, \dots, y_{2k+1}^r))$. Then $\dim(D_J : J \in \mathcal{J})C = b_k(q - 1)$, so (S) is equivalent to $b_k(q - 1) = Q_k(q)$. Eliminating $x_0 := -(y_1 + \dots + y_{2k+1})$, $b_k(r) = \dim F_p[x_0, \dots, x_{2k+1}] / \bigcap_{J \in \mathcal{J}} (I_J + (x_0^r, \dots, x_{2k+1}^r))$ with $I_J := (x_a + x_b : \{a, b\} \in J)$.*

Proof. On B put $\langle f, g \rangle :=$ the coefficient of $\Pi_i y_i^{q-1}$ in fg . Then $\langle y^\alpha, y^\beta \rangle = 1$ if $\alpha + \beta = (q-1, \dots, q-1)$ and 0 otherwise, a permutation matrix, so the pairing is non-degenerate. For an ideal $I \subseteq B$, $\langle f, gh \rangle = \langle fg, h \rangle$ shows that $f \in \text{ann}_B(I)$ iff $f \perp I$; hence $\dim \text{ann}_B(I) = \dim B - \dim I$, and $\text{ann}_B(\Sigma_J I_J) = \bigcap_J \text{ann}_B(I_J)$. So $\dim(\bar{\psi}_J : J)B = \dim B / \bigcap_J \text{ann}_B(\bar{\psi}_J)$, which equals $\dim(D_J : J)C$ by the proof of Proposition 2.5. For a fixed J , the linear forms $w := y_{k_0}, u_i := y_{j_i} + y_{k_i}, v_i := y_{k_i} (i \geq 1)$ form a basis of $F_p y_1 \oplus \dots \oplus F_p y_{2k+1}$. Since $\ell^q = \Sigma c_i y_i^q$ for every linear form $\ell = \Sigma c_i y_i$ over F_p (Frobenius, and $c^q = c$ on F_p), the ideal (y_i^q) is also generated by w^q, u_i^q, v_i^q . In these coordinates $\bar{\psi}_J = (\text{unit}) \cdot w \cdot \Pi_{i \geq 1} v_i u_i^{q-1}$, a monomial times a unit, whose annihilator in the box ring is the monomial ideal $(w^{q-1}) + (v_i^{q-1}, u_i : i \geq 1)$. Modulo u_i one has $y_{j_i} \equiv -v_i$, so this ideal is $I'_J + (y_1^{q-1}, \dots, y_{2k+1}^{q-1})$. Finally $y_i^q \in (y_i^{q-1})$, so $B / \bigcap_J (\dots) = F_p[y] / \bigcap_J (I'_J + (y_i^{q-1}))$. For the last sentence, put $\sigma := x_0 + \dots + x_{2k+1}$ and substitute $y_i = x_i (i \geq 1)$, $x_0 = -\Sigma y_i$. Two facts are used. (1) Every I_J contains σ (it is the sum of the generators $x_a + x_b$); by the correspondence between the ideals of S containing σ and the ideals of $S/(\sigma) \cong F_p[y]$, the intersection of the ideals $I_J + (x_i^r)$ maps onto the intersection of their images. (2) Modulo I_J , $x_0 \equiv -x_{k_0}$, so $x_0^r \in I_J + (x_1^r, \dots, x_{2k+1}^r)$, and the image of $I_J + (x_0^r, \dots, x_{2k+1}^r)$ is $I'_J + (y_1^r, \dots, y_{2k+1}^r)$. ■

By Matlis duality this form is **equivalent** to (S), not weaker: it is a restatement, not a reduction. It is used in §9.

2.6 A worked example: the cubic surface, $(k, q) = (1, 3)$

Here $n = 2$, $N = 4$, and $\mathcal{J}$ has three matchings, $[[0,1],[2,3]], [[0,2],[1,3]], [[0,3],[1,2]]$. The pairs avoiding 0 are $(2,3), (1,3), (1,2)$, and at $q = 3$, $D(a,b) = b - a$, so the generators are $y_3 - y_2, y_3 - y_1, y_2 - y_1$ in $C = F_3[y_1, y_2, y_3]/(y_1^2, y_2^2, y_3^2)$, of dimension 8. The ideal they generate has dimension 2 in degree 1 (two of the three are independent), 3 in degree 2 ($y_1(y_2 - y_1) = y_1 y_2, y_1(y_3 - y_1) = y_1 y_3$, and $y_2(y_3 - y_1) = y_2 y_3 - y_1 y_2$), and 1 in degree 3: in total 6. On the other side, $T = F_3^\times = \{1, 2\}$, and Γ' (§3) consists of the tuples in T^3 with one negation pair, i.e. the $2^3 - 2 = 6$ non-constant tuples. So (S) holds, $\text{rank } L(X) = 7$, and $\dim F_3[G]/(\bar{\psi}_J) = 3^3 - 6 = 21$.

3. The count

By Lemma 2.2, $Q_k(q) = N! \cdot [x^N] I_0(2x)^h$ with $N = 2k+2$ counts the N -tuples in T whose multiset is closed under the involution. In small cases:

- $Q_0(q) = q - 1$;
- $Q_1(q) = 3(q-1)(q-2)$ (the coefficient of x^4 in $I_0(2x)^h$ is $h/4 + h(h-1)/2$, times $4!$);
- $Q_k(3) = C(2k+2, k+1)$ (here $h = 1$, and $[x^{2k+2}] I_0(2x) = 1/((k+1)!)^2$);
- $Q_2(q) = 15q^3 - 90q^2 + 175q - 100$ and $Q_3(q) = 105q^4 - 1050q^3 + 3955q^2 - 6335q + 3325$, the polynomials of [DS, Remark 4.4].

$q \setminus k$	1	2	3	4	5
3	6	20	70	252	924
5	36	400	4 900	63 504	853 776
7	90	1 860	44 730	1 172 556	32 496 156
9	168	5 120	190 120	7 939 008	357 713 664
11	270	10 900	551 950	32 232 060	2 070 891 900
13	396	19 920	1 281 420	96 807 312	8 175 770 064
25	1 656	182 400	26 926 200	4 890 642 624	1 038 711 332 736
27	1 950	234 260	37 849 630	7 550 855 676	1 767 675 505 596

(By the Main Theorem, for $m = q$, $\text{rank } L(X) = Q_k(q) + 1$ and $L(X)$ is primitive in every cell of the table.)

The zero-free point set. For the proof we use a set of points of the same cardinality. For a tuple M in T let $\nu(M)$ be the largest number of pairwise disjoint pairs of indices $\{a, b\}$ with $y_b = -y_a$, and put

$$\Gamma' := \{y \in T^{2k+1} : \nu(y) = k\}.$$

Lemma 3.1. $|\Gamma'| = Q_k(q)$.

Proof. In each class $\{u, -u\}$, a tuple M has a entries equal to u and $\bar{a}$ equal to $-u$; the class contributes $\min(a, \bar{a})$ disjoint negation pairs, and $\nu(M)$ is the sum of these contributions, since negation pairs never leave a class. If $\nu(y) = k$ for $y \in T^{2k+1}$, then exactly one entry is left unpaired, with a value w . Appending $-w$ gives an N -tuple closed under the involution. Conversely, if an N -tuple is closed under the involution, deleting its last entry z leaves a $(2k+1)$ -tuple with $\nu = k$ whose unpaired value is $-z$, so the last entry is determined by the others. This is a bijection onto the closed N -tuples, which number $Q_k(q)$ by Lemma 2.2. ■

4. Warm-up: the cubic case $q = 3$

For $q = 3$ the proof of §5 applies, but there is a shorter proof that also produces an explicit basis. It works over every field; we write it over F_3 . Put $n' := 2k+1$. Then $C = F_3[y_1, \dots, y_{n'}]/(y_i^2)$ has the basis $y_U := \prod_{i \in U} y_i$, $U \subseteq [n'] := \{1, \dots, n'\}$, and $D(a, b) = b - a$, so $D_J = \prod_{(a,b) \in J} (y_b - y_a)$ over the k pairs $(a < b)$ of J not containing 0. Order the monomials lexicographically with $y_1 > y_2 > \dots > y_{n'}$.

For $U \subseteq [n']$, read $i = 1, \dots, n'$ as a path: an up-step if $i \in U$, a down-step if $i \notin U$. Let $h(i) := |U \cap [i]| - |[i]| \setminus U$, and let $\mathcal{U} := \{U : h(i) \geq -1 \text{ for all } i\}$ (paths that never go below -1).

Theorem 4.1. For every $k \geq 0$, $\dim_{F_3}(D_J : J \in \mathcal{J})C = C(2k+2, k+1) = Q_k(3)$. The lexicographic leading monomials of the ideal are exactly the y_U , $U \in \mathcal{U}$, and its graded ranks are $C(n', u) - C(n', u+2)$ in degree $u = k, \dots, n'$. Hence Conjecture 1.2 of [DS] holds for the cubic Fermat variety of every even dimension.

Proof. (i) *Leading monomials of certificate elements.* For $T \subseteq [n']$ and $J \in \mathcal{J}$ with $\{0, c\} \in J$, the element $y_T D_J$ is $y_c^{[c \in T]} \cdot \prod_{(a,b) \in J} f_{a,b}$, where $f_{a,b} = y_b - y_a$ if $\{a,b\} \cap T = \emptyset$ («untouched»), $f_{a,b} = y_a y_b$ if $a \in T \not\subseteq b$, $f_{a,b} = -y_a y_b$ if $b \in T \not\subseteq a$, and $f_{a,b} = 0$ if $\{a,b\} \subseteq T$. The factors involve disjoint sets of variables, so a non-zero $y_T D_J$ is a sum of $2^{\#\text{untouched}}$ distinct squarefree monomials with coefficients ± 1 . The lexicographically largest one chooses $-y_a$ in every untouched pair: two of these monomials first differ at the smaller element of a pair where the choices differ, and that element is an a . So $LM(y_T D_J) = y_U$ with coefficient ± 1 , where $U = \{a : (a,b) \text{ untouched}\} \cup \bigcup_{\text{touched}} \{a, b\} \cup (\{c\} \cap T)$.

(ii) *Every $U \in \mathcal{U}$ is reached.* Match the steps greedily: each down-step $i \notin U$ is matched with the nearest earlier unmatched up-step. A down-step stays unmatched exactly when the path reaches a new minimum below 0 at it, so the number of unmatched down-steps is $\max(0, -\min_i h(i)) \leq 1$.

- If there is one unmatched down-step, call it c . The number of unmatched up-steps is then $|U| - (n' - |U| - 1)$, which is even.
- If there is none, $2|U| - n'$ up-steps are unmatched. This number is odd and ≥ -1 , hence ≥ 1 ; let c be the largest unmatched up-step. An even number remain.

Pair the remaining unmatched up-steps consecutively into «full pairs». The matched pairs ($a < b$), the full pairs and $\{0, c\}$ form a matching $J \in \mathcal{J}$. Let T consist of the smaller element of each full pair, together with c if $c \in U$. Then every matched pair is untouched and every full pair is touched exactly once, so by (i) $LM(y_T D_J) = \pm y_U$. Conversely, if y_U is a leading monomial as in (i), then every element of $[n'] \setminus U$ other than possibly c is the larger element b of an untouched pair whose smaller element a lies in U . This injection shows that at most one down-step is unmatched, so $U \in \mathcal{U}$.

(iii) *Counting.* Paths of length n' with u up-steps number $C(n', u)$. By reflecting the part after the first visit to -2 (André's reflection), those that visit -2 are in bijection with the paths with $n' - u - 2$ up-steps, of which there are $C(n', u+2)$. For $u \leq k-1$ the endpoint $2u - n' \leq -3$ forces a visit to -2 . So $\#\{U \in \mathcal{U} : |U| = u\} = C(n', u) - C(n', u+2)$ for $u \geq k$ and 0 otherwise. The sum telescopes to $C(n', k) + C(n', k+1) = C(2k+2, k+1)$.

(iv) *Conclusion.* Elements of a subspace with pairwise distinct leading monomials are linearly independent, in any characteristic: the leading monomial of a non-trivial combination is the largest leading monomial that occurs with a non-zero coefficient. Hence $\dim(D_J)C \geq |\mathcal{U}| = C(2k+2, k+1) = Q_k(3)$. Equality holds by Theorem 5.9 (or by Proposition 2.5), and it forces the set of leading monomials of the ideal to be exactly $\{y_U : U \in \mathcal{U}\}$, which gives the graded ranks. ■

Remarks 4.2. (1) The graded ranks $C(n', k+j) - C(n', k+j+2)$ are the ballot numbers $C(2k+2, k+1-j)(2j+1)/(k+2+j)$ (OEIS A039599). For $k = 1, \dots, 6$ they are $2, 3, 1 \cdot 5, 9, 5, 1 \cdot 14, 28, 20, 7, 1 \cdot 42, 90, 75, 35, 9, 1 \cdot 132, 297, 275, 154, 54, 11, 1 \cdot 429, 1001, 1001, 637, 273, 77, 13, 1$. (2) The proof uses the full family: step (ii) needs matchings that are arbitrary. For a subfamily the same argument only bounds the union of the leading-monomial sets, and at $q = 3, k = 2$ there are subfamilies for which the analogue of (S) fails (§9). (3) *An alternative route, not needed here.* In the N -variable form of Proposition 2.6, the span of the generators at $q = 3$ is spanned by the Specht polynomials of the two-row shape $(k+1, k)$ of S_{2k+1} , and the degree- $(N-r)$ part of the ideal can be identified with the Specht module $S^{(N-r, r)}$, whose dimension does not depend on the field [Jam]. We checked this identification at $N = 4, 6, 8$ (dimensions 6, 20, 70), but we have not written it out as a proof, and Theorem 4.1 does not depend on it.

5. The proof of Theorem B

Throughout this section, F is an arbitrary field, $q \geq 3$ is odd, $h = (q-1)/2$, and T is a set of $q-1$ elements with a fixed-point-free involution $u \mapsto -u$ (so T has h classes $\{u, -u\}$). For $m \geq 0$ put

$$C_m := F[y_1, \dots, y_m] / (y_1^{q-1}, \dots, y_m^{q-1}), \text{ so } C_F = C_{2k+1}.$$

The field and the set T never interact: F carries the ideals, T carries the point sets, and the two are compared only through dimensions and cardinalities. Every identity used is an identity of polynomials with integer coefficients. The field reappears in §5.9, where we evaluate the generators on a copy of T inside F .

The idea in one paragraph. Peeling off y_1 splits an ideal V into $q-1$ nested slices, and a point set Z into $q-1$ nested layers (Lemma 5.1). If each layer is the point set of an ideal contained in the corresponding slice, induction on m gives $\dim V \geq |Z|$. To make this possible we index both sides by down-sets of partitions: V_Λ is generated by products of divided differences and Vandermonde determinants, and Z_Λ is the set of tuples whose unpaired part has shape in Λ . The number of values of y_1 that complete a tail depends only on the shape μ of the tail (P1, §5.4). The $q-1$ shapes obtained from μ form a chain for weak dominance (§5.5), so the layers are again down-sets (P2, §5.6). And each generator of a layer lifts to the right slice by one of three explicit constructions (P3, §5.7).

5.1 Peeling one variable

Write $C_m = C_{m-1}[y_1]/(y_1^{q-1})$, where C_{m-1} is the ring in $y_2, \dots, y_m$. For an ideal $V \subseteq C_m$ and $0 \leq j \leq q-2$ put

$$V_{\leq j} := \{f \in V : \deg_{y_1} f \leq j\}, W_j(V) := \{[y_1^j]f : f \in V_{\leq j}\} \subseteq C_{m-1}.$$

Lemma 5.1 (peeling; linear algebra and counting). (i) $W_j(V)$ is an ideal of C_{m-1} . (ii) $W_j(V) \subseteq W_{j+1}(V)$ for $0 \leq j < q-2$. (iii) $\dim V = \sum_{j=0}^{q-2} \dim W_j(V)$. (iv) For $Z \subseteq T^m$, let $F(M') := \{y \in T : (y, M') \in Z\}$ for $M' \in T^{m-1}$ and $Z_{>i} := \{M' : |F(M')| > i\}$. Then $|Z| = \sum_{i=0}^{q-2} |Z_{>i}|$.

Proof. (i) For $g \in C_{m-1}$ and $f \in V_{\leq j}$, $gf \in V_{\leq j}$ and $[y_1^j](gf) = g[y_1^j]f$. (ii) If $f \in V_{\leq j}$, then $y_1 f \in V_{\leq j+1}$ and $[y_1^{j+1}](y_1 f) = [y_1^j]f$; this uses $y_1^{j+1} \neq 0$, i.e. $j+1 \leq q-2$. (iii) The map $V_{\leq j} \rightarrow W_j(V)$, $f \mapsto [y_1^j]f$, is onto with kernel $V_{\leq j-1}$, and $V = V_{\leq q-2}$. (iv) $|Z| = \sum_{M'} |F(M')|$. Since $|F(M')| \leq q-1$, each M' lies in $Z_{>i}$ for exactly the $|F(M')|$ values $i = 0, \dots, |F(M')| - 1$, so $\sum_i |Z_{>i}| = \sum_{M'} |F(M')|$. ■

5.2 Partitions, residues and weak dominance

- **Residue.** Regard $M \in T^m$ as a multiset of values. In each class $\{u, -u\}$ let $a, \bar{a}$ be the multiplicities of u and $-u$ in M . The class contributes $\min(a, \bar{a})$ negation pairs, and $\nu(M) = \sum_{\text{classes}} \min(a, \bar{a})$, since a negation pair lies inside one class and the classes are disjoint. The **residue** keeps, in each class with $a \neq \bar{a}$, $|a - \bar{a}|$ copies of the majority value.
- $\lambda(M)$ is the partition formed by the multiplicities of the residue. Distinct residual values lie in distinct classes, so $\ell(\lambda(M)) \leq h$, and $|\lambda(M)| = \sum |a - \bar{a}| = m - 2\nu(M) \equiv m \pmod{2}$.
- $\text{Par}_m^{(h)} := \{\lambda : |\lambda| \leq m, |\lambda| \equiv m \pmod{2}, \ell(\lambda) \leq h\}$. The cap $\ell(\lambda) \leq h$ matters: for example $(1^m) \notin \text{Par}_m^{(h)}$ when $m > h$. We write Par_m when h is fixed.
- **Weak dominance.** For a partition λ let $S_t(\lambda)$ be the sum of its t largest parts, padded with zeros. Put $\lambda \leq \mu$ iff $S_t(\lambda) \leq S_t(\mu)$ for all $t \geq 1$. The order is defined also between partitions of different sizes; it is a partial order, not a total one ((2) and $(1, 1, 1)$ are incomparable).
- For a $\leq$ -down-set $\Lambda \subseteq \text{Par}_m$, $Z_\Lambda := \{M \in T^m : \lambda(M) \in \Lambda\}$.
- **Tight patterns.** For $\lambda \in \text{Par}_m$ and an index set I with $|I| = m$, a tight pattern of λ on I consists of $p = (m - |\lambda|)/2$ pairwise disjoint pairs P of elements of I , together with pairwise disjoint blocks $B_1, \dots, B_{\lambda_1}$ with $|B_c| = \lambda'_c$ (the column lengths of λ), which partition the rest of I . Its **product** is $D_P \cdot \prod_c \Delta(B_c)$, where $D_P := \prod_{(a < b) \in P} D(y_a, y_b)$ and, for $B = \{b_1 < \dots < b_r\}$, $\Delta(B) := \det(y_{b_c}^{e-1})_{e,c=1..r} = \prod_{c < c'} (y_{b_c} - y_{b_{c'}})$ (the Vandermonde determinant, with $\Delta(\emptyset) = \Delta(\{b\}) = 1$).
- The blocks are indexed by the *columns* of λ . In the proof of Theorem 5.9 a row of λ carries one value and a column carries pairwise distinct values.
- $V_\Lambda \subseteq C_m$ is the ideal generated by the products of all tight patterns on $[m]$ of all $\lambda \in \Lambda$. For $m = 0$ the empty product is 1.

Lemma 5.2 (the root; the only place where the full family enters). In Par_{2k+1} , $\Lambda = \{(1)\}$ is a down-set, $V_{(1)} = (D_J : J \in \mathcal{J})C_{2k+1}$, and $Z_{(1)} = \Gamma'$, so $|Z_{(1)}| = Q_k(q)$.

Proof. If $\lambda \leq (1)$ then $S_t(\lambda) \leq 1$ for all t , so $|\lambda| \leq 1$; since $|\lambda|$ is odd, $\lambda = (1)$. A tight pattern of (1) on $[2k+1]$ consists of k disjoint pairs and one singleton block (with $\Delta = 1$), so its product is D_P for a set P of k disjoint pairs covering $[2k+1] \setminus \{c\}$. Adding the pair $\{0, c\}$ gives a matching $J \in \mathcal{J}$, and every J arises in this way; since D is antisymmetric, $D_P = \pm D_J$. So $V_{(1)}$ contains D_J for every matching J : this is where the whole family $\mathcal{J}$ is used, and a subfamily would give a smaller ideal. Finally $\lambda(M) = (1)$ iff $\nu(M) = k$, so $Z_{(1)} = \Gamma'$, and $|\Gamma'| = Q_k(q)$ by Lemma 3.1. ■

5.3 The theorem

Theorem 5.3. For every field F , every odd $q \geq 3$, every $m \geq 0$ and every $\leq$ -down-set $\Lambda \subseteq \text{Par}_m^{(h)}$, $\dim_F V_\Lambda \geq |Z_\Lambda|$.

Corollary 5.4. *Theorem B holds, and so does (S) for every $k \geq 0$ and every $q = p^v$, p an odd prime. Hence the Main Theorem.*

Proof of the Corollary. By Lemma 5.2 and Theorem 5.3 with $m = 2k+1$ and $\Lambda = \{(1)\}$, $\dim_F(D_J : J \in \mathcal{J})C_F \geq |\Gamma'| = Q_k(q)$. The equality clause and the integral statement of Theorem B are Theorems 5.9 and 5.9' below (with Lemma 5.2, which holds verbatim over Z); for (S) the reverse inequality is also Proposition 2.5. Proposition 2.5 then turns (S) into the Main Theorem. ■

The proof of Theorem 5.3 is by induction on m , with q fixed, and occupies §5.4–§5.8. We fix $m \geq 1$, a down-set $\Lambda \subseteq \text{Par}_m$, and peel y_1 (Lemma 5.1).

5.4 (P1) The fibres depend only on the residue partition

Take $Z = Z_\Lambda$ in Lemma 5.1(iv). Fix $M' \in T^{m-1}$ (indices $2, \dots, m$) with $\lambda(M') = \mu$, residual values $u_1, \dots, u_\ell$ in distinct classes, and multiplicities $\mu_1 \geq \dots \geq \mu_\ell$. Adding a value y for the index 1 gives:

- $y = -u_j$: the class of u_j gains a negation pair, and $\lambda(y, M') = \mu - e_j$ (one box removed from row j , re-sorted);
- $y = u_j$: $\lambda(y, M') = \mu + e_j$;
- y in one of the $h - \ell$ classes absent from the residue (where M' has equal multiplicities, possibly $0 = 0$): $\lambda(y, M') = \mu \sqcup 1$ (a new part equal to 1). There are $2(h - \ell) = q - 1 - 2\ell$ such y .

These are $\ell + \ell + (q - 1 - 2\ell) = q - 1 = |T|$ values. Hence $|F(M')| = F_\Lambda(\mu)$, where

$$F_\Lambda(\mu) := \#\{j : \mu - e_j \in \Lambda\} + \#\{j : \mu + e_j \in \Lambda\} + (q - 1 - 2\ell) \cdot [\mu \sqcup 1 \in \Lambda],$$

which depends only on μ . So $Z_{>1} = Z_{\Lambda_1}$ with $\Lambda_1 := \{\mu \in \text{Par}_{m-1} : F_\Lambda(\mu) > 1\}$. All the options lie in Par_m ; their size is $|\mu| \pm 1 \leq m$, with the parity of m , and $\mu \sqcup 1$ only occurs when $\ell < h$.

5.5 (Chain) The options form a chain

Let $\bar{r}_j$ be the last row of μ of length μ_j , and r_j the first. Removing a box from row j lowers the last row of that length, and adding a box to row j raises the first one, so for all $t \geq 1$:

$$S_t(\mu - e_j) = S_t(\mu) - [t \geq \bar{r}_j], S_t(\mu + e_j) = S_t(\mu) + [t \geq r_j], S_t(\mu \sqcup 1) = S_t(\mu) + [t > \ell].$$

List the $L := q - 1$ options by position $p = 1, \dots, L$:

$$\mu - e_1 \leq \mu - e_2 \leq \dots \leq \mu - e_\ell \leq \mu \sqcup 1 = \dots = \mu \sqcup 1 \leq \mu + e_\ell \leq \dots \leq \mu + e_1$$

with $\mu - e_p$ at position $p \leq \ell$, the $q - 1 - 2\ell$ copies of $\mu \sqcup 1$ at positions $\ell < p \leq L - \ell$, and $\mu + e_j$ at position $L + 1 - j$.

Lemma 5.5 (inequality of partitions). *The options at positions $1, \dots, L$ are weakly increasing for $\leq$. Consequently the options lying in Λ form an initial segment, and $F_\Lambda(\mu)$ is the length of that segment.*

Proof. $\bar{r}_j$ is non-decreasing in j (rows are sorted), so $\mu - e_j \leq \mu - e_{j+1}$. Next, $\mu - e_\ell \leq \mu \leq \mu \sqcup 1$. Then $\mu \sqcup 1 \leq \mu + e_\ell$, because $r_\ell \leq \ell < t$ whenever $[t > \ell] = 1$. Finally r_j is non-decreasing in j , so $\mu + e_{j+1} \leq \mu + e_j$. Since Λ is a down-set, $\text{opt}_p \in \Lambda$ implies $\text{opt}_{p'} \in \Lambda$ for all $p' \leq p$. ■

In particular the removals in Λ are $\mu - e_r, \dots, \mu - e_\ell$ for some $r \leq \ell$. Since rows of equal length give equal partitions, the options at the positions of equal rows coincide, so $r = 0$ or r is the last row of its length. The additions in Λ are $\mu + e_j$ for $j_0 \leq j \leq \ell$, with j_0 (if it exists) the first row of its length. Put $\iota := [\mu \sqcup 1 \in \Lambda]$ (so $\iota = 0$ when $\ell = h$). If some addition lies in Λ , then $r = \ell$, and $\iota = 1$ whenever $\ell < h$; if $\iota = 1$ then $r = \ell$. Therefore

$$F_\Lambda(\mu) = r + (q - 1 - 2\ell) \cdot \iota + (\ell - j_0 + 1) \cdot [j_0 \text{ exists}]. \quad (5.1)$$

5.6 (P2) The layers are down-sets

Proposition 5.6 (inequality of partitions). *If $\mu, \tilde{\mu} \in \text{Par}_{m-1}$ and $\mu \leq \tilde{\mu}$, then $\text{opt}_p(\mu) \leq \text{opt}_p(\tilde{\mu})$ for every position p . Hence $F_\Lambda(\mu) \geq F_\Lambda(\tilde{\mu})$, and every Λ_1 is a $\leq$ -down-set of Par_{m-1} .*

The second sentence follows from the first: if $\text{opt}_p(\tilde{\mu}) \in \Lambda$ then $\text{opt}_p(\mu) \in \Lambda$, so $F_\Lambda(\tilde{\mu}) \leq F_\Lambda(\mu)$, and $\tilde{\mu} \in \Lambda_i$ implies $\mu \in \Lambda_i$. We prove the first sentence. Write $\ell = \ell(\mu)$, $\tilde{\ell} = \ell(\tilde{\mu})$, and mark with a tilde the quantities of $\tilde{\mu}$. Recall that $|\mu| \equiv |\tilde{\mu}| \pmod{2}$, since both lie in Par_{m-1} . Two facts are used throughout.

- **(T)** If $t \geq 1$ is **tight**, i.e. $S_t(\mu) = S_t(\tilde{\mu})$, then $\mu_t \geq \tilde{\mu}_t$ (from $S_{t-1}(\mu) \leq S_{t-1}(\tilde{\mu})$) and $\mu_{t+1} \leq \tilde{\mu}_{t+1}$ (from $S_{t+1}(\mu) \leq S_{t+1}(\tilde{\mu})$).
- **(G)** Each comparison below has the form $S_t(\mu) + \delta \leq S_t(\tilde{\mu}) + \tilde{\delta}$ with $\delta, \tilde{\delta} \in \{-1, 0, 1\}$. If $\delta - \tilde{\delta} \leq 1$, it can only fail at a tight t with $\delta = \tilde{\delta} + 1$. The one place where $\delta - \tilde{\delta} = 2$ is possible is treated explicitly, in (d1).

The positions fall into seven cases. A removal of one partition is never compared with an addition of the other, because $\ell + \tilde{\ell} \leq 2h = q - 1$ leaves the middle positions of μ in between.

case	positions	option of μ	option of $\tilde{\mu}$	only possible failure	why it cannot occur
(a)	$p \leq \min(\ell, \tilde{\ell})$	$\mu - e_p$	$\tilde{\mu} - e_p$	tight t , $\tilde{r}_p \leq t < \bar{r}_p$	forces $S_{p-1}(\tilde{\mu}) < S_{p-1}(\mu)$
(b)	$j \leq \min(\ell, \tilde{\ell})$	$\mu + e_j$	$\tilde{\mu} + e_j$	tight t , $\underline{r}_j \leq t < \bar{r}_j$	forces $S_j(\tilde{\mu}) < S_j(\mu)$
(c)	middle	$\mu \sqcup 1$	$\tilde{\mu} \sqcup 1$	$\ell < t \leq \tilde{\ell}$	the rows $\ell+1, \dots, t$ of $\tilde{\mu}$, (5.2)
(d1)	$\ell < p \leq \tilde{\ell}$	$\mu \sqcup 1$	$\tilde{\mu} - e_p$	$\delta - \tilde{\delta} = 2$	(5.2) and parity
(d2)	$\ell < j \leq \tilde{\ell}$	$\mu \sqcup 1$	$\tilde{\mu} + e_j$	none	(5.2)
(e1)	$\tilde{\ell} < p \leq \ell$	$\mu - e_p$	$\tilde{\mu} \sqcup 1$	none	chain $\mu - e_p \leq \mu \leq \tilde{\mu} \leq \tilde{\mu} \sqcup 1$
(e2)	$\tilde{\ell} < j \leq \ell$	$\mu + e_j$	$\tilde{\mu} \sqcup 1$	tight $t \leq \tilde{\ell}$	forces $S_j(\tilde{\mu}) < S_j(\mu)$

(a) *Removal against removal*, $p \leq \min(\ell, \tilde{\ell})$. We need $S_t(\mu) - [t \geq \bar{r}_p] \leq S_t(\tilde{\mu}) - [t \geq \tilde{r}_p]$. The only danger is a tight t with $\tilde{r}_p \leq t < \bar{r}_p$. Then $t \geq p$, and rows $p, \dots, t+1$ of μ all have length μ_p ; in particular $\mu_t = \mu_{t+1} = \mu_p$. Since $t+1 > \tilde{r}_p$, $\tilde{\mu}_{t+1} < \tilde{\mu}_p$. By (T), $\tilde{\mu}_{t+1} \geq \mu_{t+1} = \mu_t \geq \tilde{\mu}_t \geq \tilde{\mu}_{t+1}$, so $\tilde{\mu}_t = \tilde{\mu}_{t+1} = \mu_p$ and $\tilde{\mu}_p > \mu_p$. Hence $S_t(\tilde{\mu}) - S_{p-1}(\tilde{\mu}) = \sum_{s=p}^t \tilde{\mu}_s \geq \tilde{\mu}_p + (t-p)\mu_p > (t-p+1)\mu_p = S_t(\mu) - S_{p-1}(\mu)$. Since t is tight, $S_{p-1}(\tilde{\mu}) < S_{p-1}(\mu)$, contradicting $\mu \leq \tilde{\mu}$.

(b) *Addition against addition*, $j \leq \min(\ell, \tilde{\ell})$ (position $L+1-j$). We need $S_t(\mu) + [t \geq \underline{r}_j] \leq S_t(\tilde{\mu}) + [t \geq \tilde{r}_j]$. The only danger is a tight t with $\underline{r}_j \leq t < \tilde{r}_j$. Then $t < \tilde{r}_j \leq j$, so $\mu_t = \mu_{t+1} = \mu_j$, and $\tilde{\mu}_t > \tilde{\mu}_j$ because t lies before the first row of $\tilde{\mu}$ of length $\tilde{\mu}_j$. By (T), $\tilde{\mu}_{t+1} \geq \mu_{t+1} = \mu_j = \mu_t \geq \tilde{\mu}_t \geq \tilde{\mu}_{t+1}$, so $\tilde{\mu}_t = \tilde{\mu}_{t+1} = \mu_j > \tilde{\mu}_j$. Every $\tilde{\mu}_s$ with $s > t$ is at most μ_j , and $\tilde{\mu}_j < \mu_j$, so $\sum_{s=t+1}^j \tilde{\mu}_s < (j-t)\mu_j = \sum_{s=t+1}^j \mu_s$. With tightness at t , $S_j(\tilde{\mu}) < S_j(\mu)$: a contradiction.

(c) *Middle against middle*. We need $S_t(\mu) + [t > \ell] \leq S_t(\tilde{\mu}) + [t > \tilde{\ell}]$. If $\ell \geq \tilde{\ell}$ this is immediate. If $\ell < \tilde{\ell}$ and $\ell < t \leq \tilde{\ell}$, the rows $\ell+1, \dots, t$ of $\tilde{\mu}$ are non-empty, so

$$S_t(\tilde{\mu}) \geq S_\ell(\tilde{\mu}) + (t - \ell) \geq S_\ell(\mu) + (t - \ell) = |\mu| + (t - \ell) \geq |\mu| + 1 = S_t(\mu) + 1. \quad (5.2)$$

Layout of the positions. If $\ell = \tilde{\ell}$ only (a), (b), (c) occur. If $\ell < \tilde{\ell}$, positions $\ell+1, \dots, \tilde{\ell}$ compare a middle option of μ with a removal of $\tilde{\mu}$ (case d1), and positions $L+1-j$ with $\ell < j \leq \tilde{\ell}$ compare a middle option of μ with an addition of $\tilde{\mu}$ (case d2). These positions are middle positions of μ because $\ell + \tilde{\ell} \leq 2h = L$. If $\ell > \tilde{\ell}$, the roles are reversed (cases e1, e2).

(d1) $\ell < p \leq \tilde{\ell}$: $\mu \sqcup 1$ against $\tilde{\mu} - e_p$. We need $S_t(\mu) + [t > \ell] \leq S_t(\tilde{\mu}) - [t \geq \tilde{r}_p]$. Here $\delta - \tilde{\delta}$ can be 2, so we check every t .

- $t \leq \ell$: both brackets vanish, since $\tilde{r}_p \geq p > \ell \geq t$.
- $t \geq \tilde{\ell}$: $S_t(\tilde{\mu}) = |\tilde{\mu}| \geq |\mu| + (\tilde{\ell} - \ell) \geq |\mu| + 1$ by (5.2) at $t = \tilde{\ell}$; by parity $|\tilde{\mu}| \geq |\mu| + 2$, which suffices.
- $\ell + 2 \leq t < \tilde{\ell}$: $S_t(\tilde{\mu}) \geq |\mu| + (t - \ell) \geq |\mu| + 2$ by (5.2).
- $t = \ell + 1 < \tilde{\ell}$: if $t < \tilde{r}_p$, (5.2) suffices. If $t \geq \tilde{r}_p$, then $\tilde{r}_p = p = \ell + 1$ and we need $S_{\ell+1}(\tilde{\mu}) \geq |\mu| + 2$. If $\tilde{\mu}_{\ell+1} \geq 2$ this holds, since $S_\ell(\tilde{\mu}) \geq |\mu|$. If $\tilde{\mu}_{\ell+1} = 1$, then row $\ell+2 \leq \tilde{\ell}$ of $\tilde{\mu}$ also has length 1, so $\tilde{r}_{\ell+1} \geq \ell + 2 > t$, a contradiction.

(d2) $\ell < j \leq \tilde{\ell}$: $\mu \sqcup 1$ against $\tilde{\mu} + e_j$. We need $S_\ell(\mu) + [t > \ell] \leq S_\ell(\tilde{\mu}) + [t \geq \tilde{\ell}]$. For $t \leq \ell$ this is immediate. For $\ell < t \leq \tilde{\ell}$ it follows from (5.2), and for $t > \tilde{\ell}$ from $|\tilde{\mu}| \geq |\mu| + 1$.

(e1) $\tilde{\ell} < p \leq \ell$: $\mu - e_p$ against $\tilde{\mu} \sqcup 1$ (which exists because $\tilde{\ell} < \ell \leq h$). Here $\mu - e_p \leq \mu \leq \tilde{\mu} \leq \tilde{\mu} \sqcup 1$.

(e2) $\tilde{\ell} < j \leq \ell$: $\mu + e_j$ against $\tilde{\mu} \sqcup 1$. We need $S_\ell(\mu) + [t \geq r_j] \leq S_\ell(\tilde{\mu}) + [t > \tilde{\ell}]$. For $t > \tilde{\ell}$ the right bracket is 1, so this is immediate. For $t \leq \tilde{\ell}$ with $t \geq r_j$ we need strictness. Suppose t is tight. Then $r_j \leq t \leq \tilde{\ell} < j$, so $\mu_t = \mu_{t+1} = \mu_j \geq 1$. By (T), $\tilde{\mu}_t \leq \mu_t = \mu_j$, so $\tilde{\mu}_s \leq \mu_j$ for every $s \geq t$. Moreover $\tilde{\mu}_j = 0$, since $j > \tilde{\ell}$. Hence $\sum_{s=t+1}^j \tilde{\mu}_s < (j-t)\mu_j = \sum_{s=t+1}^j \mu_s$, and with tightness $S_j(\tilde{\mu}) < S_j(\mu)$, a contradiction.

The positions $1, \dots, L$ are covered by (a)–(e) exactly once: removals occupy positions $\leq \ell$ or $\leq \tilde{\ell}$, additions the last ℓ or $\tilde{\ell}$ positions, and the middle the rest. The degenerate cases $\ell = 0$, $\ell = h$ (no middle positions) and $\tilde{\ell} = h$ are included. ■

5.7 (P3) Every tight pattern of a layer lies in the right slice

We need three identities of polynomials, all with integer coefficients, hence valid in C_m over any field.

Lemma 5.7 (identities over $\mathbb{Z}$). (i) For $0 \leq t \leq q-2$: $[y_1^{q-2-t}] D(y_1, y_b) = (-1)^{t+1} y_b^t$. (ii) Let $S = \{b_1 < \dots < b_r\}$ and $\varepsilon_{b_c} := (-1)^{r+c}$. Then $\sum_{b \in S} \varepsilon_b \Delta(S \setminus b) y_b^t = 0$ for $0 \leq t \leq r-2$, and $= \Delta(S)$ for $t = r-1$. (iii) If $1 \notin B$ and $|B| \leq q-2$, then $\Delta(B \cup \{1\}) = \Pi_{c \in B} (y_c - y_1) \cdot \Delta(B)$ has y_1 -degree $|B|$ in C_m , with top coefficient $(-1)^{|B|} \Delta(B)$.

Proof. (i) The coefficient is $(-1)^{q-2-t} y_b^t$, and $q-2$ is odd. (ii) The left side is the expansion, along its last row, of the Vandermonde matrix of S with its last row (y_b^{r-1}) replaced by (y_b^t) . For $t \leq r-2$ two rows coincide. (iii) This is the factorisation of the Vandermonde determinant; $y_1^{|B|} \neq 0$ in C_m because $|B| \leq q-2$. ■

Proposition 5.8 (linear algebra). For every $i \in \{0, \dots, q-2\}$: $V_{\Lambda_i} \subseteq W_{q-2-i}(V_\Lambda)$. Here $V_{\Lambda_i} \subseteq C_{m-1}$ is built on the indices $2, \dots, m$.

Proof. Since $W_{q-2-i}(V_\Lambda)$ is an ideal (Lemma 5.1(i)), it suffices to treat a generator. Let $\mu \in \Lambda_i$, so $\Phi := F_\Lambda(\mu) \geq i + 1$, and let G be the product of a tight pattern of μ on $\{2, \dots, m\}$, with pairs P and column blocks B_c . By Lemma 5.1(ii) it is enough to show $G \in W_j(V_\Lambda)$ for some $j \leq q-1-\Phi$: then $j \leq q-2-i$. We use (5.1). Since $\Phi \geq 1$, at least one term of (5.1) is non-zero, and we split into three cases, which are mutually exclusive by construction: (α) j_0 exists; (β) j_0 does not exist and $\iota = 1$; (γ) neither, so that $\Phi = r \geq 1$. In case (β), $\iota = 1$ forces $\ell < h$, since $\mu \sqcup 1 \in \text{Par}_m$ requires $\ell(\mu) + 1 \leq h$.

(α) j_0 exists ($\lambda := \mu + e_{j_0} \in \Lambda$). The column $c^* := \mu_{j_0} + 1$ of μ has height $j_0 - 1$, because j_0 is the first row of its length; let $B := B_{c^*}$ (possibly empty). Put the index 1 into B . The result is a tight pattern of λ on $[m]$: column c^* of λ has height j_0 , and the number of pairs $(m - |\lambda|)/2 = ((m-1) - |\mu|)/2$ is unchanged. Its product lies in V_Λ , has y_1 -degree $j_0 - 1$, and by Lemma 5.7(iii) its top y_1 -coefficient is $\pm G$ (note $|B| = j_0 - 1 \leq h - 1 \leq q - 2$). So $G \in W_{j_0-1}(V_\Lambda)$. By (5.1), $r = \ell$ here. If $\ell < h$, then $\iota = 1$ and $\Phi = \ell + (q-1-2\ell) + (\ell - j_0 + 1) = q - j_0$. If $\ell = h$, then $q-1-2\ell = 0$ and $\Phi = 2\ell - j_0 + 1 = q - j_0$. In both cases $j_0 - 1 = q - 1 - \Phi$.

(β) no j_0 , and $\iota = 1$ ($\lambda := \mu \sqcup 1 \in \Lambda$, $\ell < h$). Put the index 1 into the first column block B_1 , of height ℓ ; column 1 of λ has height $\ell + 1$, and the number of pairs is unchanged. The product has y_1 -degree $\ell \leq q-2$ and top coefficient $\pm G$, so $G \in W_\ell(V_\Lambda)$. By (5.1), $r = \ell$ and $\Phi = q-1-\ell$, i.e. $\ell = q-1-\Phi$. (For $\mu = \emptyset$ the block is $\{1\}$, $\Delta = 1$, and the degree is 0.)

(γ) only removals ($\Phi = r \geq 1$, $\lambda := \mu - e_r \in \Lambda$). The column μ_r of μ has height r , because r is the last row of its length; call its block S , $|S| = r$, and write $G = \Delta(S) \cdot G'$, where G' involves no index of $S \cup \{1\}$. For $b \in S$, the pairs $P \cup \{(1, b)\}$ together with the blocks B_c ($c \neq \mu_r$) and $S \setminus \{b\}$ form a tight pattern of λ on $[m]$: column μ_r of λ has height $r-1$, and the number of pairs $(m - |\lambda|)/2 = ((m-1) - |\mu|)/2 + 1$ has grown by one. Hence

$$f := \sum_{b \in S} \varepsilon_b \Delta(S \setminus b) D(y_1, y_b) G' \in V_\Lambda.$$

By Lemma 5.7(i),(ii), $[y_1^{q-2-t}] f = (-1)^{t+1} (\sum_b \varepsilon_b \Delta(S \setminus b) y_b^t) G'$. This vanishes for $t \leq r-2$ and equals $\pm \Delta(S) G' = \pm G$ for $t = r-1$. So $\deg_{y_1} f \leq q-1-r$, with top coefficient $\pm G$, and $G \in W_{q-1-r}(V_\Lambda) = W_{q-1-\Phi}(V_\Lambda)$. ■

The three constructions land exactly in the slice $q - 1 - \Phi$, the largest index the proof can afford (see Remark 5.10(2)).

5.8 The induction

Proof of Theorem 5.3. Induction on m , with F , q and T fixed.

- **Base $m = 0$.** $Par_0 = \{\emptyset\}$. If $\Lambda = \{\emptyset\}$, then $V_\Lambda = C_0 = F$ (the empty product) and $Z_\Lambda = T^0$ is one point: $1 \geq 1$. If $\Lambda = \emptyset$, both sides are 0.
- **Relabelling.** A permutation of the index set sends every tight-pattern product to $\pm$ another one, since D and Δ are antisymmetric. Hence V_Λ is invariant under relabelling, and so is $|Z_\Lambda|$. So the statement for $m - 1$ variables applies to the variables $y_2, \dots, y_m$.
- **Step.** Let $m \geq 1$. By Lemma 5.1(iii), Proposition 5.8, the induction hypothesis, and Lemma 5.1(iv),

$$\dim V_\Lambda = \sum_{i=0}^{q-2} \dim W_{q-2-i}(V_\Lambda) \geq \sum_{i=0}^{q-2} \dim V_{\Lambda_i} \geq \sum_{i=0}^{q-2} |Z_{\Lambda_i}| = \sum_{i=0}^{q-2} |Z_{>i}| = |Z_\Lambda|.$$

The induction hypothesis applies because each Λ_i is a down-set of Par_{m-1} (Proposition 5.6). The identity $Z_{\Lambda_i} = Z_{>i}$ is §5.4. ■

5.9 Equality

For an ideal $I \subseteq F[y_1, \dots, y_m]$ let $gr(I)$ be the ideal of top-degree forms: its degree- d part consists of 0 and the homogeneous parts of degree d of the elements of I of degree exactly d . (It is an ideal: for $f, g \in I$ of degree d , the form $f^{top} + g^{top}$ is 0 or $(f + g)^{top}$, and $y_i f^{top} = (y_i f)^{top}$.)

Lemma 5.8' (top-degree forms). (i) If $I \subseteq F[y]$ is generated by $f_1, \dots, f_s$, then $gr(I) \supseteq (f_1^{top}, \dots, f_s^{top})$. (ii) If $\dim_F F[y]/I < \infty$, then $\dim_F F[y]/gr(I) = \dim_F F[y]/I$.

Proof. (i) $f_i \in I$, so $f_i^{top} \in gr(I)$. (ii) Fix a degree-compatible monomial order. For $f \neq 0$, the leading monomial of f is that of f^{top} . The ideal $gr(I)$ is homogeneous, so each of its non-zero elements g has $g^{top} \in gr(I)$, and $g^{top} = f^{top}$ for some $f \in I$. Hence $gr(I)$ and I have the same monomial ideal of leading monomials, and by Macaulay's basis theorem [CLO] the standard monomials are a basis of both quotients. ■

(The initial ideal of I is a monomial ideal and contains the leading monomials of the f_i , not the f_i ; it is $gr(I)$ that contains the forms themselves. The previous version of this paper wrote «initial ideal» for gr in the proofs of Theorem 5.9 and Proposition 9.1.)

Theorem 5.9. If $\text{char } F$ does not divide $q - 1$, then $\dim_F V_\Lambda = |Z_\Lambda|$ for every $m \geq 0$ and every $\leq$ -down-set $\Lambda \subseteq Par_m^{(h)}$. In particular, for such F , equality holds in Theorem B (for every field, see Theorem 5.9'), and $\dim_{F_p}(D_J : J \in \mathcal{J})C = Q_k(q)$ for $q = p^v$ without any appeal to topology.

Proof. By Theorem 5.3 it suffices to prove $\leq$. The generators of V_Λ have integer coefficients, so $\dim_F V_\Lambda$ does not change under extension of the field, and we may assume that F contains the group μ_{q-1} of $(q-1)$ -th roots of unity. Since $\text{char } F \nmid q - 1$, the polynomial $y^{q-1} - 1$ is separable, $|\mu_{q-1}| = q - 1$, and $-1 \in \mu_{q-1}$ is not 1 (because $q - 1$ is even and $\text{char } F \neq 2$). So we may take $T := \mu_{q-1}$ with the involution $u \mapsto -u$, which has no fixed point.

Evaluation. For $a, b \in T$, the identity $(a + b)D(a, b) = b^{q-1} - a^{q-1} = 0$ shows $D(a, b) = 0$ unless $b = -a$, and $D(a, -a) = \sum_u (-1)^u (-1)^{q-2-u} a^{q-2} = -(q-1)a^{q-2} \neq 0$. A Vandermonde determinant $\Delta(B)$ is non-zero at a point iff the values on B are pairwise distinct. So the product of a tight pattern $(P, B_1, \dots, B_\lambda)$ of λ is non-zero at $M \in T^m$ iff every pair of P is a negation pair of M and every block B_c carries pairwise distinct values.

Support. Suppose such a product is non-zero at M . Let M' be the multiset of the $|\lambda|$ values of M outside P , and ν' its multiplicity partition. A value occurs at most once in each column block, so the t most frequent values of M' occupy at most $\min(t, \lambda'_c)$ places of column c ; summing over c , $S_t(\nu') \leq \sum_c \min(t, \lambda'_c) = S_t(\lambda)$, so $\nu' \leq \lambda$. The pairs of P do not change the residue, and passing from M' to its residue replaces, in each class, the two multiplicities $a, \bar{a}$ by one part $|a - \bar{a}| \leq \max(a, \bar{a})$; so the parts of $\lambda(M)$ are bounded, one by one, by distinct parts of ν' , and $\lambda(M) \leq \nu'$. Hence $\lambda(M) \leq \lambda$, and if $\lambda \in \Lambda$ then $M \in Z_\Lambda$. So the union Σ of the supports of the

generators of V_Λ is contained in Z_Λ . (It is in fact equal: for $M \in Z_\Lambda$, take $\nu(M)$ disjoint negation pairs of M and put the residue into the columns of $\lambda(M)$, one value per row.)

Degeneration. Let $g_1, \dots, g_s$ be the generators of V_Λ , viewed as homogeneous polynomials in $F[y_1, \dots, y_m]$, and put $I := (g_1, \dots, g_s) + (y_1^{q-1} - 1, \dots, y_m^{q-1} - 1)$. Since $y_i^{q-1} - 1 = \prod_{u \in T}(y_i - u)$, the Chinese remainder theorem gives $F[y]/(y_i^{q-1} - 1) \cong F^{T^m}$, the ring of functions on T^m . In a finite product of fields, the ideal generated by some functions consists of the functions that vanish outside the union of their supports. So I is the ideal of the functions vanishing on $T^m \setminus \Sigma$, and $\dim_F F[y]/I = (q-1)^m - |\Sigma|$. The top-degree form of g_i is g_i and that of $y_i^{q-1} - 1$ is y_i^{q-1} . So, by Lemma 5.8'(i), $I' := (g_1, \dots, g_s) + (y_1^{q-1}, \dots, y_m^{q-1}) \subseteq \text{gr}(I)$, and by Lemma 5.8'(ii)

$$\dim_F C_m/V_\Lambda = \dim_F F[y]/I' \geq \dim_F F[y]/\text{gr}(I) = \dim_F F[y]/I = (q-1)^m - |\Sigma|,$$

that is, $\dim_F V_\Lambda \leq |\Sigma| \leq |Z_\Lambda|$. ■

Complement (the associated graded ring). Assume, as in the proof, that $\text{char } F \nmid q-1$ and $\mu_{q-1} \subseteq F$, and read $V_\Lambda + (y_i^{q-1})$ as the ideal $(g_i) + (y_i^{q-1})$ of $F[y_1, \dots, y_m]$. In the situation of the proof, Theorem 5.3 turns every inequality into an equality: $\Sigma = Z_\Lambda$ and $I' = \text{gr}(I)$. So $V_\Lambda + (y_1^{q-1}, \dots, y_m^{q-1})$ is the ideal of top-degree forms of the ideal of the finite set $T^m \setminus Z_\Lambda \subseteq \mu_{q-1}^m$, and C_m/V_Λ is the associated graded ring of the ring of functions on $T^m \setminus Z_\Lambda$, for the degree filtration. In other words, the tight-pattern products give a complete description of this associated graded ring. This is the same kind of statement as the Garsia–Procesi and Tanisaki descriptions of the associated graded rings of orbits [GP], [Ta], here for a new family of point sets, indexed by down-sets of partitions under weak dominance.

Theorem 5.9' (equality in every characteristic; freeness over $\mathbb{Z}$). *For every field F , every odd $q \geq 3$, every $m \geq 0$ and every $\leq$ -down-set $\Lambda \subseteq \text{Par}_m^{(h)}$, $\dim_F V_\Lambda = |Z_\Lambda|$. Moreover, let $V_{\Lambda, \mathbb{Z}}$ be the ideal of $C_{m, \mathbb{Z}} := \mathbb{Z}[y_1, \dots, y_m]/(y_1^{q-1}, \dots, y_m^{q-1})$ generated by the tight-pattern products of the elements of Λ . Then $C_{m, \mathbb{Z}}/V_{\Lambda, \mathbb{Z}}$ is a free abelian group of rank $(q-1)^m - |Z_\Lambda|$, each of its graded pieces is free, and the dimension over F of each graded piece of V_Λ does not depend on F .*

Proof. The ring $C_{m, \mathbb{Z}}$ is free of rank $(q-1)^m$ on the monomials, and $V_{\Lambda, \mathbb{Z}}$ is homogeneous, so $M := C_{m, \mathbb{Z}}/V_{\Lambda, \mathbb{Z}} = \bigoplus_d M_d$ with each M_d a finitely generated abelian group. The image of $V_{\Lambda, \mathbb{Z}}$ in $C_{m, \mathbb{Z}} \otimes F = C_m$ generates V_Λ (the generators have integer coefficients), so by right exactness of $\otimes$, $C_m/V_\Lambda \cong M \otimes F$, degree by degree. Write $M_d \cong \mathbb{Z}^{\rho_d} \oplus (\text{finite})$. Then $\dim_F (M_d \otimes F) = \rho_d + c_d(F)$, where $c_d(F)$ is the number of cyclic factors of order divisible by $\text{char } F$ in the finite part ($c_d(F) = 0$ if $\text{char } F = 0$). With $F = \mathbb{Q}$, where Theorem 5.9 applies because 0 does not divide $q-1$, $\sum_d \rho_d = \dim_{\mathbb{Q}} C_{m, \mathbb{Q}}/V_{\Lambda, \mathbb{Q}} = (q-1)^m - |Z_\Lambda|$. So for every field F , $\dim_F C_m/V_\Lambda \geq (q-1)^m - |Z_\Lambda|$, that is, $\dim_F V_\Lambda \leq |Z_\Lambda|$, and Theorem 5.3 gives equality. Equality forces $c_d(F) = 0$ for every d . Taking $F = F_p$ for every prime p , no M_d has torsion, so M is free and $\dim_F (C_m/V_\Lambda)_d = \rho_d$ for every F . ■

Remark 5.10. (1) *Where the full family enters.* Only at the root, Lemma 5.2. The down-set closure of $\{(1)\}$ then carries the symmetry of $\mathcal{J}$ through every layer. §9 shows that no argument valid for all subfamilies can exist. (2) *No slack.* In Proposition 5.8 the slice $q-1-F$ cannot be lowered: replacing it by $q-2-F$ makes the statement false (§12, negative controls). (3) *The characteristic.* Theorem 5.3 and, by Theorem 5.9', equality hold over every field. Theorem 5.9 is needed only over $\mathbb{Q}$. The earlier measurements of equality with $\text{char } F \mid q-1$ (§12) are instances of Theorem 5.9'. (4) *Leading monomials.* In the computed cells, the degree-lexicographic leading monomials of $(D_\nu)C$ are the reflection $e \mapsto (q-2) - e$ of the lexicographic standard monomials of Γ' in the sense of [CM] (§12). By the Complement to Theorem 5.9, the Hilbert function part of this statement holds in general: the graded dimensions of C_m/V_Λ are those of the degree filtration on the functions on $T^m \setminus Z_\Lambda$.

5.10 A worked example: $(q, m) = (9, 4)$

Here $h = 4$, T has 8 elements, and $\text{Par}_4^{(4)} = \{\emptyset, (2), (1,1), (4), (3,1), (2,2), (2,1,1), (1,1,1,1)\}$ has ten down-sets. Take $\Lambda := \{\emptyset, (1,1), (2)\}$, the down-set generated by (2) . The tails have $m-1 = 3$ coordinates, with shapes in $\text{Par}_3 = \{(1), (3), (2,1), (1,1,1)\}$, and $|Z_{(1)}| = 168$, $|Z_{(3)}| = 8$, $|Z_{(2,1)}| = 144$, $|Z_{(1,1,1)}| = 192$ (total $8^3 = 512$).

tail shape μ	its chain (positions 1...8)	options in Λ	$F_\Lambda(\mu)$	case of Prop. 5.8	slice $q - 1 - \Phi$
(1)	$\emptyset \cdot (1,1) \times 6 \cdot (2)$	all	8	$(\alpha), j_0 = 1$	0
(3)	$(2) \cdot (3,1) \times 6 \cdot (4)$	first	1	$(\gamma), r = 1$	7
(2,1)	$(1,1) \cdot (2) \cdot (2,1,1) \times 4 \cdot (2,2) \cdot (3,1)$	first two	2	$(\gamma), r = 2$	6
(1,1,1)	$(1,1) \times 3 \cdot (1,1,1,1) \times 2 \cdot (2,1,1) \times 3$	first three	3	$(\gamma), r = 3$	5

So $\Lambda_0 = \text{Par}_3$, $\Lambda_1 = \{(1), (2,1), (1,1,1)\}$, $\Lambda_2 = \{(1), (1,1,1)\}$, and $\Lambda_i = \{(1)\}$ for $3 \leq i \leq 7$: all down-sets, as P2 says. Lemma 5.1(iv) gives $|Z_\Lambda| = 512 + 504 + 360 + 5 \cdot 168 = 2216$, and the computed dimension of V_Λ is 2216 (§12). For the down-set $\{\emptyset, (1,1)\}$ the tail (1) has $F = 7$ (the addition (2) is missing) and falls into case (β) , with slice $\ell = 1 = q - 1 - F$; there $|Z_\Lambda| = 1896$. Case (β) needs a middle option in Λ and no addition, which is why it does not occur for the first down-set.

6. Odd degrees that are not prime powers: the colour reduction

In §6–§7, $m \geq 3$ is odd and p is a prime dividing m . Write $m = q \cdot r$ with $q = p^v$, $v \geq 1$, and $p \nmid r$; so r is odd, and $r = 1$ is the case treated in §2–§5. Put $F := \bar{F}_p$, $n = 2k$ with $k \geq 1$, $N = n + 2$,

$$F[G] := F[t_1, \dots, t_{n+1}] / (t_1^m - 1, \dots, t_{n+1}^m - 1), I := (\bar{\psi}_J : J \in \mathcal{J}) \cdot F[G],$$

where $\bar{\psi}_J$ denotes the image of ψ_J in $F[G]$ (in §2 it denoted the image in $F_p[G]$; the ideal of $F[G]$ it generates is the extension of that ideal).

By Proposition 2.1, $\dim_{F_p} F_p[G] / (\bar{\psi}_J : J \in \mathcal{J}) \geq m^{n+1} - |\Gamma_{\mathcal{J}}|$ for every p , with equality at every $p \mid m$ if and only if $H_n(X)/L(X)$ is torsion free. The $\bar{\psi}_J$ have coefficients in F_p , so $(\bar{\psi}_J)F[G] = (\bar{\psi}_J)F_p[G] \otimes_{F_p} F$, and dimensions do not change under this extension of the field; so

$$\dim_F I \leq |\Gamma_{\mathcal{J}}|, (6.1)$$

and **the Main Theorem at m is equivalent to $\dim_F I \geq |\Gamma_{\mathcal{J}}|$ for every prime $p \mid m$** . This section reduces that inequality to an inequality for one family of ideals in two sets of variables, the bipartite ideals; §7 proves it. All tensor products are over F , and a «unit» is an invertible element of the ring in which the statement is made. The idea is that $F[G]$ is a product of local rings indexed by the ways of colouring the coordinates with r -th roots of unity; on each factor the generators $\bar{\psi}_J$ either vanish or become pure tensors, and the pure tensors split along the colour classes.

6.1 The splitting

Lemma 6.1 (splitting). Put $R_c := \bigotimes_{i=1}^{n+1} F[t_i] / ((t_i - c_i)^q)$ for $c = (c_1, \dots, c_{n+1}) \in \mu_r^{n+1}$, where $\mu_r \subseteq F^\times$ is the group of r -th roots of unity. Then the product of the quotient maps $\pi_c : F[G] \rightarrow R_c$ is an isomorphism $F[G] \cong \prod_{c \in \mu_r^{n+1}} R_c$, and for every ideal $I' \subseteq F[G]$, $\dim_F I' = \sum_c \dim_F \pi_c(I')$, where $\pi_c(I')$ is the ideal of R_c generated by the images of any set of generators of I' .

Proof. In characteristic p , $(t^r - 1)^q = t^m - 1$. The polynomial $t^r - 1$ is separable, since $r \neq 0$ in F , so $t^r - 1 = \prod_{\zeta \in \mu_r} (t - \zeta)$ with r distinct roots, and $t^m - 1 = \prod_{\zeta} (t - \zeta)^q$. The ideals $((t - \zeta)^q)$ are pairwise comaximal, and the Chinese remainder theorem gives $F[t] / (t^m - 1) \cong \prod_{\zeta} F[t] / ((t - \zeta)^q)$. Now $F[G]$ is the tensor product of the $n + 1$ one-variable rings, and the tensor product over a field distributes over finite products of algebras. In a finite product of rings every ideal is the product of its images, which are generated by the images of its generators; dimensions add. ■

6.2 Which generators survive

Fix $c \in \mu_r^{n+1}$ and put $c_0 := (c_1 \cdots c_{n+1})^{-1}$, so that $\prod_{i=0}^{n+1} c_i = 1$. In R_c every $t_i - c_i$ is nilpotent, with $(t_i - c_i)^q = 0$.

Lemma 6.2. (i) For $1 \leq i \leq n+1$: if $c_i \neq 1$, then $t_i - 1$ is a unit of R_c ; if $c_i = 1$, it is nilpotent. (ii) Let $1 \leq j < l \leq n+1$ and $u := t_j t_l \in R_c$. If $c_j c_l \neq 1$, then $\varphi(u) = 0$ in R_c . If $c_j c_l = 1$, then $\varphi(u) = (\text{unit}) \cdot (u - 1)^{q-1}$. (iii) If some pair $\{j_i, k_i\}$ of J , $i \geq 1$, has $c_{j_i} c_{k_i} \neq 1$, then $\pi_c(\bar{\psi}_J) = 0$. If $c_{j_i} c_{k_i} = 1$ for $i = 1, \dots, k$, then also $c_0 c_{k_0} = 1$, and

$$\pi_c(\bar{\psi}_J) = (\text{unit}) \cdot (t_{k_0} - 1) \cdot \prod_{i=1}^k (t_{k_i} - 1) (t_{j_i} t_{k_i} - 1)^{q-1}. \quad (6.2)$$

Proof. (i) $t_i - 1 = (c_i - 1) + (t_i - c_i)$, a non-zero constant plus a nilpotent if $c_i \neq 1$. (ii) In $F[u]$, $(u - 1)\varphi(u) = u^m - 1 = \prod_{\xi \in \mu_r} (u - \xi)^q$, so $\varphi(u) = (u - 1)^{q-1} \cdot \prod_{\xi \neq 1} (u - \xi)^q$. In R_c , $t_j^q = c_j^q$ and $t_l^q = c_l^q$ (Frobenius), so $(u - c_j c_l)^q = 0$. If $\xi_0 := c_j c_l \neq 1$, then $\xi_0 \in \mu_r \setminus \{1\}$ and the factor $(u - \xi_0)^q$ vanishes. If $c_j c_l = 1$, then $u - 1$ is nilpotent and each $u - \xi = (1 - \xi) + (u - 1)$, $\xi \neq 1$, is a unit. (iii) The first sentence follows from (ii). For the second, $1 = \prod_{i=0}^{n+1} c_i = c_0 c_{k_0} \cdot \prod_{i \geq 1} c_{j_i} c_{k_i}$; then apply (ii) to each factor $\varphi(t_{j_i} t_{k_i})$. ■

6.3 Colour classes and the tensor decomposition

For $\zeta \in \mu_r$ put $\mathcal{C}_\zeta := \{i \in \{0, \dots, n+1\} : c_i = \zeta\}$ (the index 0 carries the colour c_0). Since r is odd, $\zeta \neq \zeta^{-1}$ for $\zeta \neq 1$, and $\mu_r \setminus \{1\}$ splits into $(r-1)/2$ pairs $\{\zeta, \zeta^{-1}\}$; fix a set $\mathcal{R}_r$ of representatives. The sets $\mathcal{C}_1$ and $\mathcal{C}_\zeta \cup \mathcal{C}_{\zeta^{-1}}$ ($\zeta \in \mathcal{R}_r$) partition $\{0, \dots, n+1\}$; we call them the **blocks** of c . We say that $J \in \mathcal{J}$ is **compatible** with c if $c_a c_b = 1$ for every pair $\{a, b\}$ of J ; by Lemma 6.2(iii) this means $c_{j_i} c_{k_i} = 1$ for $i = 1, \dots, k$, and $\pi_c(\bar{\psi}_J) = 0$ unless J is compatible.

Lemma 6.3. A pair $\{a, b\}$ has $c_a c_b = 1$ iff $\{a, b\} \subseteq \mathcal{C}_1$, or $\{a, b\}$ meets both $\mathcal{C}_\zeta$ and $\mathcal{C}_{\zeta^{-1}}$ for some $\zeta \in \mathcal{R}_r$. Hence J is compatible with c iff it consists of a perfect matching P of $\mathcal{C}_1$ and, for each $\zeta \in \mathcal{R}_r$, of the pairs of a bijection $\sigma_\zeta : \mathcal{C}_\zeta \rightarrow \mathcal{C}_{\zeta^{-1}}$; and every such choice $(P, (\sigma_\zeta))$ is a compatible matching (it is a perfect matching of $\{0, \dots, n+1\}$, hence an element of $\mathcal{J}$: this is where the full family is used, cf. Remark 6.10(3)). A compatible matching exists iff $|\mathcal{C}_1|$ is even and $|\mathcal{C}_\zeta| = |\mathcal{C}_{\zeta^{-1}}|$ for every ζ ; we then call c a **compatible colouring**. ■

Group the tensor factors of R_c by blocks: $R_c = R_{c,1} \otimes \bigotimes_{\zeta \in \mathcal{R}_r} R_{c,\zeta}$, where $R_{c,1} := \bigotimes_{i \in \mathcal{C}_1 \setminus \{0\}} F[t_i]/((t_i - 1)^q)$ and $R_{c,\zeta} := \bigotimes_{i \in (\mathcal{C}_\zeta \cup \mathcal{C}_{\zeta^{-1}}) \setminus \{0\}} F[t_i]/((t_i - c_i)^q)$ (an empty tensor product is F ; the variable t_0 does not exist). Each factor of (6.2) involves the variables of one pair of J , and a pair of a compatible J lies in one block; the factor of the pair $\{0, k_0\}$ is $t_{k_0} - 1$, which lies in the block of k_0 . So, for a compatible $J = (P, (\sigma_\zeta))$,

$$\pi_c(\bar{\psi}_J) = (\text{unit}) \cdot g_P \otimes \bigotimes_{\zeta \in \mathcal{R}_r} g_{\sigma_\zeta}$$

where $g_P \in R_{c,1}$ and $g_{\sigma_\zeta} \in R_{c,\zeta}$ are the products of the factors of (6.2) belonging to the pairs of P and of σ_ζ (an empty product is 1; the unit of (6.2) is absorbed into one of them, and units do not change the ideals below). Put $I_{c,1} := (g_P : P)$ and $I_{c,\zeta} := (g_{\sigma_\zeta} : \sigma)$, the ideals generated over all perfect matchings P of $\mathcal{C}_1$ and all bijections $\sigma : \mathcal{C}_\zeta \rightarrow \mathcal{C}_{\zeta^{-1}}$.

Lemma 6.4 (tensor products of ideals). Let R, R' be finite-dimensional commutative F -algebras and $I = (g_\alpha)$, $I' = (g'_\beta)$ ideals. Then the ideal of $R \otimes R'$ generated by the $g_\alpha \otimes g'_\beta$ is $I \otimes I'$, and $\dim(I \otimes I') = \dim I \cdot \dim I'$.

Proof. $I \otimes I'$ is an ideal of $R \otimes R'$ (pure tensors span, and $(r \otimes r')(x \otimes x') = rx \otimes r'x'$), and it contains the $g_\alpha \otimes g'_\beta$. Conversely, for $x = \sum_\alpha r_\alpha g_\alpha \in I$ and $x' = \sum_\beta r'_\beta g'_\beta \in I'$, $x \otimes x' = \sum_{\alpha, \beta} (r_\alpha \otimes r'_\beta) (g_\alpha \otimes g'_\beta)$. A basis of I tensored with a basis of I' is a basis of $I \otimes I'$. ■

Proposition 6.5 (tensor decomposition). If c is not compatible, $\pi_c(I) = 0$. If c is compatible, $\pi_c(I) = I_{c,1} \otimes \bigotimes_{\zeta \in \mathcal{R}_r} I_{c,\zeta}$ and $\dim \pi_c(I) = \dim I_{c,1} \cdot \prod_\zeta \dim I_{c,\zeta}$.

Proof. Units do not change the ideal generated, so by Lemmas 6.2 and 6.3 $\pi_c(I)$ is generated by the pure tensors $g_P \otimes \bigotimes_{\zeta \in \mathcal{R}_r} g_{\sigma_\zeta}$ over all independent choices of P and (σ_ζ) . Apply Lemma 6.4 repeatedly. ■

6.4 The block of colour 1

In $F[t]/((t - 1)^q)$ put $y := t - t^{-1}$. Lemma 2.3 holds verbatim here: its proof uses only $(t - 1)^q = 0$, $q = p^v$ and $2 \in F^\times$. So $R_{c,1} = F[y_i : i \in \mathcal{C}_1 \setminus \{0\}]/(y_i^q)$, with $y_i = (\text{unit}) \cdot (t_i - 1)$ and $t_i t_l - 1 = (\text{unit}) \cdot (y_i + y_l)$. By Lemma 2.4(i), for a pair $\{i < l\} \subseteq \mathcal{C}_1 \setminus \{0\}$, $(t_l - 1)(t_i - 1)^{q-1} = (\text{unit}) \cdot y_i y_l \cdot D(y_i, y_l)$; if $0 \in \mathcal{C}_1$, the pair $\{0, k_0\}$ contributes $t_{k_0} - 1 = (\text{unit}) \cdot y_{k_0}$. The indices in these monomial factors are the elements of $\mathcal{C}_1 \setminus \{0\}$, each once, so $g_P =$

$(\text{unit}) \cdot Y_I \cdot D_P$ with $Y_I := \prod_{i \in \mathcal{C}_I \setminus 0} y_i$ and D_P the product of the $D(y_p, y_l)$ over the pairs of P avoiding 0. By the argument of Proposition 2.5 (the kernel of multiplication by Y_I on $R_{c,I}$ is exactly $(y_i^{q-1} : i \in \mathcal{C}_I \setminus 0)$, so $f \mapsto Y_I f$ induces an injective map from the quotient $C(\mathcal{C}_I \setminus 0)$ into $R_{c,I}$),

$$\dim I_{c,I} = \dim_F(D_P : P) \cdot C(\mathcal{C}_I \setminus 0), \quad C(S) := F[y_i : i \in S] / (y_i^{q-1} : i \in S).$$

Let $N_I(c)$ be the number of tuples $(w_i)_{i \in \mathcal{C}_I \setminus 0} \in (\mu_q \setminus \{1\})^{\mathcal{C}_I \setminus 0}$ such that, with $w_0 := (\prod w_i)^{-1}$ if $0 \in \mathcal{C}_I$, the multiset $\{w_i : i \in \mathcal{C}_I\}$ lies in $\mu_q \setminus \{1\}$ and is closed under inversion. (For $|\mathcal{C}_I|$ even the condition $w_0 \neq 1$ is automatic: otherwise the other $|\mathcal{C}_I| - 1$ entries, an odd number, would form a multiset in $\mu_q \setminus \{1\}$ closed under the fixed-point-free inversion.)

Lemma 6.6. $\dim I_{c,I} \geq N_I(c)$.

Proof. If $|\mathcal{C}_I|$ is odd, both sides are 0: there is no perfect matching, and a multiset closed under the fixed-point-free inversion of $\mu_q \setminus \{1\}$ has even size. Let $|\mathcal{C}_I| = 2k'$.

- If $0 \in \mathcal{C}_I$, relabel $\mathcal{C}_I \setminus \{0\}$ increasingly as $\{1, \dots, 2k' - 1\}$. Write $\mathcal{J}_j$ for the set $\mathcal{J}$ in dimension $2j$. The perfect matchings P of $\mathcal{C}_I$ are the matchings $J \in \mathcal{J}_{k'-1}$, and $D_P = \pm D_J$. So $\dim I_{c,I} = \dim(D_J : J \in \mathcal{J}_{k'-1}) C_{2k'-1} \geq Q_{k'-1}(q)$ by Lemma 5.2 and Theorem 5.3 (over every field, here F). By Lemma 2.2 (whose proof uses only that q is odd), $N_I(c) = Q_{k'-1}(q)$. (For $k' = 1$ both sides are $q - 1$.)
- If $0 \notin \mathcal{C}_I$, relabel $\mathcal{C}_I$ as $\{1, \dots, 2k'\}$. In $\text{Par}_{2k'}^{(h)}$ the set $\{\emptyset\}$ is a down-set, since $\lambda \leq \emptyset$ forces $\lambda = \emptyset$. A tight pattern of $\emptyset$ on $[2k']$ has k' pairs and no blocks: it is a perfect matching P , with product D_P . So $(D_P : P) C_{2k'} = V_{\{\emptyset\}}$, and Theorem 5.3 gives $\dim \geq |Z_{\{\emptyset\}}|$, the number of $2k'$ -tuples in T whose multiset is closed under the involution. A bijection $\mu_q \setminus \{1\} \rightarrow T$ carrying inversion to $u \mapsto -u$ (both are sets of $q - 1$ elements with a fixed-point-free involution) identifies this number with $N_I(c)$. (For $k' = 0$, $I_{c,I} = R_{c,I} = F$ and $N_I(c) = 1$.) ■

6.5 The pair blocks

Fix $\zeta \in \mathcal{R}_r$, and put $A := \mathcal{C}_\zeta$, $B := \mathcal{C}_{\zeta^{-1}}$, $A^* := A \setminus \{0\}$, $B^* := B \setminus \{0\}$. For $i \in A^*$ put $x_i := \zeta^{-1} t_i - 1$, and for $l \in B^*$ put $z_l := (\zeta t_l)^{-1} - 1$.

Lemma 6.7. (i) $R_{c,\zeta} = F[x_i (i \in A^*), z_l (l \in B^*)] / (x_i^q, z_l^q)$. (ii) For $i \in A^*, l \in B^*$: $t_i t_l - 1 = (\text{unit}) \cdot (x_i - z_l)$. (iii) Let $\alpha := |A^*|$ and $\beta := |B^*|$ (the numbers of x - and z -variables, as in §7), and relabel $A^* = \{1, \dots, \alpha\}$, $B^* = \{1, \dots, \beta\}$. If $0 \notin A \cup B$ and $\alpha = \beta =: a$, then $I_{c,\zeta} \cong I_a^{\text{bal}} := (\prod_{i=1}^a (x_i - z_{\sigma(i)})^{q-1} : \sigma \in S_a)$. If $0 \in B$ and $\alpha = \beta + 1$, then $I_{c,\zeta} \cong I_a^{\text{ph}}$, where $I_a^{\text{ph}} := (\prod_{i \neq i_0} (x_i - z_{\sigma(i)})^{q-1} : i_0 \in [a+1], \sigma : [a+1] \setminus \{i_0\} \rightarrow [a] \text{ bijective})$; if $0 \in A$ and $\beta = \alpha + 1$, then $I_{c,\zeta} \cong I_a^{\text{ph}}$ after exchanging the names x and z . In all other cases $I_{c,\zeta} = 0$.

Proof. (i) $x_i = \zeta^{-1}(t_i - \zeta)$ is a unit multiple of the generator of the maximal ideal of the local ring $F[t_i] / ((t_i - \zeta)^q)$, so $x_i^q = 0$ and $1, x_p, \dots, x_i^{q-1}$ is a basis. Similarly $\varepsilon_l := \zeta t_l - 1$ is nilpotent with $\varepsilon_l^q = 0$, and $z_l = (1 + \varepsilon_l)^{-1} - 1 = -\varepsilon_l(1 + \varepsilon_l)^{-1}$ is a unit multiple of ε_l . (ii) $t_i = \zeta(1 + x_i)$ and $t_l = \zeta^{-1}(1 + z_l)^{-1}$, so $t_i t_l - 1 = (x_i - z_l)(1 + z_l)^{-1}$. (iii) Every $t_s - 1$ with $s \in A^* \cup B^*$ is a unit (Lemma 6.2(i)), since $c_s \neq 1$. By (6.2) and (ii), the factor of a pair $\{i, l\}$, $i \in A^*, l \in B^*$, is $(\text{unit}) \cdot (x_i - z_l)^{q-1}$, and the factor of the pair $\{0, k_0\}$ is a unit. If $0 \notin A \cup B$, the bijections $A \rightarrow B$ are the $\sigma \in S_a$, $a = \alpha = \beta$. If $0 \in B$, a bijection $\sigma : A \rightarrow B$ sends exactly one $i_0 \in A$ to 0, and the other pairs give the product over $i \neq i_0$. If $0 \in A$, the same holds with the roles of A and B exchanged; since $q - 1$ is even, $(x - z)^{q-1} = (z - x)^{q-1}$, so exchanging the names of the variables carries this ideal onto I_a^{ph} . If $|A| \neq |B|$, there is no bijection. (For $\alpha = \beta = 0$, $I_0^{\text{bal}} = (1)$ is the ring F ; for $\beta = 0$ in the second case, $I_0^{\text{ph}} = (1)$ is the ring $F[x_1] / (x_1^q)$; and in the third case with $\alpha = 0$, for instance $\mathcal{C}_\zeta = \{0\}$ and $\mathcal{C}_{\zeta^{-1}} = \{l\}$, it is the ring $F[z_l] / (z_l^q)$, of dimension q .) ■

6.6 The count factors in the same way

Let Ω be a set of q elements. For $a \geq 0$ put

$$N_{\text{bal}}(a, q) := \#\{(\xi, \eta) \in \Omega^a \times \Omega^a : \text{the multisets of } \xi \text{ and } \eta \text{ are equal}\} = \sum_{c_1 + \dots + c_q = a} (a! / (c_1! \dots c_q!))^2,$$

$$N_{\text{ph}}(a, q) := \#\{(\xi, \eta) \in \Omega^{a+1} \times \Omega^a : \text{the multiset of } \eta \text{ is contained in that of } \xi\}.$$

Lemma 6.8. $|\Gamma_{\mathcal{J}}| = \sum_c N_1(c) \cdot \prod_{\zeta \in \mathcal{R}_r} N_{\zeta}(c)$, the sum over the compatible colourings, where $N_{\zeta}(c) := N_{\text{bal}}(\alpha, q)$ in the first case of Lemma 6.7(iii) and $N_{\zeta}(c) := N_{\text{ph}}(\min(\alpha, \beta), q)$ in the other two (α, β as there).

Proof. By the first paragraph of the proof of Lemma 2.2 (which uses nothing about m), $\Gamma_{\mathcal{J}}$ is the set of $g = (g_1, \dots, g_{n+1}) \in \mu_m^{n+1}$ such that, with $g_0 := (g_1 \cdots g_{n+1})^{-1}$, all $g_i \neq 1$ and the multiset $\{g_0, \dots, g_{n+1}\}$ splits into inverse pairs; since m is odd, inversion has no fixed point on $\mu_m \setminus \{1\}$, and «splits into inverse pairs» means «closed under inversion». Here $\mu_m \subset C$ is the group of complex m -th roots of unity (the rest of §6 works over $\bar{F}_p$). As $\gcd(q, r) = 1$, $\mu_m = \mu_q \times \mu_r$; write $g_i = c_i w_i$ with $c_i \in \mu_r$, $w_i \in \mu_q$ ($i = 0, \dots, n+1$). Then $c_0 = (c_1 \cdots c_{n+1})^{-1}$, $w_0 = (w_1 \cdots w_{n+1})^{-1}$, $g_i g_l = 1$ iff $c_i c_l = 1$ and $w_i w_l = 1$, and $g_i \neq 1$ iff $(c_i, w_i) \neq (1, 1)$. If $g \in \Gamma_{\mathcal{J}}$ its pairing into inverse pairs is a matching compatible with $c := (c_1, \dots, c_{n+1})$, so c is a compatible colouring. Fix a compatible c . By Lemma 6.3, a tuple g with colouring c lies in $\Gamma_{\mathcal{J}}$ iff (1) the multiset $\{w_i : i \in \mathcal{C}_1\}$ lies in $\mu_q \setminus \{1\}$ and is closed under inversion, and (2) for each $\zeta \in \mathcal{R}_r$ the multiset $\{w_i : i \in \mathcal{C}_{\zeta}\}$ equals the multiset $\{w_l^{-1} : l \in \mathcal{C}_{\zeta^{-1}}\}$. (Condition (2) allows the value $w_i = 1$, rightly: for $i \in \mathcal{C}_{\zeta} \cup \mathcal{C}_{\zeta^{-1}}$ one has $g_i \neq 1$ whatever w_i is, since $c_i \neq 1$.) Each condition involves the w_i of one block. The only coupling between blocks is w_0 : every block that satisfies its condition and does not contain 0 has $\prod w_i = 1$ over its indices, so, when all these blocks satisfy their conditions, $w_0 = (\prod_{i \in \mathcal{X} \setminus 0} w_i)^{-1}$ for the block $\mathcal{X} \ni 0$, a function of the w_i of $\mathcal{X}$ alone. Conversely, given w -values on each block that satisfy its condition, where for the block $\mathcal{X} \ni 0$ the value w_0 is computed from that block alone, the blocks not containing 0 have product 1, so this value is the true $(w_1 \cdots w_{n+1})^{-1}$ and the assembled tuple lies in $\Gamma_{\mathcal{J}}$. Hence the count over c is the product of the block counts. (For a compatible c , exactly one of the three cases of Lemma 6.7(iii) holds for each ζ .) The block $\mathcal{C}_1$ gives $N_1(c)$, and a balanced pair block gives $N_{\text{bal}}(\alpha, q)$ ($\Omega = \mu_q$, $\xi := (w_i)_{i \in \mathcal{C}_{\zeta}}$, $\eta := (w_l^{-1})_{l \in \mathcal{C}_{\zeta^{-1}}}$). For a pair block with $0 \in \mathcal{C}_{\zeta^{-1}}$, let ξ be the w_i on $\mathcal{C}_{\zeta}$ ($\beta + 1$ of them) and η the w_l^{-1} on $\mathcal{C}_{\zeta^{-1}} \setminus 0$; the condition says $\text{multiset}(\xi) = \text{multiset}(\eta) \cup \{w_0^{-1}\}$, with $w_0^{-1} = \prod \xi (\prod \eta)^{-1}$. If $\text{multiset}(\eta) \subseteq \text{multiset}(\xi)$, the difference is one element ξ^* and $\prod \xi (\prod \eta)^{-1} = \xi^*$, so the condition holds; conversely it implies $\text{multiset}(\eta) \subseteq \text{multiset}(\xi)$. So this block gives $N_{\text{ph}}(\beta, q)$, and so does the case $0 \in \mathcal{C}_{\zeta}$ by the symmetry $\xi \leftrightarrow \eta$, $w \mapsto w^{-1}$. ■

6.7 The reduction

Proposition 6.9 (colour reduction). Let $k \geq 1$, let $m = q \cdot r$ be odd, $q = p^v$, $p \nmid r$, and suppose that $\dim_F I_a^{\text{bal}} \geq N_{\text{bal}}(a, q)$ and $\dim_F I_a^{\text{ph}} \geq N_{\text{ph}}(a, q)$ for every $a \geq 0$, where $F = \bar{F}_p$. Then $\dim_{F_p} F_p[G]/(\bar{\psi}_{\mathcal{J}} : \mathcal{J} \in \mathcal{J}) = m^{n+1} - |\Gamma_{\mathcal{J}}|$, so $H_n(X)/L(X)$ has no p -torsion.

Proof. By Lemma 6.1, Proposition 6.5, Lemmas 6.6 and 6.7 and the hypothesis, $\dim_F I = \sum_c \dim \pi_c(I) \geq \sum_c N_1(c) \cdot \prod_{\zeta} N_{\zeta}(c)$, the sum over the compatible colourings, and this is $|\Gamma_{\mathcal{J}}|$ by Lemma 6.8. With (6.1), $\dim_F I = |\Gamma_{\mathcal{J}}|$, and the per-prime clause of Proposition 2.1 concludes. ■

Remarks 6.10. (1) Only the inequality $\geq$ is needed on each block; the reverse inequality is global, (6.1). (2) For $r = 1$ there is one colouring, $\mathcal{C}_1 = \{0, \dots, n+1\}$, and Proposition 6.9 is Corollary 5.4. (3) For a subfamily $K \subsetneq \mathcal{J}$ the splitting of Lemma 6.1 still holds, but the matchings of K compatible with c are no longer a product set, and Proposition 6.5 fails; this is consistent with §9. (4) The splitting of Lemma 6.1 over F_p , with two factors per variable (the $(t-1)$ -primary part and the rest), was used earlier by the author as a memory-saving device for machine computations of these ranks, and to prove that any torsion is confined to one block [Rep, *The Localization Theorem, The Block Decomposition*]. Proposition 6.9 refines it over $\bar{F}_p$ to one factor per colouring and identifies every factor.

Example 6.11 ($k = 1$, $m = 15$). Here $N = 4$, $\mathcal{J}$ has three matchings, and $Q_1(15) = 3 \cdot 14 \cdot 13 = 546$. The compatible colourings, grouped by the shape of their blocks, contribute as follows (i, l, s are indices other than 0; $Q_0(q) = q - 1$):

blocks of the colouring	$p = 3$ ($q = 3, r = 5$): colourings $\times$ product	$p = 5$ ($q = 5, r = 3$): colourings $\times$ product
$\mathcal{C}_1 = \{0, 1, 2, 3\}$	$1 \times Q_1(3) = 6$	$1 \times Q_1(5) = 36$
$\mathcal{C}_1 = \{0, i\}$, and one balanced block $1 + 1$	$12 \times Q_0(3) \cdot N_{bal}(1, 3) = 12 \times 6$	$6 \times Q_0(5) \cdot N_{bal}(1, 5) = 6 \times 20$
$\mathcal{C}_1 = \{i, l\}$, and a phantom block $\{0\}$ against $\{s\}$	$12 \times Q_0(3) \cdot N_{ph}(0, 3) = 12 \times 6$	$6 \times Q_0(5) \cdot N_{ph}(0, 5) = 6 \times 20$
$\mathcal{C}_1 = \emptyset$, and one phantom block $2 + 2$ containing 0	$12 \times N_{ph}(1, 3) = 12 \times 15$	$6 \times N_{ph}(1, 5) = 6 \times 45$
$\mathcal{C}_1 = \emptyset$, a balanced block $1 + 1$ and a phantom block $\{0\}$ against $\{s\}$	$24 \times N_{bal}(1, 3) \cdot N_{ph}(0, 3) = 24 \times 9$	none (μ_3 has one pair of inverse colours)
total	61 colourings, 546	19 colourings, 546

At both primes the sum is $Q_1(15)$, as Lemma 6.8 says; the literal computation of §12.3 gives the same 546. (At $k = 1$ the Main Theorem is Degtyarev's theorem [De14]; the example only shows the bookkeeping.)

7. The bipartite count

This section proves the hypothesis of Proposition 6.9, and so completes the proof of the Main Theorem. The method is that of §5, with two changes: the box is q instead of $q - 1$, and the single partition of §5 is replaced by a *pair* of partitions, one for each of the two sets of variables. Throughout, $q \geq 3$ is odd and F is a field in which the binomial coefficients $C(q - 1, t)$, $0 \leq t \leq q - 1$, are non-zero: for instance $\text{char } F = 0$, or $\text{char } F = p$ and $q = p^v$ (then $C(q - 1, t) \equiv (-1)^t \pmod{p}$, because $(1 + u)^{q-1} = (1 + u^q)(1 + u)^{-1}$ in $F_p[[u]]$). Let Ω be a set of q elements. For finite index sets A and B put $\alpha := |A|$, $\beta := |B|$ and

$$R_{A,B} := F[x_i (i \in A), z_l (l \in B)] / (x_i^q, z_l^q).$$

(The symbols $A, B, R_{A,B}$ are local to this section.) Points are pairs $(\xi, \eta) \in \Omega^A \times \Omega^B$.

THEOREM C (the bipartite count). *For every $a \geq 0$:*

$$\dim_F I_a^{bal} \geq N_{bal}(a, q) \text{ and } \dim_F I_a^{ph} \geq N_{ph}(a, q) = N_{bal}(a + 1, q),$$

where $I_a^{bal} \subseteq R_{[a],[a]}$ and $I_a^{ph} \subseteq R_{[a+1],[a]}$ are the ideals of Lemma 6.7(iii) and N_{bal}, N_{ph} are the counts of §6.6.

Theorem C is the root case of Theorem 7.6 below. With Proposition 6.9 it gives the Main Theorem for m not a prime power (Corollary 7.7). Over $\bar{F}_p$ both inequalities are equalities (Corollary 7.8); in characteristic 0 they are strict in general (Remarks 7.9).

7.1 Peeling with the box q

Write $R_{A,B} = R_{A \setminus \{i_1\}, B}[[w]] / (w^q)$ with $w := x_{i_1}$ for some $i_1 \in A$. For an ideal $V \subseteq R_{A,B}$ and $0 \leq j \leq q - 1$ put $V_{\leq j} := \{f \in V : \deg_w f \leq j\}$ and $W_j(V) := \{[w^j]f : f \in V_{\leq j}\} \subseteq R_{A \setminus \{i_1\}, B}$.

Lemma 7.1 (peeling). (i) $W_j(V)$ is an ideal. (ii) $W_j(V) \subseteq W_{j+1}(V)$ for $0 \leq j < q - 1$. (iii) $\dim V = \sum_{j=0}^{q-1} \dim W_j(V)$. (iv) For $Z \subseteq \Omega^A \times \Omega^B$ and a tail M' (the coordinates other than i_1) let $F(M') := \{u \in \Omega : (u, M') \in Z\}$ and $Z_{> i_1} := \{M' : |F(M')| > i\}$; then $|Z| = \sum_{i=0}^{q-1} |Z_{> i}|$.

Proof. As for Lemma 5.1, with q in place of $q - 1$: in (ii) one uses $w^{j+1} \neq 0$, i.e. $j + 1 \leq q - 1$, and in (iv) $|F(M')| \leq q$. ■

7.2 Pairs of partitions and the bipartite down-set family

We use the partial sums S_i and the weak dominance order $\preceq$ of §5.2 (defined also between partitions of different sizes), and the rows $\bar{r}_j, \underline{r}_j$ of §5.5.

- **Residue shape.** For (ξ, η) , let X and Y be the multisets of values of ξ and η . Let R_+ keep, for each value u with $\text{mult}_X(u) > \text{mult}_Y(u)$, the difference $\text{mult}_X(u) - \text{mult}_Y(u)$ of copies of u , and let R_- be defined symmetrically. The **shape** of (ξ, η) is $\lambda(\xi, \eta) := (\lambda_+, \lambda_-)$, the multiplicity partitions of R_+ and R_- . Their supports are disjoint subsets of Ω , so $\ell(\lambda_+) + \ell(\lambda_-) \leq q$; moreover $|\lambda_+| - |\lambda_-| = \alpha - \beta$ and $|\lambda_+| + |\lambda_-| \leq \alpha + \beta$.
- $BPar_q(\alpha, \beta)$ is the set of pairs (λ_+, λ_-) of partitions with $|\lambda_+| - |\lambda_-| = \alpha - \beta$, $|\lambda_+| + |\lambda_-| \leq \alpha + \beta$ and $\ell(\lambda_+) + \ell(\lambda_-) \leq q$. It is ordered by the product order: $(\lambda_+, \lambda_-) \preceq (\mu_+, \mu_-)$ iff $\lambda_+ \preceq \mu_+$ and $\lambda_- \preceq \mu_-$. Every element of $BPar_q(\alpha, \beta)$ is the shape of some point (give the $\ell(\lambda_+) + \ell(\lambda_-) \leq q$ rows pairwise distinct values and the pairs arbitrary values); this is not used below.
- For a $\preceq$ -down-set $\Lambda \subseteq BPar_q(\alpha, \beta)$ put $Z_\Lambda := \{(\xi, \eta) \in \Omega^A \times \Omega^B : \lambda(\xi, \eta) \in \Lambda\}$.
- **Tight patterns.** A tight pattern of $\lambda = (\lambda_+, \lambda_-) \in BPar_q(\alpha, \beta)$ on (A, B) consists of a set P of $\alpha - |\lambda_+| = \beta - |\lambda_-|$ pairwise disjoint pairs $(i, l) \in A \times B$, of blocks $B^+_{i,l}, \dots, B^+_{\lambda_+, l}$ partitioning $A \setminus A(P)$ with $|B^+_{i,l}| = (\lambda_+)_i$, and of blocks $B^-_{i,l}, \dots, B^-_{\lambda_-, l}$ partitioning $B \setminus B(P)$ with $|B^-_{i,l}| = (\lambda_-)_l$. Its **product** is

$$\prod_{(i,l) \in P} (x_i - z_l)^{q-1} \cdot \prod_c \Delta(x_{B^+_{i,l}}) \cdot \prod_c \Delta(z_{B^-_{i,l}}),$$

with Δ the Vandermonde determinant of §5.2.

- $V_\Lambda \subseteq R_{A,B}$ is the ideal generated by the products of all tight patterns on (A, B) of all elements of Λ . For $\alpha = \beta = 0$ the empty product is 1.

Lemma 7.2 (the roots). (i) $\{(\emptyset, \emptyset)\}$ is a down-set of $BPar_q(a, a)$, $V_{\{(\emptyset, \emptyset)\}} = I_a^{\text{bal}}$, and $|Z_{\{(\emptyset, \emptyset)\}}| = N_{\text{bal}}(a, q)$. (ii) $\{((1), \emptyset)\}$ is a down-set of $BPar_q(a+1, a)$, $V_{\{((1), \emptyset)\}} = I_a^{\text{ph}}$, and $|Z_{\{((1), \emptyset)\}}| = N_{\text{ph}}(a, q)$. (iii) $N_{\text{ph}}(a, q) = N_{\text{bal}}(a+1, q)$.

Proof. (i) $\lambda_\pm \preceq \emptyset$ forces $\lambda_\pm = \emptyset$. A tight pattern of $(\emptyset, \emptyset)$ is a perfect matching of A with B , i.e. a bijection σ , with product $\prod_i (x_i - z_{\sigma(i)})^{q-1}$; and the shape is $(\emptyset, \emptyset)$ iff $X = Y$. (ii) In $BPar_q(a+1, a)$, $|\lambda_+| = |\lambda_-| + 1$; $\lambda_- \preceq \emptyset$ forces $\lambda_- = \emptyset$ and then $\lambda_+ \preceq (1)$ with $|\lambda_+| = 1$ forces $\lambda_+ = (1)$. A tight pattern has a pairs and one x -block of size 1 (with $\Delta = 1$), so its product is a generator of I_a^{ph} , and every generator arises. The shape is $((1), \emptyset)$ iff $Y \subseteq X$. (iii) Given $(\xi, \eta) \in \Omega^{a+1} \times \Omega^a$ with $Y \subseteq X$, append to η the one element of $X - Y$; given a balanced point in $\Omega^{a+1} \times \Omega^{a+1}$, delete the last coordinate of η . These maps are mutually inverse. ■

For instance, $N_{\text{bal}}(a, 3) = 3, 15, 93, 639, 4653$ and $N_{\text{bal}}(2, q) = 15, 45, 91, 153, 231$ for $a = 1, \dots, 5$ and $q = 3, 5, 7, 9, 11$.

Lemma 7.2' (§5.5 and §5.6 for arbitrary partitions). Let $\lambda \preceq \tilde{\lambda}$ be partitions of arbitrary sizes and lengths. Then the identities of §5.5, the chain of Lemma 5.5 for the removals and the additions, (T) of §5.6, the inequality (5.2) of §5.6 (if $\ell(\lambda) < t \leq \ell(\tilde{\lambda})$, then $S_t(\tilde{\lambda}) \geq |\lambda| + (t - \ell(\lambda))$) and the comparisons (a), (b), (c), (e2) of §5.6, recalled in the proof of Proposition 7.4, hold for λ and $\tilde{\lambda}$.

Proof. The identities of §5.5 and the chain sums are statements about one partition. The proofs of (T), (5.2), (a), (b), (c) and (e2) in §5.6 compare the partial sums S_i of the two partitions term by term and use only $\lambda \preceq \tilde{\lambda}$. In §5.6 the parity of $|\mu|$ is used only in case (d1), and the cap $\ell \leq h$ only in the layout of the positions and in case (e1); none of these is borrowed here. ■

7.3 (P1) Fibres and (Chain)

Fix a down-set $\Lambda \subseteq BPar_q(\alpha, \beta)$ with $\alpha \geq 1$, and peel $w = x_{i_1}$. Let M' be a tail with shape $\mu = (\mu_+, \mu_-) \in BPar_q(\alpha - 1, \beta)$, residual values $u_1, \dots, u_{e_+}$ of R_+ (u_j with multiplicity μ_{+j}) and $v_1, \dots, v_{e_-}$ of R_- . Adding the value u at the index i_1 changes the shape as follows:

- $u = v_j$: $(\mu_+, \mu_- - e_j)$ (one box removed from row j of μ_- , re-sorted);

- $u = u_j: (\mu_+ + e_p, \mu_-)$;
- u any of the $q - \ell_+ - \ell_-$ other values (the values whose multiplicities in the two coordinate sets of the tail agree, possibly $0 = 0$): $(\mu_+ \sqcup 1, \mu_-)$.

These are $q = |\Omega|$ values, and every resulting shape lies in $BPar_q(\alpha, \beta)$: one value is added to the x-side, and a middle value exists only when $\ell_+ + \ell_- < q$, which gives $\ell(\mu_+ \sqcup 1) + \ell(\mu_-) \leq q$. List them by position $p = 1, \dots, q$: $opt_p(\mu) := (\mu_+, \mu_- - e_p)$ for $p \leq \ell_-$; $(\mu_+ \sqcup 1, \mu_-)$ for $\ell_- < p \leq q - \ell_+$ (the **middle** positions); $(\mu_+ + e_{q+1-p}, \mu_-)$ for $p > q - \ell_+$. Hence $|F(M)| = F_\Lambda(\mu) := \#\{p : opt_p(\mu) \in \Lambda\}$, and $Z_{>i} = Z_{\Lambda_i}$ with $\Lambda_i := \{\mu \in BPar_q(\alpha - 1, \beta) : F_\Lambda(\mu) > i\}$.

Lemma 7.3 (chain). $opt_1(\mu) \leq opt_2(\mu) \leq \dots \leq opt_q(\mu)$. Hence the options lying in Λ form an initial segment, of length $F_\Lambda(\mu)$, and

$$F_\Lambda(\mu) = r + (q - \ell_+ - \ell_-) \cdot \iota + (\ell_+ - j_0 + 1) \cdot [j_0 \text{ exists}], \quad (7.1)$$

where $r = 0$ or r is the last row of μ_- of its length (the removals in Λ are those at positions $\leq r$), $\iota := [(\mu_+ \sqcup 1, \mu_-) \in \Lambda]$ ($\iota := 0$ if $\ell_+ + \ell_- = q$), and j_0 , if it exists, is the first row of μ_+ of its length such that $(\mu_+ + e_j, \mu_-) \in \Lambda$ exactly for $j_0 \leq j \leq \ell_+$. If j_0 exists, then $r = \ell_-$, and either $\iota = 1$ or $\ell_+ + \ell_- = q$; if $\iota = 1$, then $r = \ell_-$.

Proof. By the identities of §5.5 (Lemma 7.2'), $\mu_- - e_1 \leq \dots \leq \mu_- - e_{\ell_-} \leq \mu_-$ and $\mu_+ \leq \mu_+ \sqcup 1 \leq \mu_+ + e_{\ell_+} \leq \dots \leq \mu_+ + e_1$. Along the removals the first component is μ_+ , and along the middle and the additions the second component is μ_- ; so in the product order the list is weakly increasing, including the step from the last removal to the first middle or addition option. Since Λ is a down-set, the options in Λ form an initial segment. Rows of equal length give equal options, which gives the description of r and j_0 . ■

In particular, with $\Phi := F_\Lambda(\mu)$: $\Phi = q - j_0 + 1$ if j_0 exists, $\Phi = q - \ell_+$ if j_0 does not exist and $\iota = 1$, and $\Phi = r$ otherwise.

7.4 (P2) The layers are down-sets

Proposition 7.4 (inequality of pairs of partitions). If $\mu, \tilde{\mu} \in BPar_q(\alpha - 1, \beta)$ and $\mu \leq \tilde{\mu}$, then $opt_p(\mu) \leq opt_p(\tilde{\mu})$ for every p . Hence $F_\Lambda(\mu) \geq F_\Lambda(\tilde{\mu})$ and every Λ_i is a down-set.

Proof. The second sentence follows from the first as in Proposition 5.6. Write $\ell_\pm := \ell(\mu_\pm)$, $\tilde{\ell}_\pm := \ell(\tilde{\mu}_\pm)$. Since $|\mu_+| - |\mu_-| = |\tilde{\mu}_+| - |\tilde{\mu}_-|$, any strict inequality $|\tilde{\mu}_-| > |\mu_-|$ gives $|\tilde{\mu}_+| > |\mu_+|$. We use (T) of §5.6 and the inequality (5.2) (Lemma 7.2'): if $\lambda \leq \tilde{\lambda}$ and $\ell(\lambda) < t \leq \ell(\tilde{\lambda})$, then $S_\ell(\tilde{\lambda}) \geq |\lambda| + (t - \ell(\lambda))$. We also use four comparisons of §5.6, valid for arbitrary partitions by Lemma 7.2': **(a)** $\lambda - e_p \leq \tilde{\lambda} - e_p$ for $p \leq \min(\ell(\lambda), \ell(\tilde{\lambda}))$; **(b)** $\lambda + e_j \leq \tilde{\lambda} + e_j$ for $j \leq \min(\ell(\lambda), \ell(\tilde{\lambda}))$; **(c)** $\lambda \sqcup 1 \leq \tilde{\lambda} \sqcup 1$; **(e2)** $\lambda + e_j \leq \tilde{\lambda} \sqcup 1$ for $\ell(\tilde{\lambda}) < j \leq \ell(\lambda)$. Each position p is a removal (R), a middle position (M) or an addition (A) for μ , and likewise for $\tilde{\mu}$, which gives nine cases; put $j := q + 1 - p$ when p is an addition.

case	the position p is	option of μ	option of $\tilde{\mu}$	used
RR	a removal for both	$(\mu_+, \mu_- - e_p)$	$(\tilde{\mu}_+, \tilde{\mu}_- - e_p)$	$\mu_+ \leq \tilde{\mu}_+$; (a)
AA	an addition for both	$(\mu_+ + e_p, \mu_-)$	$(\tilde{\mu}_+ + e_p, \tilde{\mu}_-)$	(b); $\mu_- \leq \tilde{\mu}_-$
MM	middle for both	$(\mu_+ \sqcup 1, \mu_-)$	$(\tilde{\mu}_+ \sqcup 1, \tilde{\mu}_-)$	(c); $\mu_- \leq \tilde{\mu}_-$
RM, RA	a removal for μ	$(\mu_+, \mu_- - e_p)$	middle or addition	the chain through $\mu \leq \tilde{\mu}$
AM	an addition for μ , middle for $\tilde{\mu}$	$(\mu_+ + e_p, \mu_-)$	$(\tilde{\mu}_+ \sqcup 1, \tilde{\mu}_-)$	(e2); $\mu_- \leq \tilde{\mu}_-$
MA	middle for μ , an addition for $\tilde{\mu}$	$(\mu_+ \sqcup 1, \mu_-)$	$(\tilde{\mu}_+ + e_p, \tilde{\mu}_-)$	dominance, (5.2), sizes; $\mu_- \leq \tilde{\mu}_-$
MR	middle for μ , a removal for $\tilde{\mu}$	$(\mu_+ \sqcup 1, \mu_-)$	$(\tilde{\mu}_+, \tilde{\mu}_- - e_p)$	(5.2) and the transfer of sizes
AR	an addition for μ , a removal for $\tilde{\mu}$	$(\mu_+ + e_p, \mu_-)$	$(\tilde{\mu}_+, \tilde{\mu}_- - e_p)$	(T), (5.2) and the transfer of sizes

The «transfer of sizes» is the one genuinely bipartite step: since $|\mu_+| - |\mu_-| = |\tilde{\mu}_+| - |\tilde{\mu}_-|$, a strict inequality on the z -side gives one on the x -side.

- **RR** ($p \leq \min(\ell_-, \tilde{\ell}_-)$): first components $\mu_+ \leq \tilde{\mu}_+$; second components by (a).
- **AA** ($j \leq \min(\ell_+, \tilde{\ell}_+)$): (b), and $\mu_- \leq \tilde{\mu}_-$.
- **MM**: (c), and $\mu_- \leq \tilde{\mu}_-$.
- **RM**: $(\mu_+, \mu_- - e_p) \leq \mu \leq \tilde{\mu} \leq (\tilde{\mu}_+ \sqcup 1, \tilde{\mu}_-)$.
- **RA**: $(\mu_+, \mu_- - e_p) \leq \mu \leq \tilde{\mu} \leq (\tilde{\mu}_+ + e_p, \tilde{\mu}_-)$.
- **AM** ($j \leq \ell_+$ and $\tilde{\ell}_+ < j$): (e2), and $\mu_- \leq \tilde{\mu}_-$.
- **MA** (p middle for μ , so $j > \ell_+$; addition for $\tilde{\mu}$, so $j \leq \tilde{\ell}_+$): we need $\mu_+ \sqcup 1 \leq \tilde{\mu}_+ + e_j$. For $t \leq \ell_+$, $S_t(\mu_+ \sqcup 1) = S_t(\mu_+) \leq S_t(\tilde{\mu}_+) \leq S_t(\tilde{\mu}_+ + e_j)$. For $\ell_+ < t \leq \tilde{\ell}_+$, $S_t(\tilde{\mu}_+) \geq |\mu_+| + 1 = S_t(\mu_+ \sqcup 1)$ by (5.2). For $t > \tilde{\ell}_+ \geq j \geq r_j(\tilde{\mu}_+)$, $S_t(\tilde{\mu}_+ + e_j) = |\tilde{\mu}_+| + 1 \geq |\mu_+| + 1$. The second components are $\mu_- \leq \tilde{\mu}_-$.
- **MR** (p middle for μ , so $p > \ell_-$; removal for $\tilde{\mu}$, so $p \leq \tilde{\ell}_-$): by (5.2) at $t = \tilde{\ell}_-$, $|\tilde{\mu}_-| \geq |\mu_-| + 1$, hence $|\tilde{\mu}_+| \geq |\mu_+| + 1$. *First components*: $\mu_+ \sqcup 1 \leq \tilde{\mu}_+$: for $t \leq \ell_+$ this is dominance; for $t > \ell_+$ we need $S_t(\tilde{\mu}_+) \geq |\mu_+| + 1$, which holds by (5.2) if $t < \tilde{\ell}_+$ and by $|\tilde{\mu}_+| \geq |\mu_+| + 1$ if $t \geq \tilde{\ell}_+$. *Second components*: $\mu_- \leq \tilde{\mu}_- - e_p$: for $t < \tilde{r}_p(\tilde{\mu}_-)$ this is dominance; for $t \geq \tilde{r}_p(\tilde{\mu}_-) \geq p > \ell_-$, $S_t(\mu_-) = |\mu_-|$, while the rows $\ell_- + 1, \dots, p$ of $\tilde{\mu}_-$ are non-empty, so $S_t(\tilde{\mu}_-) \geq S_{\ell_-}(\tilde{\mu}_-) + 1 \geq |\mu_-| + 1$.
- **AR** (p addition for μ , so $j \leq \ell_+$; removal for $\tilde{\mu}$, so $p \leq \tilde{\ell}_-$): since $\ell_+ + \ell_- \leq q$ and $\tilde{\ell}_+ + \tilde{\ell}_- \leq q$, we have $\ell_- \leq q - \ell_+ < p$ and $\tilde{\ell}_- \leq q - p = j - 1$. *Second components*: exactly as in MR, $\mu_- \leq \tilde{\mu}_- - e_p$ and $|\tilde{\mu}_-| \geq |\mu_-| + 1$, so $|\tilde{\mu}_+| \geq |\mu_+| + 1$. *First components*: we need $S_t(\mu_+) + [t \geq r_j(\mu_+)] \leq S_t(\tilde{\mu}_+)$. This is dominance for $t < r_j(\mu_+)$, and it follows from $|\tilde{\mu}_+| \geq |\mu_+| + 1$ for $t \geq \tilde{\ell}_+$. It remains to show $S_t(\mu_+) < S_t(\tilde{\mu}_+)$ for $r_j(\mu_+) \leq t < \tilde{\ell}_+$. Suppose t is tight. The rows $r_j(\mu_+), \dots, j$ of μ_+ have the same length $c := \mu_{+,j} \geq 1$, and $t < t + 1 \leq \tilde{\ell}_+ \leq j - 1$, so $\mu_{+,t} = \mu_{+,t+1} = c$. By (T), $c = \mu_{+,t} \geq \tilde{\mu}_{+,t} \geq \tilde{\mu}_{+,t+1} \geq \mu_{+,t+1} = c$, so every row of $\tilde{\mu}_+$ after the t -th has length $\leq c$. As $\tilde{\mu}_+$ has only $\tilde{\ell}_+ \leq j - 1$ rows, $S_j(\tilde{\mu}_+) \leq S_t(\mu_+) + (j - 1 - t)c < S_t(\mu_+) + (j - t)c = S_j(\mu_+)$, contradicting $\mu_+ \leq \tilde{\mu}_+$.

For each of μ and $\tilde{\mu}$ the three types partition $\{1, \dots, q\}$, so the nine cases cover every position. ■

7.5 (P3) Every tight pattern of a layer lies in the right slice

Proposition 7.5. For every $i \in \{0, \dots, q - 1\}$, $V_{\Lambda_i} \subseteq W_{q-1-i}(V_{\Lambda})$, where $V_{\Lambda_i} \subseteq R_{\Lambda} \setminus \{i_1\}, B^*$.

Proof. It suffices to treat a generator G : the product of a tight pattern $(P, (B^+_{\phi}, (B^-_{\phi}))$ of some $\mu \in \Lambda_i$, so that $\Phi := F_{\Lambda}(\mu) \geq i + 1$. By Lemma 7.1(ii) it is enough to find $f \in V_{\Lambda}$ with $\deg_w f \leq q - \Phi$ and $[w^{q-\Phi}]f = \pm \kappa \cdot G$ for a non-zero constant κ ; then $G \in W_{q-\Phi}(V_{\Lambda}) \subseteq W_{q-1-i}(V_{\Lambda})$. By (7.1) exactly one of the following holds.

- **(α)** j_0 exists and $\lambda := (\mu_+ + e_{j_0}, \mu_-) \in \Lambda$, $\Phi = q - j_0 + 1$. The column $c^* := \mu_{+,j_0} + 1$ of μ_+ has height $j_0 - 1$, because j_0 is the first row of its length; let $S := B^+_{c^*}$ (if $j_0 = 1$, c^* is a new column and $S = \emptyset$). Putting i_1 into S gives a tight pattern of λ : column c^* of λ_+ has height j_0 , and the number of pairs $\alpha - |\lambda_+| = (\alpha - 1) - |\mu_+|$ is unchanged. Its product lies in V_{Λ} ; by the factorisation $\Delta(x_S \cup i_1) = \pm \prod_{s \in S} (w - x_s) \cdot \Delta(x_S)$ it has w -degree $|S| = j_0 - 1 = q - \Phi \leq q - 1$ (so $w^{|S|} \neq 0$ in the box) and top coefficient $\pm G$.
- **(β)** j_0 does not exist, $i = 1$, $\lambda := (\mu_+ \sqcup 1, \mu_-) \in \Lambda$, $\Phi = q - \ell_+$. Put i_1 into $B^+_{\ell_+}$, of height ℓ_+ (empty if $\mu_+ = \emptyset$): this is a tight pattern of λ , whose product has w -degree $\ell_+ = q - \Phi \leq q - 1$ (a middle value exists, so $\ell_+ < q$) and top coefficient $\pm G$.
- **(γ)** j_0 does not exist, $i = 0$, and $\Phi = r \geq 1$, $\lambda := (\mu_+, \mu_- - e_r) \in \Lambda$. The column $c_0 := \mu_{-,r}$ of μ_- has height r , because r is the last row of its length; let $S := B^-_{c_0}$ and write $G = \Delta(z_S) \cdot G'$. For $b \in S$, the pairs $P \cup \{(i_1, b)\}$, the blocks $B^+_{c_0}$, the blocks $B^-_{c_0}$ ($c \neq c_0$) and $S \setminus \{b\}$ form a tight pattern of λ : column c_0 of λ_- has height $r - 1$, and the number of pairs grows by one, as $\alpha - |\lambda_+| = ((\alpha - 1) - |\mu_+|) + 1$. So, with the signs ε_b of Lemma 5.7(ii),

$$f := \sum_{b \in S} \varepsilon_b \Delta(z_S \setminus b) \cdot (w - z_b)^{q-1} \cdot G' \in V_{\Lambda}.$$

Expanding, $[w^{q-1-t}]f = C(q-1, t)(-1)^t \cdot (\sum_{b \in S} \varepsilon_b \Delta(z_{S \setminus b}) z_b^t) \cdot G'$, which is 0 for $t \leq r-2$ and $\pm C(q-1, r-1) \cdot G$ for $t = r-1$, by Lemma 5.7(ii) (an identity over Z , valid in $R_A \setminus \{i_0, B\}$). So $\deg_w f \leq q-r = q-\Phi$, and the top coefficient is $\pm C(q-1, r-1) \cdot G$ with $C(q-1, r-1) \neq 0$ in F (here $r \leq \ell_- \leq q$). ■

The three constructions land exactly in the slice $q-\Phi$, and in general the slice cannot be lowered: the statement with W_{q-2-i} in place of W_{q-1-i} fails at the cells of §12.4, for some generators G .

7.6 The induction

Theorem 7.6 (bipartite down-set theorem). *For all $\alpha, \beta \geq 0$ and every $\leq$ -down-set $\Lambda \subseteq \text{BPar}_q(\alpha, \beta)$, $\dim_F V_\Lambda \geq |Z_\Lambda|$.*

Proof. Induction on $\alpha + \beta$. **Base:** $\text{BPar}_q(0, 0) = \{(\emptyset, \emptyset)\}$; for $\Lambda = \emptyset$ both sides are 0, and for $\Lambda = \{(\emptyset, \emptyset)\}$, $V_\Lambda = F$ and Z_Λ is one point. **Symmetry:** the isomorphism $R_{A,B} \cong R_{B,A}$ that exchanges the names x and z carries tight-pattern products of (λ_+, λ_-) on (A, B) to tight-pattern products of (λ_-, λ_+) on (B, A) (as $(x-z)^{q-1} = (z-x)^{q-1}$), and $(\xi, \eta) \mapsto (\eta, \xi)$ exchanges λ_+ and λ_- . A relabelling of A or B changes each product by a sign and preserves $|Z_\Lambda|$. So we may assume $\alpha \geq 1$ and peel any x -variable. **Step:** by Lemma 7.1(iii), Proposition 7.5, the induction hypothesis (each Λ_i is a down-set of $\text{BPar}_q(\alpha-1, \beta)$ by Proposition 7.4), §7.3 and Lemma 7.1(iv),

$$\dim V_\Lambda = \sum_{i=0}^{q-1} \dim W_{q-1-i}(V_\Lambda) \geq \sum_{i=0}^{q-1} \dim V_{\Lambda_i} \geq \sum_{i=0}^{q-1} |Z_{\Lambda_i}| = \sum_{i=0}^{q-1} |Z_{>i}| = |Z_\Lambda|. \blacksquare$$

Proof of Theorem C. Lemma 7.2 and Theorem 7.6. ■

7.7 Consequences

Corollary 7.7 (Main Theorem' and the Main Theorem). *Main Theorem' holds, and Conjecture 1.2 of [DS] holds for every odd $m \geq 3$ and every even dimension.*

Proof. Let $k \geq 1$. For every prime $p \mid m$, $\dim_{F_p} F_p[G]/(\bar{\psi}_J : J \in \mathcal{J}) = m^{n+1} - |\Gamma_{\mathcal{J}}|$; for $m = p^v$ by Corollary 5.4 and Proposition 2.5; otherwise by Proposition 6.9, whose hypothesis is Theorem C over $F = \bar{F}_p$, $q = p^v$. For $p \nmid m$ the same equality is Theorem 0(b) with $K_p = \bar{F}_p$. The rank of $R/(\psi_J : J \in \mathcal{J})$ is $m^{n+1} - |\Gamma_{\mathcal{J}}|$ (Theorem 0(b) with $p = 0$), and $|\Gamma_{\mathcal{J}}| = Q_k(m)$ (Lemma 2.2, valid for odd m); so $R/(\psi_J : J \in \mathcal{J})$ has no torsion, which is Main Theorem'. No step of this uses topology. With Theorem 0(a) (Proposition 2.1) it gives the Main Theorem; for $k = 0$ the Main Theorem is trivial. ■

Corollary 7.8 (equality over $\bar{F}_p$). *Let $q = p^v$ with p odd. Then $\dim_{\bar{F}_p} I_a^{\text{bal}} = N_{\text{bal}}(a, q)$ and $\dim_{\bar{F}_p} I_a^{\text{ph}} = N_{\text{ph}}(a, q)$ for every $a \geq 0$.*

Proof. For $a = 0$, $I_0^{\text{bal}} = F$ and $I_0^{\text{ph}} = F[x_1]/(x_1^q)$ have dimensions $1 = N_{\text{bal}}(0, q)$ and $q = N_{\text{ph}}(0, q)$. Let $a \geq 1$, choose an odd prime $r \neq p$ and put $m := qr$ and $n := 2a$, so that $k = a \geq 1$ and (6.1) applies. Colour a coordinates by a primitive r -th root ζ , a coordinates by ζ^{-1} , and one coordinate s by 1; then $c_0 = 1$, the block $\mathcal{C}_1 = \{0, s\}$ contributes $\dim I_{c,1} = q-1 = N_1(c)$, and the pair block contributes $\dim I_a^{\text{bal}}$. If $\dim I_a^{\text{bal}} > N_{\text{bal}}(a, q)$, this colouring has $\dim \pi_c(I) > N_1(c) \cdot N_{\text{bal}}(a, q)$, while every other compatible colouring has $\dim \pi_c(I) \geq N_1(c) \cdot \Pi_\zeta N_\zeta(c)$ (Proposition 6.5, Lemmas 6.6 and 6.7, and Theorem C for every $a \leq n$). Summing, $\dim_F I > |\Gamma_{\mathcal{J}}|$ by Lemma 6.8, contradicting (6.1). For I_a^{ph} , take $n := 2a$, colour $a+1$ coordinates by ζ and a coordinates by ζ^{-1} ; then $c_0 = \zeta^{-1}$, $\mathcal{C}_1 = \emptyset$, and the only block is I_a^{ph} . ■

Remarks 7.9. (1) Corollary 7.8 is an algebraic statement with an algebraic proof: the upper bound (6.1) is Theorem 0(b), a Chinese-remainder count. The proof is global, through the sum over the colourings of $F[G]$; we do not know a direct proof of $\leq$ on one block: the evaluation argument of Theorem 5.9 fails, because $(x-z)^{q-1}$ vanishes at $x=z$ rather than off it. (2) In characteristic 0 the inequalities of Theorem C are strict in general: $\dim_Q((x-z)^2) = 4 > 3 = N_{\text{bal}}(1, 3)$ in $Q[x, z]/(x^3, z^3)$, and $\dim_Q I_2^{\text{bal}} = 24 > 15$ at $q = 3$ (§12). So, unlike Theorem 5.9', in these cells (where Corollary 7.8 gives $\dim_{F_3} I_a^{\text{bal}} = N_{\text{bal}}(a, 3) < \dim_Q I_a^{\text{bal}}$) the quotient of the integral box ring by the bipartite ideal has 3-torsion: the bipartite ideals are genuinely of characteristic p . (3) By Lemma 7.2(iii) the phantom and balanced counts are related. On the ideal side, if $\sigma \in S_{a+1}$ and $i_0 := \sigma^{-1}(a+1)$, the generator $\Pi_i(x_i - z_{\sigma(i)})^{q-1}$ of I_{a+1}^{bal} has z_{a+1} -degree $q-1$, with top coefficient the generator of I_a^{ph} indexed by (i_0, σ) , so the top slice $W_{q-1}(I_{a+1}^{\text{bal}})$ (peeling z_{a+1} , that is, Lemma 7.1 with the names x and z

exchanged, as in the symmetry step of Theorem 7.6) contains I_a^{ph} . With Corollary 7.8, $W_j(I_{a+1}^{bal}) = 0$ for $j < q - 1$, and $f \mapsto [z_{a+1}^{q-1}]f$ is an isomorphism of graded vector spaces $I_{a+1}^{bal} \rightarrow I_a^{ph}$ of degree $-(q-1)$ over $\bar{F}$.

8. A different count: Theorem A

For a finite set L of variables over a field F and an odd $q \geq 1$, put $I_q(L) := (e_j(L) : j \text{ odd}) + (x^q : x \in L)$ and

$$N_q(n) := n! \cdot [y^n] e^y I_q(2y)^{(q-1)/2}.$$

Theorem A. *Let F be a field with $\text{char } F \neq 2$, let $q \geq 1$ be odd, and let $|L| = n$. Then $\dim_F F[L]/I_q(L) = N_q(n)$.*

The proof is in Appendix A. The lower bound degenerates the ideal of an explicit finite set of points. The upper bound is an induction on n over a family of ideals $K_\mu(L)$ indexed by partitions μ ; the ideals are built from the coefficients of one generating function, and each inductive step shifts one index in that function. The proof uses the characteristic only through $2 \in F^\times$, and it uses no relation between q and $\text{char } F$.

Corollary 8.1. *Let p be an odd prime and $q = p^v$. For every $k \geq 0$, $A_k(q) := \dim F_p[x_0, \dots, x_{2k+1}]/(E + (x_0^q, \dots, x_{2k+1}^q)) = P_k(q)$, where $P_k(q)$ is the number of points of F_q^{2k+2} at which every odd elementary symmetric polynomial vanishes, i.e. the number of F_q -points of $\bigcup_{J \in \mathcal{J}} V(I_J)$.*

Corollary 8.2. *For every odd q and every n , the finitely generated abelian group $M := Z[x_1, \dots, x_n]/(e_j : j \text{ odd}; x_i^q)$ has rank $N_q(n)$, and its torsion subgroup is a 2-group.*

Remarks 8.3. (1) In characteristic 2 the statement fails: at $n = 4$ the dimensions are 21, 65, 133, 225 for $q = 3, 5, 7, 9$, against $N_q(4) = 19, 61, 127, 217$. (2) For $q = 3$ and all k , Corollary 8.1 was known earlier through Steinberg modules and tilting theory [Jan, II.4.16], [Mat, Theorem 1(1)], [DCP, Theorems 5.6–5.7]; for $k \leq 4$ and all $q = 3^v$ it was known through case-by-case arguments in earlier notes of this project. Theorem A contains all of these, in every characteristic $\neq 2$ and for every odd q . (3) **Theorem A is not used anywhere in the proof of the Main Theorem, and it does not imply it (§9).**

9. Why the full family is needed

9.1 Two counts

Let p be an odd prime, $q = p^v$, $S := F_p[x_0, \dots, x_{2k+1}]$, $N = 2k+2$ and $I_J := (x_a + x_b : \{a, b\} \in J)$. The ideal E vanishes on $\bigcup_J V(I_J)$: a point lies there iff its multiset of coordinates is closed under negation iff $\prod_i (1 + x_i t)$ is even in t (unique factorisation in $\bar{F}_p[t]$). By Proposition 2.6, (S) says

$$b_k(q-1) := \dim S / \bigcap_{J \in \mathcal{J}} (I_J + (x_i^{q-1})) = Q_k(q),$$

an **intersection** at the **even** box $q - 1$, counting the points of the union of the matching spaces on the **zero-free** grid T^N . Corollary 8.1 says $\dim S/(E + (x_i^q)) = P_k(q)$, a **sum** at the **odd** box q , counting points on the grid **with 0**.

Proposition 9.1 (the sandwich). $b_k(q-1) \leq Q_k(q) \leq \dim S/(E + (x_i^{q-1}))$.

Proof. Dimensions do not change under extension of scalars, so we may work over F_q . Let $\Gamma \subseteq T^N$ be the set of zero-free points of $\bigcup_J V(I_J)$; $|\Gamma| = Q_k(q)$ by Lemma 2.2, and $S/I(\Gamma)$ has dimension $|\Gamma|$. We use Lemma 5.8' (with N variables). The ideal $I'' := E + (x_i^{q-1} - 1)$ is contained in $I(\Gamma)$, and $\text{gr}(I'') \supseteq E + (x_i^{q-1})$ because E is homogeneous; so $\dim S/(E + (x_i^{q-1})) \geq \dim S/\text{gr}(I'') = \dim S/I'' \geq |\Gamma|$, the right-hand inequality. For each J , the ideal of the finite set $V(I_J) \cap T^N$ is $I_J + (x_i^{q-1} - 1)$: modulo I_J one free variable per pair remains, and $x^{q-1} - 1$ is separable with roots exactly T . Its gr contains $I_J + (x_i^{q-1})$, and is equal to it, since both quotients have dimension $(q-1)^{k+1}$. Finally $I(\Gamma) = \bigcap_J I(V(I_J) \cap T^N)$, and gr of an intersection is contained in the intersection of the gr 's, so $\text{gr}(I(\Gamma)) \subseteq \bigcap_J (I_J + (x_i^{q-1}))$, which gives the left-hand inequality. ■

So the Main Theorem is the **left** equality. It is not a consequence of Corollary 8.1, which is an equality of the right-hand kind at a different box.

9.2 A counterexample for subfamilies

Proposition 2.1 and Proposition 2.5 hold for every non-empty $K \subseteq \mathcal{J}$. So, for a subfamily, $L_K(X)$ is primitive iff $\dim(D_J : J \in K)C = |\Gamma_K|$.

Fact 9.2 (computed; not used in any proof). Let $q = 9$, $k = 2$, and let $K := \mathcal{J} \setminus \{[0,1],[2,3],[4,5], [0,2], [1,5],[3,4]\}$ (13 of the 15 matchings). Then $\dim(D_J : J \in K)C = 4730 < 4736 = |\Gamma_K|$. So $H_4(X)/L_K(X)$ has 3-torsion for the Fermat fourfold of degree 9. For the same K , the sum-type count holds: with $E_K := \bigcap_{J \in K} I_J$, $\dim S/(E_K + (x_i^9)) = 7089$, which is the number of F_9 -points of $\bigcup_{J \in K} V(I_J)$.

The value 4730 was obtained by four independent routes (§12): the literal form of [DS] in the ring $F_3[G]$ ($\dim F_3[G]/(\bar{\psi}_J : J \in K) = 54\,319 = 9^5 - 4730$, against $9^5 - |\Gamma_K| = 54\,313$), the y -form of §2, the intersection form of Proposition 2.6, and an engine written separately (a kernel basis $D \cdot y_a^i$ per pair, exact rank modulo 3 degree by degree). $|\Gamma_K| = 4736$ was obtained by direct enumeration.

At $q = 3$, $k = 2$, the analogue of (S) fails for 800 of the 32 767 non-empty subfamilies (worst deficit 2, e.g. 18 < 20), while every subfamily satisfies $\leq$. So 3-torsion for subfamilies occurs already for the cubic fourfold.

Consequences.

- Any argument that proves the analogue of (S) for every subfamily proves a false statement. A proof of the Main Theorem must use a property of the full family $\mathcal{J}$; in §5 this property enters at the root (Lemma 5.2), and in §6–§7 through the product structure of the compatible matchings (Proposition 6.5) and the roots of Lemma 7.2.
- $A = P$ cannot imply the Main Theorem through any argument that is valid for subfamilies.
- This is the natural first guess, and it is the error described in §1.8.

10. The integral Hodge conjecture

In this section $k \geq 1$, $n = 2k$, $m \geq 3$ is odd, and $X \subset P^{n+1}$ is the Fermat variety of degree m .

10.1 Classical facts

- **(H1) Integral cohomology.** $H^*(X; Z)$ is torsion free and vanishes in odd degrees. By the Lefschetz hyperplane theorem [Mil, §7], $H^{2j}(X; Z) = Z \cdot h^j$ for $2j < n$, where h is the hyperplane class, and $H_{2l}(X; Z) \rightarrow H_{2l}(P^{n+1}; Z) \cong Z$ is an isomorphism for $2l < n$.
- **(H2) Hodge classes.** Through Poincaré duality we call a class in $H_n(X; Z)$ a Hodge class if its dual lies in $H^{k,k}(X)$, and write $\text{Hdg}(X) \subseteq H_n(X; Z)$ for the group of Hodge classes. It is the set of integral classes on which every class of $F^{k+1}H^n(X, C)$ integrates to zero, so it is a primitive subgroup. The class of an algebraic subvariety of dimension k is a Hodge class, so $L(X) \subseteq \text{Hdg}(X)$. The class P_X of Appendix B, Step 3 (the dual of h^k) is a Hodge class, and $H^n(X; Q) = H^n_{\text{prim}}(X; Q) \oplus Q \cdot h^k$.
- **(H3) Characters [SK], [Shi79], [Ran, Proposition 1.7].** Let $\mathcal{A}$ be the set of $\alpha = (a_0, \dots, a_{n+1}) \in (Z/m \setminus \{0\})^{n+2}$ with $a_0 + \dots + a_{n+1} = 0$, and for $\alpha \in \mathcal{A}$ put $|\alpha| := \sum_i \langle a_i \rangle / m$, where $\langle a \rangle \in \{1, \dots, m-1\}$ is the representative of a . Under the action of μ_m^{n+2} on the coordinates, $H^n_{\text{prim}}(X; C) = \bigoplus_{\alpha \in \mathcal{A}} V(\alpha)$ with every $V(\alpha)$ of dimension one, and $(H^{k,k}(X) \cap H^n_{\text{prim}}(X; Q)) \otimes C = \bigoplus_{\alpha \in \mathcal{B}} V(\alpha)$, where $\mathcal{B} := \{\alpha \in \mathcal{A} : |t\alpha| = k+1 \text{ for every } t \in (Z/m)^\times\}$ is the set of **Hodge characters**. So $\text{rank Hdg}(X) = |\mathcal{B}| + 1$. An element $\alpha \in \mathcal{A}$ is of **pair type** if the multiset $\{a_0, \dots, a_{n+1}\}$ splits into pairs $\{a, -a\}$; we write $\mathcal{D} \subseteq \mathcal{A}$ for the set of these.
- **(H4) Prime degree [Ran, Proposition 1.8(i)], based on [KO].** If m is prime, every Hodge character is of pair type. We include a short proof, so that (H4) does not depend on [Ran] or [KO].
- **(H5) Aoki's theorem [Aok, Theorem A, p. 24].** $\mathcal{B} = \mathcal{D}$ if and only if m is a prime or 4, or every prime divisor of m is greater than $n+2$. (Aoki's sets $\mathcal{A}_m^n$, $\mathcal{B}_m^n$, $\mathcal{D}_m^n$ [Aok, §0] are the sets $\mathcal{A}$, $\mathcal{B}$, $\mathcal{D}$ above, with

$a_i \in \{1, \dots, m-1\}$ and $\sum a_i \equiv 0 \pmod{m}$; the proof of his Theorem A is in [Aok, §7], pp. 41–43, and the erratum [Aok] concerns only his Theorem B.) For odd m and even n , «greater than $n + 2$ » means «at least $2k + 3$ ».

Proof of (H4). Let $m = p$ and $\alpha \in \mathfrak{B}$, and for $c \in F_p^\times$ let $N(c)$ be the number of i with $a_i = c$. Then $|\alpha| = \sum_c N(c) \langle tc \rangle / p = k + 1$ for every $t \in F_p^\times$. Let χ be an odd character of $F_p^\times$, that is, $\chi(-1) = -1$. Multiply by $\chi(t)$ and sum over t . The constant contributes 0, and the substitution $u = tc$ gives $B_{1,\chi} \cdot \sum_c N(c) \chi(c) = 0$, where $B_{1,\chi} = (1/p) \sum_{u=1}^{p-1} \chi(u)u$. An odd character is non-principal, hence primitive with conductor p since p is prime; so $B_{1,\chi} \neq 0$, by the classical formula $L(1, \chi) = \pi i \tau(\chi) B_{1,\chi} / p$ together with $L(1, \chi) \neq 0$ [Was, Theorem 4.9]. Hence $\sum_c N(c) \chi(c) = 0$ for every odd χ . The function $f(c) = N(c) - N(-c)$ is odd, so $\sum_c f(c) \chi(c)$ equals $2 \sum_c N(c) \chi(c) = 0$ for odd χ , and vanishes trivially for even χ . By Fourier inversion on $F_p^\times$, $f = 0$. Since p is odd, $c \neq -c$, and the entries split into pairs $\{c, -c\}$. ■ (We also checked numerically that $B_{1,\chi} \neq 0$ for all 2 090 odd characters of the odd primes $p < 200$.)

10.2 Pair-type characters

Lemma 10.1. (i) Every pair-type $\alpha \in \mathfrak{A}$ is a Hodge character: $\mathfrak{D} \subseteq \mathfrak{B}$. (ii) $|\mathfrak{D}| = Q_k(m)$.

Proof. (i) For $t \in (Z/m)^\times$ and $a \neq 0$, $ta \neq 0$ and $\langle ta \rangle + \langle -ta \rangle = m$. Summing over the $k + 1$ pairs, $|\alpha| = k + 1$. (ii) Identify Z/m with μ_m by $a \mapsto e^{2\pi i a/m}$. The pair-type elements of $\mathfrak{A}$ become the N -tuples in $\mu_m \setminus \{1\}$ whose multiset is closed under inversion; the condition $\sum a_i = 0$ is then automatic. By Lemma 2.2, whose proof uses only that m is odd, there are $Q_k(m)$ of them. ■

10.3 Proof of Corollary H

By (H3), (H2), Theorem 0(c) and Lemmas 2.2 and 10.1, $\text{rank Hdg}(X) = |\mathfrak{B}| + 1$ and $\text{rank } L(X) = |\Gamma_{\mathcal{J}}| + 1 = |\mathfrak{D}| + 1$, with $L(X) \subseteq \text{Hdg}(X)$ and $\mathfrak{D} \subseteq \mathfrak{B}$.

(ii). $\text{Hdg}(X)/L(X)$ is a subgroup of $H_n(X; Z)/L(X)$, which is finitely generated and torsion free by the Main Theorem; so $\text{Hdg}(X)/L(X)$ is free, of rank $|\mathfrak{B}| - |\mathfrak{D}| = |\mathfrak{B}| - Q_k(m)$. It is zero iff $\mathfrak{B} = \mathfrak{D}$, which by (H5) happens, for odd m , iff m is prime or every prime factor of m is at least $2k + 3$. (For prime m , (H4) gives $\mathfrak{B} = \mathfrak{D}$ without appeal to [Aok].)

(i). Under the hypothesis of (i), $\text{Hdg}(X) = L(X)$ by (ii). (Only the primitivity of $L(X)$ is used here; that $\text{Hdg}(X)$ is primitive is not needed.) For the other degrees, by (H1) (the Lefschetz hyperplane theorem) and Poincaré duality: odd degrees vanish; for $2j < n$, $H^{2j}(X; Z)$ is generated by h^j , the class of the intersection of X with j general hyperplanes; for $2j > n$, $H^{2j}(X; Z) \cong H_{2l}(X; Z) \cong Z$ with $l = n - j < k$, and it is generated by the class of any linear subspace $P^l \subseteq X$, whose image in $H_{2l}(P^{n+1}; Z)$ is a generator. Such subspaces exist, since X contains the k -spaces $L_{j,\beta}$. So every integral Hodge class of X , in every degree, is the class of an algebraic cycle with integer coefficients. ■

Remark 10.2. (1) Under the hypothesis of (i), Corollary H says more than the integral Hodge conjecture for X : the integral Hodge classes of middle degree are generated by the $(2k+1)!! \cdot m^{k+1}$ standard k -spaces alone. (2) In the same degrees, the elementary divisors and the discriminant of the lattice of integral Hodge classes of middle degree, computed in [AMV] for small (n, m) , are those of the lattice generated by the standard spaces. For $k = 1$ and $m = p \geq 5$ prime, closed forms are stated in the author's *Watermark* and *Double Ladder* theorems [Rep], which are cited and not proved here (the *Double Ladder*, which gives the structure of the group, is not complete: four of its steps have been verified at three or four primes and are not proved for every prime): $|\text{disc}| = p^{3(p-3)^2}$, and the discriminant group is $(Z/p)^{3p^2-24p+59} \times (Z/p^2)^{3p-16}$ for $p \geq 7$ and $(Z/5)^{10} \times Z/25$ for $p = 5$. They agree with the surface rows $m = 5, 7, 11, 13$ of [AMV, Table 1]. For $k \geq 2$ this is open (§13).

Remark 10.3 (the degrees excluded by (i)). When m is not prime and has a prime factor $p \leq 2k + 1$, Corollary H(ii) says that $L(X)$ is a primitive sublattice of $\text{Hdg}(X)$ of corank $|\mathfrak{B}| - Q_k(m) > 0$, so the standard linear cycles do not generate the integral Hodge classes, and other algebraic cycles would be needed (for $k =$

1 they exist, by the Lefschetz (1,1) theorem); we do not claim that any particular construction supplies them. Counting $\mathfrak{B}$ directly (§12), $|\mathfrak{B}| - Q_k(m)$ is 48, 2 880 and 152 880 for $m = 9$ and $k = 1, 2, 3$; it is 192 for $(k, m) = (1, 27)$, 624 for $(1, 81)$, 288 for $(1, 15)$, 432 for $(1, 21)$, 528 for $(1, 45)$, 2 880 for $(2, 25)$, 4 320 for $(2, 35)$ and 45 360 for $(2, 15)$; and it is 0 for $(k, m) = (1, 25), (1, 35), (1, 49), (1, 55), (1, 77), (1, 121)$, as (H5) predicts. For $m = 9$ the count has a closed form.

Lemma 10.4 ($m = 9$). For $m = 9$ and every $k \geq 1$, $|\mathfrak{B}| = C(2k+2, k+1)^3$. Hence $\text{Hdg}(X)/L(X)$ is free of rank $C(2k+2, k+1)^3 - Q_k(9)$ (48, 2 880, 152 880, 8 064 000 for $k = 1, 2, 3, 4$).

Proof. Since $(Z/9)^\times = \{\pm 1, \pm 2, \pm 4\}$ and $\langle -x \rangle = 9 - \langle x \rangle$, the condition $|t\alpha| = k + 1$ for t implies it for $-t$, and $\alpha \in \mathfrak{A}$ is a Hodge character iff $\sum_i \delta(a_i) = 0$, where $\delta(a) := (2\langle a \rangle - 9, 2\langle 2a \rangle - 9, 2\langle 4a \rangle - 9) \in Z^3$ (note $\sum_i 9 = 9N = 18(k+1)$). Let $s(a) \in \{\pm 1\}^3$ be the vector of signs of $\delta(a)$. Then $\delta(1) = (-7, -5, -1)$, $\delta(2) = (-5, -1, 7)$, $\delta(4) = (-1, 7, 5)$, $\delta(3) = (-3, 3, -3)$, and $\delta(-a) = -\delta(a)$; so $a \mapsto s(a)$ is a bijection from $Z/9 \setminus \{0\}$ onto $\{\pm 1\}^3$, and $\delta(a) = M \cdot s(a)$ for the matrix M with rows $(4, 2, 1), (-1, 4, 2), (-2, -1, 4)$ (check on $a = 1, 2, 4, 3$, then use $\delta(-a) = -\delta(a)$). As $\det M = 81 \neq 0$, $\sum_i \delta(a_i) = 0$ iff $\sum_i s(a_i) = 0$ in Z^3 , i.e. iff each of the three coordinates of the sign vectors is balanced over the N entries. The condition $\sum a_i = 0$ follows from $\sum \langle a_i \rangle = 9(k+1)$. So $|\mathfrak{B}|$ is the number of N -tuples of sign vectors with three balanced coordinates, $C(N, N/2)^3$. ■

(The direct counts 216, 8 000, 343 000 at $k = 1, 2, 3$ suggested the lemma; the direct count at $k = 4$, 16 003 008 = 252^3 , agrees with it, §12.)

11. Partial Fermat varieties

In this section $m \geq 3$ is odd and $0 \leq s \leq d$. We prove Corollary W of §1.4. For $d = 0$ both statements are trivial (X and W_0 are sets of m points), so let $d \geq 1$.

Proof of Corollary W(ii). Let X be the Fermat variety of dimension $2d$, $R = Z[t_1, \dots, t_{2d+1}]/(t_i^m - 1)$, and $K := \mathcal{J}(s, d)$. A matching $J \in K$ is $J = J' \sqcup \{2i, 2i+1\} : s < i \leq d\}$ with J' a matching of $\{0, \dots, 2s+1\}$, and $J' \mapsto J$ is a bijection from $\mathcal{J}$ in dimension $2s$ onto K . By §2,

$$\psi_J = \psi_{J'} \cdot \prod_{i=s+1}^d g_i, \quad g_i := (t_{2i+1} - 1) \cdot \varphi(t_{2i} t_{2i+1}),$$

where $\psi_{J'}$ is the generator of $[\text{DS}]$ in dimension $2s$, a polynomial in $t_1, \dots, t_{2s+1}$. Let p be a prime and $F := F_p$. The ring $R \otimes F$ is the tensor product of $F[t_1, \dots, t_{2s+1}]/(t_i^m - 1)$ and the rings $F[t_{2i}, t_{2i+1}]/(t_{2i}^m - 1, t_{2i+1}^m - 1)$, $i > s$. By Lemma 6.4 (which holds over any field), the ideal generated by the $\bar{\psi}_J$, $J \in K$, is $I_{2s} \otimes \bigotimes_{i>s} (\bar{g}_i)$, where I_{2s} is the ideal of Conjecture 1.2 in dimension $2s$, and its dimension is the product of the dimensions. In $F[t_{2i}, t_{2i+1}]/(t^m - 1)$, put $u := t_{2i} t_{2i+1}$; the ring is $F[u, t_{2i+1}]/(u^m - 1, t_{2i+1}^m - 1)$, and $\bar{g}_i = (t_{2i+1} - 1) \cdot \varphi(u)$. The ideal $(\varphi(u))$ of $F[u]/(u^m - 1)$ is the line spanned by $\varphi(u)$, since $u\varphi(u) = \varphi(u)$; the ideal $(t - 1)$ of $F[t]/(t^m - 1)$ has dimension $m - 1$. By Lemma 6.4 again, $\dim(\bar{g}_i) = m - 1$, over every field. Hence

$$\dim_{F_p}(\bar{\psi}_J : J \in K) = \dim_{F_p} I_{2s} \cdot (m - 1)^{d-s}.$$

By the Main Theorem in dimension $2s$ and Proposition 2.1, $\dim_{F_p} I_{2s} = |\Gamma_{\mathcal{J}}^{(2s)}| = Q_s(m)$ for every prime p dividing m (for $s = 0$, $I_0 = (t_1 - 1)$ has dimension $m - 1 = Q_0(m)$ directly). On the other hand $|\Gamma_K| = Q_s(m)(m - 1)^{d-s}$: by [DS, Definition 1.3], a tuple $(a_1, \dots, a_{2d+1})$ lies in Γ_K iff $(a_1, \dots, a_{2s+1}) \in \Gamma_{\mathcal{J}}^{(2s)}$ and each pair (a_{2i}, a_{2i+1}) , $i > s$, is (a, a^{-1}) with $a \neq 1$. So $\dim_{F_p}(\bar{\psi}_J : J \in K) = |\Gamma_K|$ at every $p \mid m$, and by Proposition 2.1 (which holds for every non-empty $K \subseteq \mathcal{J}$) $H_{2d}(X)/L_K(X)$ is torsion free. Its rank is $|\Gamma_K| + 1$ by Theorem 0(c). ■

Proof of Corollary W(i). Consider the family of hypersurfaces $f_0(z_0, z_1) + \dots + f_d(z_{2d}, z_{2d+1}) = 0$ with $f_0 = \dots = f_s = x^m + y^m$ and $f_{s+1}, \dots, f_d$ binary forms of degree m without repeated factors. By Euler's identity its members are smooth; its parameter space is the complement of a discriminant hypersurface in an affine space, hence connected; it contains X (all $f_i = x^m + y^m$) and W_s ; and along it the m roots of each f_i , $i > s$, move continuously. The path must stay in this subfamily: the spaces $L_{J,\beta}$ whose first $2s + 2$ coordinates are paired by a matching other than $\{0, 1\}, \dots, \{2s, 2s+1\}$ involve $f_0, \dots, f_s$ and need not deform if these forms move. By [DS, §4.6], along such a path from the Fermat variety X to W_s , the subspaces $L_{J,\beta}$ with $J \in \mathcal{J}(s, d)$ deform to the d -spaces of W_s ,

and parallel transport (constant with respect to the Gauss–Manin connection) carries $L_{\mathcal{H}(s,d)}(X)$ onto the subgroup of $H_{2d}(W_s; \mathbb{Z})$ generated by these d -spaces; transport preserves rank and primitivity. So (i) follows from (ii), with the same rank $Q_s(m)(m-1)^{d-s} + 1$. ([DS, §4.6] reach the same point through the module $\mathfrak{M}_{\mathcal{H}(s)}(2d) = \mathfrak{M}_{\mathcal{H}(2s)}(2s) \otimes_{\mathbb{Z}} \tilde{S}(s, d)$ of their Theorem 1.1(d), which is free «if and only if Conjecture 1.2 holds for Fermat varieties of dimension $2s$ ».) ■

Remarks 11.1. (1) We have read the argument of [DS, §4.6] in the original and made the choice of the path explicit, but we have not re-derived the transport statement (the continuity of the family of subspaces and the invariance of primitivity under parallel transport), on which Corollary W(i) depends; part (ii) is proved above from Theorem 0, Proposition 2.1, Lemma 6.4 and the Main Theorem. (2) The rank in (ii), $Q_s(m)(m-1)^{d-s} + 1$, was also checked by enumerating Γ_K from [DS, Definition 1.3] at $m = 3, 5$ and $d \leq 3$ (14 cells; the enumeration is in the repository [Rep]). (3) We make no claim about the Hodge classes of W_s ; its rational Hodge classes are not computed here, so Corollary W is a primitivity statement only.

12. Verification record

No step of the proofs of the Main Theorem, Theorem B, Theorem C or Theorem A uses a computer. The computations below are corroboration: a proof that has never been confronted with data has not been tested. Every run was made inside a memory and time watchdog. All engines and logs are in the repository [Rep]. Main Theorem' has, in addition, been verified by a proof assistant (§12.8).

12.1 Direct computations of (S)

(k, q) , field	$Q_k(q)$	graded ranks of $(D_{\mathcal{J}})C$ from degree $k(q-2)$	independent engines
$(1,3), F_3$	6	2, 3, 1	5 (sum form; literal t -form; y -form; intersection form; rank engine)
$(2,3), F_3$	20	5, 9, 5, 1	5
$(3,3), F_3$	70	14, 28, 20, 7, 1	6
$(4,3), F_3$	252	42, 90, 75, 35, 9, 1	4
$(5,3), F_3$	924	132, 297, 275, 154, 54, 11, 1	4
$(6,3), F_3$	3 432	429, 1001, 1001, 637, 273, 77, 13, 1	3
$(1,9), F_3$	168	3, 6, 9, 12, 15, 18, 20, 21, 18, 15, 12, 9, 6, 3, 1	5
$(2,9), F_3$	5 120	15, 45, 90, 150, 224, 310, 395, 470, 515, 530, 516, 475, 411, 329, 245, 170, 110, 65, 34, 15, 5, 1	5
$(1,27), F_3$	1 950	3, 6, ..., 72, 74, 75, 72, ..., 3, 1	4
$(3,9), F_3$	190 120	—	1
$(1,5), F_5$	36	—	3, and the literal form of [DS]
$(2,5), F_5$	400	—	3, and the literal form of [DS]
$(3,5), F_5$	4 900	—	3
$(1,25), F_5$	1 656	—	3
$(2,25), F_5$	182 400	—	1
$(1,125), F_5$	45 756	—	1
$(1,7), F_7$	90	—	3, and the literal form of [DS]
$(2,7), F_7$	1 860	—	3
$(1,49), F_7$	6 768	—	2
$(3,7), F_7$	44 730	—	1
$(2,11), F_{11}$	10 900	—	1
$(2,13), F_{13}$	19 920	—	1

The literal form of [DS] (the ring $F_p[t]/(t_i^q - 1)$ modulo the $\bar{\psi}_J$) gives $q^{2k+1} - Q_k(q)$: 21, 223, 2117 at $(1,3)$, $(2,3)$, $(3,3)$ over F_3 , and 89, 2725, 253 at $(1,5)$, $(2,5)$, $(1,7)$ over F_5, F_5, F_7 . The cell $(2,25)$ is the Fermat fourfold of degree 25, which lies outside the range computed in [DS]. The cell $(3,9)$ was computed in May 2026, four months before the proof existed, as the F_3 -rank 190 120 of the generators ρ_J of [DS, §5] in their ring $\bar{R} = F_3[t_1, \dots, t_7]/(\varphi(t_i))$; in the same line of work, the cell $(5,3)$ gave $1124 = 2^{11} - 924$ for the quotient of $\bar{R}$. Equality was also measured over fields whose characteristic is not the one attached to q : at $q = 5$ over F_3 and $q = 9$ over F_5 , which Theorem 5.9 covers, and at $q = 7$ over F_3 , where $3 \mid q - 1$; these are instances of Theorem 5.9'. Over $\mathbb{Z}$, the integer Smith normal form of the literal ring $A_{\mathcal{J}} = R/(\psi_{\mathcal{J}})$ has no torsion and rank $q^{2k+1} - Q_k(q)$ at five cells, and the quotient $C_{\mathbb{Z}}/V_{\Lambda, \mathbb{Z}}$ of Theorem 5.9' is free of rank $(q-1)^m - |Z_{\Lambda}|$ at the root for $(k, q) = (1,5)$, $(1,7)$, $(2,3)$, $(1,9)$ and for every down-set at $(q, m) = (5,3)$, $(7,3)$, $(9,3)$, $(5,4)$. For composite odd q (Theorem B is a statement about every odd q), it was confirmed at $k = 1$ for $q = 15, 21, 45$ over several fields. The degrees m of the Main Theorem that are not prime powers are treated in §12.3.

12.2 The ingredients of §5

- **Every down-set, pure combinatorics.** All $\leq$ -down-sets of Par_m (not a sample) at $(q, m) = (3; 4, 6, 8, 12), (9; 3, \dots, 11), (27; 4, 5, 8, 9), (81; 4, 7), (243; 6)$, up to 940 down-sets per cell: P1, P2 and P3 hold, with 0 failures. The counts $\#\{M : \lambda(M) = \lambda\}$ were obtained by a class-by-class recursion, a route different from the fibres, with the check $\Sigma_\lambda = (q-1)^m$. P1 and (5.1) were also checked by brute force over all tails at seven cells up to $(27, 4)$ (238 257 tests), and P2 on all comparable pairs with h up to 60 and $m \leq 12$.
- **Every down-set, algebra.** $\dim V_\lambda = |Z_\lambda|$ for every down-set at $(q, m) = (3, 4), (3, 5), (3, 6), (9, 2), (9, 3), (9, 4)$ over F_3 (at $(9, 4)$: 0, 168, 1896, 2216, 2280, 2600, 3752, 3896, 4088, 4096), at $(27, 3)$ and $(81, 2)$, and at $q = 5$ and $q = 7$ over F_3, F_5, F_7 . The membership of every generator of every layer in its slice (Proposition 5.8) was checked at $(3, 3), (9, 3), (9, 4), (27, 2), (27, 3), (81, 2)$.
- **Theorem 5.9, its two ingredients.** The evaluation of D on $T \times T$ over F_5, F_7 and F_9 , and the support statement (a non-zero pattern of λ at M forces $\lambda(M) \leq \lambda$, and the pattern of $\lambda(M)$ is non-zero at M) on every $M \in T^m$ for $(q, m) = (5, 3), (5, 4), (7, 3), (9, 3), (9, 4)$: 0 failures.
- **Theorem 5.9'.** $\dim V_\lambda = |Z_\lambda|$ for every down-set at $(q, p, m) = (5, 2, 3), (5, 2, 4), (7, 2, 3), (7, 3, 3), (7, 2, 4), (9, 2, 3)$, all with $p \mid q - 1$ (34 down-sets, 0 failures); a second reader measured equality at 42 (cell, characteristic) pairs with $\text{char } F \mid q - 1$. The graded dimensions of $(D_j)C_F$ at $(q, 2k+1) = (5, 3), (7, 3), (5, 5)$ coincide for $\text{char } F = 2, 3, 5, 7, 10007$ (totals 36, 90, 400).
- **Hodge characters (§10).** The sets $\mathfrak{B}$ were counted by enumerating multisets at 19 cells (m, n) , and again, with a separate program, at the cells of Remark 10.3. For the ten cells with m prime — $(3, 2), (5, 2), (7, 2), (3, 4), (5, 4), (7, 4), (11, 4), (3, 6), (5, 6), (3, 8)$ — $|\mathfrak{B}|$ equals the number of pair-type characters, $Q_k(m)$. For prime powers the numbers are those of Remark 10.3, and no pair-type character failed to be a Hodge character (Lemma 10.1(i)). At $(k, m) = (1, 35), (1, 55), (1, 77)$, and at the prime cells $(k, m) = (3, 7), (3, 11), (2, 13)$, $|\mathfrak{B}| = Q_k(m)$, and at $(1, 15), (1, 21), (1, 45), (2, 15), (2, 35)$, $|\mathfrak{B}| > Q_k(m)$, as (H5) predicts. For $m = 9$ the direct counts are 216, 8 000, 343 000, 16 003 008 for $k = 1, \dots, 4$ (Lemma 10.4).
- **Negative controls, which fire as they should.** Counting one copy of $\mu \sqcup 1$ per empty class instead of two gives 27 failures of P1 and 61 of P2 at $(9, 6)$. Tightening P3 by one slice gives 604 failures at $(9, 7)$ and fails at every tested cell with $m \geq 2$. In characteristic 2, Theorem A fails in 8 of 12 cells.
- **The cubic case.** The ballot basis of Theorem 4.1 was recomputed as the set of lexicographic leading monomials for $k \leq 7$ (certificate) and $k \leq 6$ (echelon form), and the Specht identification of Remark 4.2(3) at $N = 4, 6, 8$.

12.3 Degrees that are not prime powers

Two engines were used, written separately: the constructor's (numpy, incremental echelon forms) and the auditor's (Singular). The auditor's predictions were computed from the closed forms and recorded before each run.

- **The literal ring of [DS], $k = 1$.** $\dim_{F_p} F_p[t_1, t_2, t_3]/(t_i^m - 1, \bar{\psi}_j) = m^3 - Q_1(m)$ at $m = 15, 21, 33, 35, 39, 45, 55$, each over both primes dividing m (14 runs), and at $m = 15, 21$ over F_3, F_5, F_31 and F_3, F_7, F_43 by the other engine. At $k = 1$ this is Degtyarev's theorem [De14]; the runs check our conventions for ψ_j at composite m .
- **The Fermat fourfolds of degree 15 and 21.** For every colouring $c \in \mu_r^5$ the ideal generated by the images of the literal generators ψ_j in R_c was computed directly, over a field containing μ_r ; this uses only Lemma 6.1. At $(k, m) = (2, 15)$ the sum over the colourings is $32\,900 = Q_2(15)$ both at $p = 3$ (over $F_81, 3\,125$ colourings, 1 001 with non-zero image) and at $p = 5$ (over $F_25, 243$ colourings, 141 with non-zero image). At $(2, 21)$ it is $102\,800 = Q_2(21)$ at $p = 3$ (over $F_729, 16\,807$ colourings, 3 301 with non-zero image) and at $p = 7$ (over $F_7, 243$ colourings, 141 with non-zero image). So $H_4(X; Z)/L(X)$ is torsion free for these two fourfolds, independently of §6–§7. They lie outside the range of [DS, §5]. The author's earlier computations [Rep] had decided the characteristic-5 half of the degree-15 fourfold and had

stopped on the characteristic-3 half at a memory limit. At $k = 1$ the colourings with non-zero image are exactly the compatible colourings (61 and 19 at $m = 15$, 19 at (21, 7)).

- **The ingredients of §6.** Lemma 6.8 (the count factors) was checked by enumerating $\Gamma_{\mathcal{J}}$ at $(m, n) = (15, 2), (21, 2), (45, 2), (15, 4)$ against the sums of products of block counts, for both primes of each m (8 sums), and Proposition 6.5 with Lemmas 6.6–6.7 by computing every block ideal directly at $(15; 3, 5)$ and $(21; 3, 7)$, $k = 1$: incompatible colourings give the zero ideal, and each compatible colouring gives exactly its block count.
- **Main Theorem' at $k = 1$, every characteristic.** In the literal ring $F_p[t_1, t_2, t_3]/(t_i^m - 1)$, the quotient by the $\bar{\psi}_J$ has dimension $m^3 - Q_I(m)$ for $m = 3, 5, 7, 9$ over F_2 and over primes dividing and not dividing m (12 cells, a third engine).
- **Controls.** The literal engine gives $5\ 120 = Q_2(9)$ at $(k, m) = (2, 9)$, and it detects torsion when there is torsion: for the subfamily of Fact 9.2 it gives $4\ 730 < 4\ 736 = |\Gamma_K|$.
- **Corollary W(ii).** In the literal ring at $(d, s) = (2, 1)$, with the three matchings containing $\{4, 5\}$, the ideal has dimension $144 = Q_I(5) \cdot 4$ at $m = 5$ and $1\ 344 = Q_I(9) \cdot 8$ at $m = 9$, the values of §11.

12.4 The bipartite count

- **The roots (Theorem C).** Over F_p , $\dim I_a^{bal} = N_{bal}(a, q)$ at the 18 cells $(a, q) = (1, 3), \dots, (5, 3), (2, 5), (3, 5), (4, 5), (2, 7), (3, 7), (2, 9), (3, 9), (2, 11), (2, 13), (1, 25), (2, 25), (1, 27), (2, 27)$, and $\dim I_a^{ph} = N_{ph}(a, q)$ at the 14 cells $(1, 3), \dots, (4, 3), (1, 5), (2, 5), (3, 5), (2, 7), (1, 9), (2, 9), (2, 11), (2, 13), (1, 25), (1, 27)$; for instance $\dim I_5^{bal} = \dim I_4^{ph} = 4\ 653$ at $q = 3$, $\dim I_4^{bal} = 7\ 885$ at $q = 5$, and $\dim I_2^{ph} = 11\ 713$ at $q = 13$. Twelve of these cells are new to this record, and their values were predicted before they were computed. In the 9 pairs measured, the graded ranks of I_a^{ph} and I_{a+1}^{bal} coincide after a shift of $q - 1$ (Remark 7.9(3)).
- **Theorem 7.6, every down-set.** $\dim V_\Lambda = |Z_\Lambda|$ for every down-set $\Lambda \subseteq BPar_q(\alpha, \beta)$ (equality, not only $\geq$) at the 18 cells $(\alpha, \beta, q) = (1, 1, 3), (2, 1, 3), (2, 2, 3), (3, 2, 3), (3, 3, 3), (4, 3, 3), (4, 4, 3), (5, 4, 3), (2, 1, 5), (2, 2, 5), (3, 2, 5), (3, 3, 5), (2, 1, 7), (2, 2, 7), (2, 2, 9), (3, 1, 9), (3, 2, 9), (2, 2, 25)$: 220 down-sets, 0 failures (auditor's engine; the constructor's engine, run separately, reported equality for 142 down-sets at 11 cells).
- **(P1), (Chain), (P2).** The chain property and the membership of the options in $BPar_q(\alpha, \beta)$ for every tail shape, and Proposition 7.4 on every comparable pair of tail shapes, at 16 cells up to $(8, 7, 5), (6, 6, 9), (7, 5, 27)$ and $(3, 3, 81)$ (up to 7 214 comparable pairs); (P1) by brute force over every tail at 8 cells up to $(2, 2, 27)$ (19 683 tails): 0 failures.
- **(P3).** Every generator of every layer lies in its slice (Proposition 7.5), 3 428 memberships at 7 cells. Negative control: lowering the slice by one gives 17, 58 and 10 failures at $(3, 2, 3), (3, 3, 3)$ and $(2, 2, 5)$.
- **The characteristic.** In characteristic 0, $\dim I_a^{bal} = 4, 24, 169$ for $a = 1, 2, 3$ at $q = 3$ (against 3, 15, 93), $\dim I_a^{ph} = 18, 126$ for $a = 1, 2$ (against 15, 93), and $\dim I_2^{bal} = 129, 416, 1\ 025$ at $q = 5, 7, 9$ (against 45, 91, 153): Theorem C holds with strict inequality (Remark 7.9(2)).
- **The full family is needed.** With a single bijection instead of all of S_2 , the ideal at $(a, q) = (2, 3)$ has dimension $9 < 15$.

12.5 Theorem A

$\dim F_p[L]/K_\mu(L) = N_\mu(n)$ at 178 cells; Theorem A in characteristics 0, 3, 5, 7, 11 for $n = 2, 4, 6$ and $q = 3, 5, 7, 9$ (60 of 60 cells); the row inclusions of Theorem A.12 as ideal memberships at 7 parents; the case bookkeeping of Theorem A.12 for all μ with $|\mu| \leq 14$ at $q = 3, 9, 27, 81$ (1 673 721 objects); the identities of Lemmas A.8–A.10 exactly at 166 random instances; and two negative controls (Remark A.14).

12.6 Independent readings, and their limits

The proofs of §2, §5 and Appendix A were checked in a first pass, then re-derived cold, step by step, by a separate reader with its own code, before this paper was written. An earlier version of this paper (for $m = 3^n$)

was then read cold by three further readers, independently, each with a list of attacks and its own engines. None found a fatal error, a gap or a mathematical error; all their findings were of presentation, and they are incorporated here. One of them audited the proofs inside [DS] of the results collected in Theorem 0, from the source of arXiv:1405.4683v3, and found them sound for every $m \geq 3$ and every even $n \geq 2$; the Picard number of the cubic surface independently confirms the signs of [DS, Theorem 2.2] (with $1 + t$ in place of $1 - t$ in ψ_j , the rank at $(n, m) = (2, 3)$ would be 20, not 7).

The version for odd prime powers was read cold by a further reader of the same system and, for the first time, by AI systems of two other providers (ChatGPT, OpenAI; Gemini, Google). None found a fatal error. Two of them found, independently, that one sentence in the proof of Theorem 5.9, and the same phrase twice in the proof of Proposition 9.1, was literally false: an initial ideal is a monomial ideal and does not contain the non-monomial generators. One classified this as a gap and the other as an error of wording, and both gave the argument used here (Lemma 5.8'). Two readers also pointed out Theorem 5.9' and Corollary H. ChatGPT then read the Hodge part of the version with Corollary H (§1.3, §8 and §11 there) and found one error, a novelty claim contradicted by the cells of [DS, §5]; it is corrected here (§1.3).

The part of this version on the degrees that are not prime powers (§6–§7) was constructed by one instance of the same system, with its own engines, and then audited cold by another, which re-derived every step, including the nine cases of Proposition 7.4 and the three constructions of Proposition 7.5, compared the colour reduction with the author's earlier block decomposition [Rep], and ran separate engines with predictions recorded before the runs (§12.3–§12.4). The new sections (§6, §7, §10, §11) were then read cold by a further instance, which re-derived them and recomputed the counts with its own code; it found no fatal error, no gap and no mathematical error, and its findings of presentation are incorporated. A further cold reader then read the whole of §6, §7, §10, §11 and the chain of the Main Theorem, with eleven section-referees and adversarial verifiers of its own. It recomputed the Fermat fourfolds of degree 15 and 21 at both primes from the literal generators of [DS], colouring by colouring, and checked Theorem 7.6 at 117 down-sets of 11 cells and Theorem C at 10 new root cells. It found no fatal error and no gap in the proofs; one error of self-description (the upper bound of Proposition 2.1 is algebraic, which is now stated in §1.2 as Main Theorem'); one sentence missing in the deformation argument of Corollary W(i), now supplied; and items of presentation, incorporated. ChatGPT (OpenAI) then read the previous version (v7) cold, attacking the new parts with its own computations and with the source of [DS]. It found no fatal error, no gap and no mathematical error; its one point of presentation is incorporated here (the proof of Lemma 6.8).

The limits of this record must be stated plainly.

- **All these readers are AI systems** (Claude, Anthropic, for all but four readings; ChatGPT for three of them and Gemini for one; see the note at the end). They worked separately and without access to each other's code, but they are not independent in the way that different human experts are.
- **No human expert has refereed this work.**
- **Pham's theorem [Ph] is not re-proved here**, and neither are the intersection numbers [DS, Theorem 2.2]; Appendix B reduces Theorem 0 to these two results.
- The numbering of [DS] was checked against arXiv:1405.4683v3, not against the printed version.
- **The Lean proof covers Main Theorem' only** (§12.8). Its proofs were written by an AI system and are checked by the Lean kernel; that the Lean statement says what Main Theorem' says is an audit, set out line by line in [LC, §1.3], which a reader can repeat on about a dozen short definitions.

12.7 A referee's guide

The joints deserve attention in this order.

1. **Proposition 5.6 (P2)** and **Proposition 7.4**, the only case analyses in the proof that are not one-line identities. A sceptical reader should try long, very uneven partitions at $q = 3$ ($h = 1$) and $q = 9$ ($h = 4$),

where lengths hit the cap $\ell \leq h$, and, for Proposition 7.4, the cases AR and MR, where a removal of one pair of partitions is compared with an option of another kind.

2. **The translation (§2):** the conventions of [DS] ($j_0 = 0$, the variable t_0 never appears, $N = 2k+2$) and Lemma 2.4.
3. **The colour reduction (§6):** Proposition 6.5 (the surviving generators are pure tensors; this is where the full family is used, as Lemma 5.2 is in §5) and Lemma 6.8, where the index 0 couples the blocks only through w_0 ; and Corollary 7.8, whose proof passes through the sum over the colourings.
4. **The coverage formulas (5.1) and (7.1), and Propositions 5.8 and 7.5,** which have no slack (Remark 5.10(2), §12.4).
5. **Theorem 0** and Appendix B. Only Theorem 0(a), (c) and (d) are topological; Theorem 0(b), which carries every upper bound of the paper, is a Chinese-remainder count.
6. **Corollary H:** the classical input (H3) and Aoki's theorem (H5) of §10.1, which are quoted and not re-proved here; (H4) is proved in §10.1.

12.8 A machine-checked proof of Main Theorem'

Main Theorem' has been proved in the proof assistant Lean 4 [dMU] with the Mathlib library [Mlib]; the certificate is [LC]. The final theorem, *ColAssembly.mainTheorem'*, states: *for every odd $m \geq 1$ and every $k \geq 0$, $R/(\psi_J : J \in \mathcal{J})$ is a free abelian group of rank $m^{2k+1} - Q_k(m)$, and for every field F , $\dim_F F[G]/(\bar{\psi}_J : J \in \mathcal{J}) = m^{2k+1} - Q_k(m)$.* This is slightly more general than Main Theorem' as stated in §1.2 ($m \geq 3$, $k \geq 1$, prime fields). In Lean, $Q_k(m)$ is defined by the formula of §1.2, and two lemmas identify it with the number of closed tuples of Lemma 2.2 and with $|\Gamma_{\mathcal{J}}|$.

- **What was formalized.** The chain of the proof as written here: Lemma 2.2 and the translation of §2 in its local form (Lemmas 2.3, 2.4 and Proposition 2.5); Theorem 5.3 and the induction of §5; the colour reduction of §6 (Lemmas 6.1–6.8, Proposition 6.9); Theorems 7.6 and C; and the upper bound [DS, Claim 4.3] (Theorem 0(b), Appendix B, Step 6), which is algebraic. Theorem 4.1 and Theorems 5.9 and 5.9' were formalized too, although Main Theorem' does not need them.
- **How.** The Lean proofs were written by the AI system Aristotle (Harmonic), in 29 pieces, each written from this paper. Every returned file was compiled again on the author's machine (Lean v4.28.0, Mathlib at the tag v4.28.0) and audited against the paper before the next piece was sent: the difference with the previous project, a search for forbidden constructs, the build, the statements as Lean prints them, and their axioms.
- **Result.** 113 files, 21 126 lines, 937 theorems. There is no *sorry*, *admit*, *native decid* or added axiom, and the final theorem depends only on the axioms *propext*, *Classical.choice* and *Quot.sound*. A rebuild from scratch compiles all 113 files. A program that collects the declarations used by the final theorem finds 2 367 of them, in 90 of the 113 files; the files it does not use are those of §4 and of Theorems 5.9 and 5.9', which §1.2 marks as off the path of the proof.
- **What is not formalized.** Theorem 0(a), (c) and (d): the topology of [DS] that identifies the torsion of $R/(\psi_J)$ with that of $H_{2k}(X; Z)/L(X)$ and gives the rank of $L(X)$. Mathlib has no singular homology of complex Fermat varieties. So the Main Theorem, as a statement about homology, is machine-checked only in its algebraic form. Theorem A, Corollaries H and W, §9 and the computations of §12.1–§12.5 are not formalized either.
- **Where.** The certificate [LC] gives the Lean statement and definitions verbatim, compares them with this paper line by line, and lists the trust base, the data of every piece and how to reproduce the build. The Lean project, the pieces, the checks and every log are in [Rep], folder *lean/*.

13. Open problems

1. **Even degrees.** For $m = 2^v$, and more generally for even m , 2 is not a unit, the coordinate $y = t - t^{-1}$ of Lemma 2.3 does not exist, the involution on $\mu_m \setminus \{1\}$ has a fixed point, and Theorem A fails in characteristic 2. The colour reduction of §6 still splits off the odd part of m , but the block of the prime 2 needs a different method. A first step would be to compute the leading monomials of $(\bar{\psi}_J)$ at $(k, m) = (1, 4), (2, 4), (1, 8), (2, 8)$.
2. **The remaining integral Hodge classes.** When m is not prime and has a prime factor at most $2k + 1$, find generators of $Hdg(X)$ beyond $L(X)$. By Corollary H(ii) the missing part is a free group of rank $|\mathfrak{B}| - Q_k(m)$ (Remark 10.3, Lemma 10.4), and the integral question is whether algebraic cycles fill it. For $k = 1$ they do, by the Lefschetz $(1,1)$ theorem; the question is for $k \geq 2$. The rational Hodge conjecture for these X is known in many cases [Shi79], [Aok], [Jum].
3. **Elementary divisors.** For $m = p$ prime, compute the elementary divisors and the discriminant of $Hdg(X) = L(X)$ in closed form for $k \geq 2$; [AMV, Table 1] gives the data for small cells. For $k = 1$ closed forms are stated in the author's *Watermark* and *Double Ladder* theorems (Remark 10.2); a complete proof of the second is part of the question. In every cell of [AMV, Table 1] with m prime, $|disc\ Hdg(X)|$ is a power of m ; whether this holds for every k is part of the question.
4. **A direct proof of Corollary 7.8.** Prove $\dim_{\mathbb{F}_p} I_a^{bal} \leq N_{bal}(a, q)$ on the bipartite block itself, without passing through the sum over the colourings of §6; since the dimension over $\mathbb{Q}$ is larger (Remark 7.9(2)), such a proof must use the characteristic. Equality $\dim V_\Lambda = |Z_\Lambda|$ for every down-set of Theorem 7.6, measured at every down-set we tested (§12.4) and never contradicted, would give it.
5. **Representation theory.** The Specht module route at $q = 3$ (Remark 4.2(3)), and more generally a representation-theoretic description of the rings C_m/V_Λ (Complement to Theorem 5.9) and of the bipartite quotients.

14. Exact status

- **PROVED, with no topology:** Main Theorem' — for every odd $m \geq 3$ and every $k \geq 1$, the group $R/(\psi_J : J \in \mathcal{J})$ of [DS] is free abelian of rank $m^{2k+1} - Q_k(m)$. It uses §5–§7 and [DS, Claim 4.3], a Chinese-remainder count re-proved in Appendix B, Step 6.
- **MACHINE-CHECKED in Lean 4 with Mathlib (§12.8):** Main Theorem', for every odd $m \geq 1$ and every $k \geq 0$, over $\mathbb{Z}$ and over every field; no *sorry*, and no axiom beyond *propext*, *Classical.choice*, *Quot.sound*. Not formalized: the topology of Theorem 0(a), (c) and (d), Theorem A, and Corollaries H and W.
- **PROVED, modulo Pham's theorem [Ph] and the intersection numbers [DS, Theorem 2.2]:** the Main Theorem, i.e. Conjecture 1.2 of Degtyarev–Shimada for the Fermat varieties of every odd degree $m \geq 3$ and every even dimension. Through Theorem 0 it uses [DS, Theorem 1.1(a), Claim 4.3, Theorem 1.4, Corollary 1.5], which Appendix B re-derives from those two results and classical topology.
- **PROVED, modulo in addition the standard topological facts (H1)–(H2), the classical description of Hodge classes (H3) [SK], [Shi79], [Ran], and Aoki's theorem (H5) [Aok] (for prime m , (H4) replaces (H5); its proof uses the formula for B_{L_X} and Dirichlet's theorem $L(1, \chi) \neq 0$ [Was]):** Corollary H — for odd m , $L(X)$ is a primitive sublattice of $Hdg(X)$ with free quotient of rank $|\mathfrak{B}| - Q_k(m)$, and the integral Hodge conjecture holds for X when m is prime or every prime factor of m is at least $2k + 3$. In the seven prime cells computed in [DS, §5] it was already a consequence of that computation (§1.3).
- **PROVED, modulo the same two results as the Main Theorem:** Corollary W(ii), and, modulo in addition the transport argument of [DS, §4.6], Corollary W(i).

- **PROVED:** Theorem B (the dimension count, with equality over every field and for every odd q , and the freeness over $\mathbb{Z}$), Theorems 5.3, 5.9 and 5.9', Theorem 4.1; the colour reduction (Proposition 6.9); Theorem C and Theorem 7.6; Corollary 7.8; Lemma 10.4.
- **PROVED:** Theorem A, for every field of characteristic $\neq 2$, every odd q and every n , with Corollaries 8.1 and 8.2. It is a different statement from the Main Theorem.
- **CITED, NOT PROVED HERE:** the author's *Watermark* and *Double Ladder* theorems (the case $k = 1$ of problem 3 of §13), deposited in [Rep]; the *Double Ladder* is not yet proved for every prime (Remark 10.2).
- **MEASURED, NOT CLAIMED:** the leading-monomial description of Remark 5.10(4) outside $q = 3$; the Specht-module route of Remark 4.2(3); equality $\dim V_\Lambda = |Z_\Lambda|$ in Theorem 7.6 for down-sets other than the roots.
- **COMPUTED EXACTLY (finite enumerations, §12):** the counts of Hodge characters of Remark 10.3.
- **FALSE:** the analogue of the Main Theorem for arbitrary subfamilies $K \subseteq \mathcal{J}$ (Fact 9.2); the identification «Conjecture 1.2 $\iff A_k(q) = P_k(q)$ »; the equality $\text{Hdg}(X) = L(X)$ for odd m not prime with a prime factor at most $2k + 1$ (Corollary H(ii)).
- **OPEN:** even m (§13).
- **NOT YET REFEREED by a human expert.**

Acknowledgements and use of AI

The mathematics of this paper was developed with extensive use of Claude, an AI system made by Anthropic, working under the author's direction in separate roles: instances that proposed and wrote proofs, instances that re-derived them cold and ran their own checks, and instances that read the finished text as referees. The previous versions were also read by AI systems of two other providers (ChatGPT, OpenAI; Gemini, Google). The degrees that are not prime powers (§6–§7) were constructed by one instance and audited cold by another. The Lean 4 proof of Main Theorem' (§12.8) was written by Aristotle, an AI system made by Harmonic, from pieces written from this paper, and was compiled and audited on the author's machine. The author directed the project and takes responsibility for the content.

The work took more than two and a half months. It went through several hundred written working sessions, recorded more than three hundred unsuccessful routes together with the reason each one failed, and produced more than two hundred standalone write-ups of intermediate results. The full record — the engines and their logs, the audits and the cold readings, the dead routes, and the history of the project — is public at github.com/tretoef-estrella/chaise-longue-theorem [Rep]. Every result of the author on which this paper relies or which it cites — the earlier block decomposition quoted in §6, the *Watermark* and *Double Ladder* theorems quoted in §10, and the computer verdicts quoted in §1.3 and §12 — is deposited in that repository, together with every engine and log behind §12 and the Lean project of §12.8.

Appendix A. Proof of Theorem A

This appendix is independent of §2–§5. **Its symbols are local:** here L is a finite set of variables, A and B are subsets of L , C is a set of field elements (Lemma A.4), J is an ideal (Proposition A.5), $R_a(G)$ are the «rows» of Lemma A.6, $K_\mu(L)$ are ideals, and Θ is a generating function (it has nothing to do with Γ_K of [DS]).

A.0 Conventions

- F is a field with $2 \in F^\times$, $q \geq 1$ is odd, and $h = (q-1)/2$.

- For $X \subseteq L$, $E_X(t) := \prod_{x \in X} (1 + xt)$ and $e_f(X) := [t^f] E_X(t)$, so $e_0 = 1$ and $e_j = 0$ for $j < 0$ or $j > |X|$. $I_q(L) := (e_f(L) : j \text{ odd}) + \text{box}_L$ with $\text{box}_L := (x^d : x \in L)$.
- $[t^D]$ of a power series is 0 when $D < 0$.
- $I_d(2y) := \sum_{b \geq 0} y^{2b+d} / (b!(b+d)!)$ for $d \geq 0$, and $N_q(n) := n! \cdot [y^n] e^y I_0(2y)^h$.
- A **partition** is a finite multiset of positive integers $\mu_1 \geq \dots \geq \mu_\ell$, with $\ell = \ell(\mu)$ and $|\mu| = \sum \mu_i$. For $w \geq 0$, $\beta_w(\mu) := \sum_i \max(\mu_i - w, 0)$ (the number of boxes of the Young diagram to the right of column w), and $c_w(\mu) := \beta_w(\mu) - \beta_{w+1}(\mu) = \#\{i : \mu_i \geq w+1\}$. So $\beta_0(\mu) = |\mu|$, $c_0(\mu) = \ell(\mu)$, and $c_w(\mu) \leq \ell(\mu)$ for all w .

Strategy. Adjoin one variable z to L . Every ideal G of $F[L \cup z]$ containing z^q has q rows $R_0(G) \subseteq \dots \subseteq R_{q-1}(G)$, and $\dim F[L \cup z]/G = \sum_a \dim F[L]/R_a(G)$ (Lemma A.6). The count satisfies the matching recursion $N_\mu(n+1) = \sum_a N_{\text{child}_a(\mu)}(n)$ (Lemma A.3), which is the derivative of a product of Bessel functions. So it suffices to attach to each μ an ideal $K_\mu(L)$ with $K_\emptyset(L) = I_q(L)$ such that every row of $K_\mu(L \cup z)$ contains the ideal of the corresponding child (Theorem A.12). The ideals come from one generating function Θ , and adjoining z to one of three sets shifts an index of Θ in a controlled way (Lemma A.8). This is the same peeling idea as §5.1, with the box q in place of $q-1$.

A.1 The count and the dictionary

For a partition μ with $\ell(\mu) \leq h$, put $F_\mu(y) := e^y \cdot \prod_{i=1}^\ell I_{\mu_i}(2y) \cdot I_0(2y)^{h-\ell}$ and $N_\mu(n) := n! \cdot [y^n] F_\mu(y)$, so that $N_q(n) = N_\emptyset(n)$.

Lemma A.1. For $d \geq 1$, $(d/dy) I_d(2y) = I_{d-1}(2y) + I_{d+1}(2y)$, and $(d/dy) I_0(2y) = 2 I_1(2y)$.

Proof. Termwise, with $(2b+d) = b + (b+d)$:

$$\sum_b (2b+d) y^{2b+d-1} / (b!(b+d)!) = \sum_{b \geq 1} y^{2b+d-1} / ((b-1)!(b+d)!) + \sum_b y^{2b+d-1} / (b!(b+d-1)!).$$

The first sum is $I_{d+1}(2y)$ (substitute $b \mapsto b+1$). The second is $I_{d-1}(2y)$ if $d \geq 1$, and for $d = 0$ it is again $I_1(2y)$. ■

Definition A.2 (the dictionary). Let $\ell = \ell(\mu) \leq h$. For $a \in \{0, \dots, q-1\}$, $\text{child}_a(\mu)$ is the partition:

- **lower**, $0 \leq a \leq \ell-1$: μ with one copy of μ_{a+1} replaced by $\mu_{a+1} - 1$; parts are lowered **largest first**, and a part 0 is deleted;
- **value-0**, $a = \ell$: μ ;
- **new-class**, $\ell+1 \leq a \leq q-\ell-1$: $\mu \cup (1)$;
- **raise**, $a = q-j$ with $1 \leq j \leq \ell$: μ with one copy of μ_j replaced by $\mu_j + 1$; row $q-1$ raises the largest part.

New-class rows exist iff $\ell+1 \leq q-\ell-1$, i.e. iff $\ell+1 \leq h$. Hence every child has at most h parts.

Lemma A.3. $N_\mu(0) = [\mu = \emptyset]$, and $N_\mu(n+1) = \sum_{a=0}^{q-1} N_{\text{child}_a(\mu)}(n)$ for every $n \geq 0$.

Proof. The first claim holds because $F_\mu(0) = [\mu = \emptyset]$. For the second, $N_\mu(n+1) = n! \cdot [y^n] F'_\mu$. By the product rule and Lemma A.1, F'_μ is a sum of:

- F_μ , from e^y ;
- for each i , F of μ with μ_i replaced by $\mu_i - 1$, plus F of μ with μ_i replaced by $\mu_i + 1$ (a part 0 is a factor I_0);
- $2(h-\ell)$ copies of $F_{\mu \cup (1)}$, from $I_0^{h-\ell}$.

These are exactly the $\ell + 1 + 2(h-\ell) + \ell = q$ children of Definition A.2. ■

Lemma A.4. Let C be a set of q elements of a field of characteristic $\neq 2$ with $0 \in C = -C$. Then $N_q(n)$ is the number of $c \in C^n$ such that $\#\{i : c_i = v\} = \#\{i : c_i = -v\}$ for every $v \in C \setminus \{0\}$.

Proof. $C \setminus \{0\}$ splits into h pairs $\{v, -v\}$ with $v \neq -v$. By the exponential formula:

- the zero coordinates contribute the factor e^y ;
- each pair with b coordinates equal to v and b equal to $-v$ contributes $\sum_b y^{2b} / (b! \cdot b!) = I_0(2y)$. ■

A.2 The lower bound

Proposition A.5. $\dim_F F[L]/I_q(L) \geq N_q(|L|)$.

Proof.

Reduction to a large field. Let F' be an infinite extension of F . The dimension is unchanged by extension of scalars, because $F'[L]/I_q(L)F'[L] = (F[L]/I_q(L)) \otimes_F F'$. Choose $c_1, \dots, c_h \in F^\times$ with $c_i \neq \pm c_j$ for $i \neq j$, and put $C := \{0, \pm c_1, \dots, \pm c_h\}$. These are q distinct elements, since $c \neq -c$ for $c \neq 0$.

A radical ideal with the right count. Let $f(x) := \prod_{c \in C} (x - c)$ and $J := (e_j(L) : j \text{ odd}) + (f(x) : x \in L)$.

- By the Chinese remainder theorem, $F'[L]/(f(x) : x \in L) \cong F'^{C^L}$, a product of copies of F' .
- Every quotient of a finite product of fields by an ideal is the product over a subset of the factors. Hence $F'[L]/J \cong F'^Z$, where Z is the set of $c \in C^L$ at which every odd e_j vanishes.
- For such c , all odd $e_j(c)$ vanish iff $\prod_x (1 + c_x t)$ is an even polynomial, i.e. $\prod_x (1 + c_x t) = \prod_x (1 - c_x t)$.
- By unique factorisation in $F'[t]$, this holds iff the multiset $\{c_x \neq 0\}$ is closed under negation.
- So $\dim F'[L]/J = |Z| = N_q(n)$ by Lemma A.4.

Degeneration. Let $\text{in}(J)$ be the span of the top-degree forms of the elements of J .

- $(e_j : j \text{ odd}) \subseteq \text{in}(J)$, since the e_j are homogeneous.
- $x^q \in \text{in}(J)$, since x^q is the top form of $f(x)$.

Hence $I_q(L) \subseteq \text{in}(J)$. For a degree-compatible monomial order, $LM(g) = LM(\text{top}(g))$. Moreover, for fixed d the non-zero top forms of degree- d elements of J form, with 0 , a vector space. So $LM(J) = LM(\text{in}(J))$, and by Macaulay's basis theorem $\dim F'[L]/\text{in}(J) = \dim F'[L]/J$. Therefore

$$\dim F'[L]/I_q(L) \geq \dim F'[L]/\text{in}(J) = N_q(n). \blacksquare$$

A.3 Rows

Lemma A.6. Let $z \notin L$, and let $G \subseteq F[L \cup z]$ be an ideal with $z^q \in G$. For $0 \leq a \leq q-1$ put

$$R_a(G) := \{f \in F[L] : z^a f \in G + z^{a+1}F[L \cup z]\}.$$

Then:

- each $R_a(G)$ is an ideal of $F[L]$;
- $R_a(G) \subseteq R_{a+1}(G)$;
- $\dim_F F[L \cup z]/G = \sum_{a=0}^{q-1} \dim_F F[L]/R_a(G)$, in $\mathbb{N} \cup \{\infty\}$.

Proof.

- **Ideal:** $R_a(G)$ is an additive group, and for $g \in F[L]$ we have $z^a gf = g \cdot z^a f$.
- **Monotone:** multiplying $z^a f \in G + (z^{a+1})$ by z gives $z^{a+1} f \in G + (z^{a+2})$.
- **Dimension:** put $G_a := G + z^a F[L \cup z]$. Then $G_0 = F[L \cup z]$ and $G_q = G$. The map $F[L] \rightarrow G_a/G_{a+1}$, $f \mapsto z^a f$, is surjective because $z^a F[L \cup z] = z^a F[L] + z^{a+1} F[L \cup z]$, and its kernel is $R_a(G)$. Summing over a gives the formula. $\blacksquare$

A.4 The generating function

Fix disjoint $A, B \subseteq L$. Put:

- $P := L \setminus (A \cup B)$;
- $r := |A|$;
- $w := |B| - |A|$, the **imbalance**.

For $a \in A$ let $\varphi_a(t) := (1 - at)/(1 + at) = 1 + \sum_{i \geq 1} 2(-1)^i a^i t^i$.

Definition A.7. For $s \in \mathbb{Z}$, let $D_L(A, B; s) := r(q-1) + |P| + 1 - s$ and

$$\Theta^s_L(A;B) := [t^{D_L(A,B;s)}] E_P(-t) \cdot \Pi_{a \in A} \varphi_a(t) \in F[L].$$

We call s the **level**.

Remark. With $H_{A;B}(t) := \Pi_{a \in A}(1+at)^{-1} \Pi_{b \in B}(1-bt)^{-1}$ one has $E_L(-t)H_{A;B}(t) = E_P(-t)\Pi_{a \in A}\varphi_a(t)$: the factors of B cancel. So B enters Θ only through P , that is, through the degree D .

Lemma A.8 (the three moves, and vanishing). *Let $z \notin L$ and $L^+ := L \cup \{z\}$. For disjoint $A, B \subseteq L$ and $s \in \mathbb{Z}$:*

- **(P)** $\Theta^s_{L^+}(A;B) = \Theta^{s-1}_L(A;B) - z \cdot \Theta^s_L(A;B)$;
- **(B)** $\Theta^s_{L^+}(A;B \cup \{z\}) = \Theta^s_L(A;B)$;
- **(A)** $\Theta^s_{L^+}(A \cup \{z\};B) = \Theta^{s-q+1}_L(A;B) + \sum_{i \geq 1} 2(-1)^i z^i \cdot \Theta^{s-q+1+i}_L(A;B)$, a finite sum;
- **(V)** if $s \leq 0$, then $\Theta^s_L(A;B) \in \text{box}_A$, and $\Theta^s_L(A;B) = 0$ if $A = \emptyset$.

Proof.

- **(P)** In L^+ , z lies in the set $P^+ = P \cup \{z\}$, and $E_{P^+}(-t) = E_P(-t)(1-zt)$. Since $D_{L^+}(A,B;s) = D_L(A,B;s-1)$ and $D_{L^+}(A,B;s) - 1 = D_L(A,B;s)$, the identity follows.
- **(B)** Neither P nor D changes.
- **(A)** One more factor $\varphi_z(t)$, and $D_{L^+}(A \cup \{z\}, B; s) = D_L(A, B; s) + (q-1)$. The coefficient of $z^i t^i$ in φ_z is 1 for $i = 0$ and $2(-1)^i$ for $i \geq 1$. Finally, level u of $(A;B)$ in L has degree $D_L(A,B;s) + (q-1) - i$ exactly when $u = s - q + 1 + i$.
- **(V)** $\Theta^s_L(A;B) = \sum_p (-1)^p e_p(P) \cdot [t^{D-P}] \Pi_A \varphi_a$. The coefficient of t^m in $\Pi_A \varphi_a$ is homogeneous of degree m in the r letters of A . For $s \leq 0$ every $p \leq |P|$ gives $D - p \geq r(q-1) + 1$. So every monomial has some exponent $\geq q$ when $r \geq 1$, and $[t^{D-p}]1 = 0$ when $r = 0$. ■

Lemma A.9 (descent). *For $a \geq 0$:*

$$z^a \cdot \Theta^s_L(A;B) = \Theta^{s-a}_{L^+}(A;B \cup \{z\}) - \sum_{i=1}^a z^{i-1} \cdot \Theta^{s-a+i}_{L^+}(A;B).$$

Proof. The case $a = 0$ is (B). By (P) and (B), $z \cdot \Theta^s_L(A;B) = \Theta^{s-1}_{L^+}(A;B \cup \{z\}) - \Theta^s_{L^+}(A;B)$, and applying (B) again to the first term gives $z \Theta^s_L = \Theta^{s-1}_L(A;B) - \Theta^s_{L^+}(A;B)$. Hence

$$z^a \Theta^s_L = z^{a-1} \Theta^{s-1}_L(A;B) - z^{a-1} \Theta^s_{L^+}(A;B),$$

and the formula follows by induction on a . ■

Lemma A.10 (the odd identity). *Let $|A| = |B| = r$, let $c \in \mathbb{Z}$, $M := D_L(A,B;c)$, and $h_m := [t^m] H_{A;B}(t)$. For every $\ell \geq 0$,*

$$z^{\ell+1} \Theta^{c+1}_L(A;B) \equiv \Omega - \frac{1}{2} z^\ell \cdot ((-1)^M \Theta^c_L(B;A) - \Theta^c_L(A;B)) \mod z^{\ell+2} F[L^+],$$

where $\Omega := z^\ell \sum_{j \text{ odd}} e_j(L^+) h_{M-j} - z^{\ell+1} \sum_{j \text{ odd}} e_j(L^+) h_{M-1-j}$ lies in $(e_j(L^+) : j \text{ odd})$.

Proof. Put $\alpha_m := [t^m] E_L(t) H_{A;B}(t)$ and $\beta_m := [t^m] E_L(-t) H_{A;B}(t)$. By definition and the Remark after Definition A.7, $\beta_M = \Theta^c_L(A;B)$ and $\beta_{M-1} = \Theta^{c+1}_L(A;B)$.

Next, $E_L(t) H_{A;B}(t) = E_P(t) \Pi_B(1+bt)/(1-bt)$, which at $t = -u$ is $E_L(-u) H_{B;A}(u)$. Since $|A| = |B|$, the degree M corresponds to the same level c for $(B;A)$, so $\alpha_M = (-1)^M \Theta^c_L(B;A)$.

Let $X_m := [t^m] O(t) H_{A;B}(t)$ with $O(t) := \sum_{j \text{ odd}} e_j(L^+) t^j = \frac{1}{2}(E_{L^+}(t) - E_{L^+}(-t))$, so that $\Omega = z^\ell X_M - z^{\ell+1} X_{M-1}$. From $E_{L^+}(\pm t) = E_L(\pm t)(1 \pm zt)$:

$$X_m = \frac{1}{2}(\alpha_m + z\alpha_{m-1} - \beta_m + z\beta_{m-1}).$$

Hence

$$\Omega = \frac{1}{2} z^\ell (\alpha_M - \beta_M) + z^{\ell+1} \beta_{M-1} - \frac{1}{2} z^{\ell+2} (\alpha_{M-2} + \beta_{M-2}).$$

Rearranging gives the claim. ■

A.5 The ideals

Definition A.11. For a partition μ with $\ell(\mu) \leq h$ and a finite set L ,

$$K_\mu(L) := I_q(L) + (\Theta_L^s(A;B) : A, B \subseteq L \text{ disjoint}, w = |B| - |A| \geq 0, 1 \leq s \leq \beta_w(\mu)).$$

Remarks.

1. $K_\emptyset(L) = I_q(L)$, since $\beta_w(\emptyset) = 0$.
2. If $\mu \neq \emptyset$ and $L = \emptyset$, then $\Theta_\emptyset^1(\emptyset; \emptyset) = [t^0]1 = 1$, so $K_\mu(\emptyset) = F$.
3. For $A = \emptyset$, the generators are $\pm e_{|S|+1-s}(S)$ with $S = L \setminus B$, allowed for $s \leq \beta_{n-|S|}(\mu)$. For $S \subsetneq L$ this is Tanisaki's rule for the partition $\mu \cup (1^{n-|\mu|})$ of n [Ta], [GP]; for $S = L$ it adds the $e_j(L)$ with $j \geq n - |\mu| + 1$.
4. The case $s = 1$, $A \neq \emptyset$ gives $\pm 2^r (\Pi_A a^{q-1}) \cdot \Pi_P x$ modulo box .
5. The generators with $A \neq \emptyset$ and higher s are not suggested by Tanisaki's rule. They were first found by computation, as the corrections needed for $\mu = (1,1)$, $(1,1,1)$ and $(2,2)$, and the generating function Θ organises them.

A.6 The row inclusions

Theorem A.12. Let μ be a partition with $\ell := \ell(\mu) \leq h$, L a finite set, $z \notin L$ and $K^+ := K_\mu(L \cup z)$. For every $a \in \{0, \dots, q-1\}$:

$$R_a(K^+) \supseteq K_{\text{child}_a(\mu)}(L).$$

Lemma A.13. Let $A, B \subseteq L$ be disjoint with $w := |B| - |A| \geq 0$.

- (i) If $s \geq 1$ and $s+1 \leq \beta_w(\mu)$, then $\Theta_L^s(A;B) \in R_0(K^+)$.
- (ii) If $s \in \mathbb{Z}$, $s \leq \beta_w(\mu)$, $a \geq 0$ and $s-a \leq \beta_{w+1}(\mu)$, then $z^a \cdot \Theta_L^s(A;B) \in K^+$. (For $s \leq 0$ this holds already by (V).)
- (iii) $I_q(L) \subseteq R_0(K^+)$.

Proof.

- (i) $\Theta^{s+1}_L(A;B)$ is a generator of K^+ : its imbalance is w and its level $s+1 \leq \beta_w(\mu)$. By (P) it equals $\Theta_L^s(A;B) - z\Theta^{s+1}_L(A;B)$.
- (ii) Use Lemma A.9.
 - The first term has imbalance $w+1$ and level $s-a \leq \beta_{w+1}(\mu)$.
 - The terms of the sum have imbalance w and levels in $[s-a+1, s] \subseteq (-\infty, \beta_w(\mu)]$.
 - Each term is a generator of K^+ if its level is ≥ 1 , and lies in box by (V) otherwise.
- (iii) $e_j(L) = e_j(L^+) - z \cdot e_{j-1}(L)$, and $x^q \in K^+$ for $x \in L$. ■

Proof of Theorem A.12. By Lemma A.6 the rows increase, so it suffices to place every generator of $K_{\text{child}_a(\mu)}(L)$ in the row a assigned by the dictionary (or in an earlier row). Generators from $I_q(L)$ are handled by (iii). So consider a child generator $\Theta_L^s(A;B)$ with $w = |B| - |A| \geq 0$ and $1 \leq s \leq \beta_w(\text{child})$. Recall $\beta_w - \beta_{w+1} = c_w \leq \ell$.

Lower, $0 \leq a \leq \ell-1$. Here $\beta_w(\text{child}) = \beta_w(\mu) - [\mu_{a+1} > w]$.

- If $\mu_{a+1} > w$, then $s+1 \leq \beta_w(\mu)$, and (i) applies.
- If $\mu_{a+1} \leq w$, then all parts $> w$ are among $\mu_1, \dots, \mu_a$ (parts are ordered decreasingly), so $c_w(\mu) \leq a$. Hence $s-a \leq \beta_w(\mu) - c_w(\mu) = \beta_{w+1}(\mu)$, and (ii) gives $z^a \Theta_L^s \in K^+$.

Value-0, $a = \ell$. Here $s \leq \beta_w(\mu)$ and $s-\ell \leq \beta_w(\mu) - c_w(\mu) = \beta_{w+1}(\mu)$, so (ii) applies.

New-class, $\ell+1 \leq a \leq q-\ell-1$. Here $\beta_w(\mu \cup (1)) = \beta_w(\mu) + [w=0]$.

- Every generator other than $(w, s) = (0, |\mu|+1)$ is a generator of $K_\mu(L)$, hence lies in row $\ell \subseteq \text{row } a$ by the value-0 case.
- For $w=0$ (so $|A| = |B|$) and $s = |\mu| + 1$, apply Lemma A.10 with $c = |\mu|$:

- $\Omega \in K^+$;
- by (ii) with $a = \ell$, both $z^\ell \Theta^{|\mu|}_L(A;B)$ and $z^\ell \Theta^{|\mu|}_L(B;A)$ lie in K^+ . The pairs $(A;B)$ and $(B;A)$ have imbalance 0; $|\mu| = \beta_0(\mu)$; and $|\mu| - \ell = \beta_0 - c_0 = \beta_1(\mu)$.
- Hence $z^{\ell+1} \Theta^{|\mu|+1}_L(A;B) \in K^+ + (z^{\ell+2})$, i.e. the generator lies in $R_{\ell+1} \subseteq R_a$.

Raise, $a = q - j$, $1 \leq j \leq \ell$. Here $\beta_w(\text{child}) = \beta_w(\mu) + [\mu_j \geq w]$, and $q - j \geq q - \ell \geq \ell + 1$ because $2\ell \leq q - 1$.

- *Old levels, $s \leq \beta_w(\mu)$:* $s - (q - j) \leq \beta_w(\mu) - \ell - 1 < \beta_{w+1}(\mu)$, so (ii) applies.
- *New level at $w = 0$, $s = |\mu| + 1$:* the new-class argument places it in $R_{\ell+1} \subseteq R_{q-j}$.
- *New level at $w \geq 1$ (only when $\mu_j \geq w$):* $s = t := \beta_w(\mu) + 1$. Put $s' := \beta_{w-1}(\mu)$ and consider $\Pi := \Theta^{s'}_{L^+}(A \cup \{z\}; B)$.
 - Π has imbalance $w - 1 \geq 0$ and level $s' \geq c_{w-1}(\mu) \geq j \geq 1$, so it is a generator of K^+ .
 - By (A), $\Pi = \sum_{i \geq 0} \kappa_i z^i \Theta^{u_i}_L(A;B)$, with $u_i := s' - q + 1 + i$, $\kappa_0 = 1$ and $\kappa_i = 2(-1)^i$ for $i \geq 1$, all units.
 - Put $i^* := q - c_{w-1}(\mu)$. Then $u_{i^*} = \beta_w(\mu) + 1 = t$, and $i^* \leq q - j$ because $\mu_1, \dots, \mu_j \geq w$ gives $c_{w-1}(\mu) \geq j$.
 - For $i < i^*$: $u_i \leq \beta_w(\mu)$, and $u_i - i = s' - q + 1 = \beta_{w+1}(\mu) + c_w(\mu) + c_{w-1}(\mu) - (q - 1) \leq \beta_{w+1}(\mu)$, since $c_w + c_{w-1} \leq 2\ell \leq q - 1$. So $z^i \Theta^{u_i}_L(A;B) \in K^+$ by (ii), or lies in *box* if $u_i \leq 0$. (This includes $i = 0$, which matters when $s' \geq q$.)
 - Therefore $\kappa_{i^*} z^{i^*} \Theta^t_L(A;B) \in K^+ + (z^{i^*+1})$. Since κ_{i^*} is a unit, $\Theta^t_L(A;B) \in R_{i^*} \subseteq R_{q-j}$. ■

Remark A.14 (the order of the lower rows is used). The lower case uses «largest part first». With the opposite order, the inclusion fails already for $\mu = (3, 1)$, $q = 9$, $|L| = 5$: 65 generators of the child (3) lie outside row 0. The count of Lemma A.3 does not see the order; the inclusions do.

A.7 Proof of Theorem A

Theorem A.15. For every $n \geq 0$, every set L of n variables and every partition μ with $\ell(\mu) \leq h$: $\dim_F F[L]/K_\mu(L) \leq N_\mu(n)$.

Proof. Induction on n .

Base, $n = 0$. $F[\emptyset] = F$. For $\mu = \emptyset$, $K_\emptyset(\emptyset) = 0$ and $N_\emptyset(0) = 1$. For $\mu \neq \emptyset$, $K_\mu(\emptyset) = F$ (Remark 2 after Definition A.11) and $N_\mu(0) = 0$.

Step. Let $|L| = n$ and $z \notin L$. Then $z^q \in K_\mu(L \cup z)$, and:

$$\begin{aligned} \dim F[L \cup z]/K_\mu(L \cup z) &= \sum_{a=0}^{q-1} \dim F[L]/R_a(K_\mu(L \cup z)) \quad (\text{Lemma A.6}) \\ &\leq \sum_a \dim F[L]/K_{\text{child}_a(\mu)}(L) \quad (\text{Theorem A.12}) \\ &\leq \sum_a N_{\text{child}_a(\mu)}(n) \quad (\text{induction; every child has } \leq h \text{ parts}) \\ &= N_\mu(n+1) \quad (\text{Lemma A.3}). \quad \blacksquare \end{aligned}$$

Proof of Theorem A. Take $\mu = \emptyset$ in Theorem A.15: $K_\emptyset(L) = I_q(L)$, so $\dim F[L]/I_q(L) \leq N_\emptyset(n)$. Combine with Proposition A.5. ■

A.8 Proofs of Corollaries 8.1 and 8.2

Corollary 8.1. Take $F = F_p$, $q = p^v$, $n = 2k+2$. Then $A_k(q) = N_q(2k+2)$ by Theorem A. Also $N_q(2k+2) = P_k(q)$: in the proof of Proposition A.5 with $F' = \bar{F}_p$ and the set of values $C = F_q \subset F'$, the set Z is $\{x \in F_q^{2k+2} : e_j(x) = 0, j \text{ odd}\}$. ■

Corollary 8.2. M is spanned by the monomials with exponents $< q$, so it is finitely generated. Write $M \cong Z^p \oplus T$ with T finite. For a field F , $M \otimes F \cong F[L]/I_q(L)$, by right exactness of $\otimes$. So:

- $\rho = \dim_Q = N_q(n)$;
- for an odd prime p , $\dim_{F_p} = \rho + (\text{number of cyclic } p\text{-primary summands of } T) = N_q(n)$.

Hence T has no odd primary part. ■

Remark A.16. In characteristic 2 the statement fails, and the proof uses $\frac{1}{2}$ in exactly one place (Lemma A.10). The measured dimensions at $n = 4$ are 21, 65, 133, 225 for $q = 3, 5, 7, 9$, against $N_q(4) = 19, 61, 127, 217$.

Appendix B. Theorem 0, modulo Pham's theorem and the intersection numbers

This appendix proves Theorem 0 for $n = 2k \geq 2$, $m \geq 3$ and $\emptyset \neq K \subseteq \mathcal{J}$. It follows [DS, §4] closely, with every step written out. We use, without proof:

- **(T1) Classical topology [Mil, §7].** $H_*(X)$ and $H_*(Y_0)$ are torsion free, where $Y_0 := X \cap \{z_0 = 0\}$ is the Fermat hypersurface of dimension $n - 1$ in P^n ; the intersection form $\langle \cdot, \cdot \rangle$ on $H_n(X)$ is unimodular; and by the Lefschetz hyperplane theorem $H_j(Y_0) \rightarrow H_j(P^n)$ is an isomorphism for $j < n - 1$. We also use three standard facts: the Gysin map $H_n(X) \rightarrow H_{n-2}(Y_0)$ sends the class of a cycle meeting Y_0 transversally to the class of the intersection; two complex submanifolds of complementary dimension meeting transversally in one point have intersection number $+1$; and the action of G by holomorphic automorphisms preserves the intersection form and is compatible with Poincaré–Lefschetz duality [DS, §2].
- **(T2) Pham's theorem [Ph], [DS, Theorem 2.1].** Let S be the Pham cycle of [DS, §2], an n -sphere in $X \setminus Y_0$, and let G act on $H_n(X \setminus Y_0)$. The map $R \rightarrow H_n(X \setminus Y_0)$, $1 \mapsto [S]$, is surjective.
- **(T3) The intersection numbers [DS, Theorem 2.2].** Fix $\eta := e^{\pi i/m}$, and put $s(\eta) := 1$, $s(\eta^{-1}) := -1$, and $s(\beta) := 0$ for the other m -th roots β of -1 . Then $\langle L_{J,\beta}, S \rangle = \text{sgn}(\sigma_J) \cdot \prod_{i=0}^k s(\beta_i)$, where σ_J is the permutation $(0, 1, \dots, n+1) \mapsto (j_0, k_0, \dots, j_k, k_k)$.

The group G , generated by $\gamma_i : z_i \mapsto \zeta z_i$ ($\zeta := e^{2\pi i/m}$, $i = 1, \dots, n+1$), acts on X by holomorphic automorphisms; we identify R with its group ring through $t_i \mapsto \gamma_i$. For an R -module M put $M^\vee := \text{Hom}_R(M, Z)$ with the contragredient action $(g \cdot \varphi)(x) = \varphi(g^{-1}x)$.

Step 1 (the Pham image is primitive, of corank 1). Let $V := \text{Im}(H_n(X \setminus Y_0) \rightarrow H_n(X))$. The cohomology sequence of the pair (X, Y_0) , with Poincaré–Lefschetz duality, gives an exact sequence of R -modules $H_n(X \setminus Y_0) \rightarrow H_n(X) \rightarrow H_{n-2}(Y_0)$ [DS, §2]. The second map is the Gysin map, which sends the class of a cycle meeting Y_0 transversally to the class of the intersection. By (T1), $H_{n-2}(Y_0) \cong H_{n-2}(P^n) \cong Z$ (note $n - 2 < n - 1$), so $H_n(X)/V$ embeds in Z : V is primitive. The map $H_n(X) \rightarrow Z$ is non-zero: a standard space $L_{J,\beta}$ meets Y_0 transversally in the linear $(k-1)$ -space $L_{J,\beta} \cap \{z_0 = 0\}$ (on $L_{J,\beta}$, $z_{k_0} = \beta_0 z_0$, and z_0 is a non-zero linear form). The intersection is transverse: at a point x , with $H := \{z_0 = 0\}$, $T_x Y_0 = T_x X \cap T_x H$, so $T_x L_{J,\beta} \cap T_x Y_0 = T_x L_{J,\beta} \cap T_x H$, of dimension $k - 1$. Its class generates $H_{n-2}(Y_0)$, since its image in $H_{n-2}(P^n)$ is the class of a linear $(k-1)$ -space. Hence $\text{rank } V = \text{rank } H_n(X) - 1$. By (T2), $R \rightarrow V$, $1 \mapsto [S]$, is surjective.

Step 2 (duals of surjections). On R put $[t^\nu, t^{\nu'}] := \delta_{\nu,\nu'} (\nu, \nu' \in (Z/m)^{n+1})$. This form is unimodular and G -invariant, so $x \mapsto [x, \cdot]$ is an isomorphism $R \cong R^\vee$ of R -modules. Let $f : R \rightarrow M$ be a surjective homomorphism of R -modules. Then $f^\vee : M^\vee \rightarrow R^\vee \cong R$ is injective, its image is the ideal $\text{Ker}(f)^\perp := \{x \in R : [x, y] = 0 \text{ for all } y \in \text{Ker } f\}$, and $R/\text{Ker}(f)^\perp$ is torsion free.² Indeed, an element of the image vanishes on $\text{Ker } f$; conversely, if $x \in \text{Ker}(f)^\perp$, then $[x, \cdot]$ vanishes on $\text{Ker } f$ and factors through $R/\text{Ker } f \cong M$, so it lies in the image. Finally, if $c x \in \text{Ker}(f)^\perp$ with $0 \neq c \in Z$, then $c[x, y] = 0$ for $y \in \text{Ker } f$, so $x \in \text{Ker}(f)^\perp$.

²As stated in [DS, §4.2], this description of the image of $f^\vee$ requires the additional hypothesis that $M/\text{Im } f$ be torsion free (for instance, f surjective); this hypothesis holds in the case where [DS] apply it, the surjection of Step 1, so the result we use is not affected.

Apply this to the surjection $f : R \rightarrow V$ of Step 1. We obtain an ideal $V^\vee \subseteq R$ with $R/V^\vee$ torsion free. Explicitly, the composite $H_n(X) \cong H_n(X)^\vee \rightarrow V^\vee \subseteq R$, where the first map is the (G -invariant) intersection form and the second is restriction, sends τ to the element r_τ with $[r_\tau, t^\nu] = \langle \tau, f(t^\nu) \rangle$, that is

$$r_\tau = \sum_{\nu \in (Z/m)^{n+1}} \langle \tau, \gamma^\nu [S] \rangle t^\nu.$$

Step 3 (the kernel is generated by the class of a linear section, which lies in L_K). The restriction $H_n(X)^\vee \rightarrow V^\vee$ is surjective, because V is a direct summand of $H_n(X)$ (a primitive subgroup of a free group); so $H_n(X) \rightarrow V^\vee$ is surjective with kernel $V^\perp$ (orthogonal for $\langle, \rangle$), which is primitive of rank 1. Let $P_X := [X \cap \Lambda]$ for a linear subspace $\Lambda \subset P^{n+1}$ of dimension $k+1$ meeting X properly; this class does not depend on Λ , since $[X \cap \Lambda] = h^k \cap [X]$ whenever the intersection is proper. Taking $\Lambda \subset \{z_0 = 0\}$ shows $P_X \perp V$, since every class of V is carried by a cycle in $X \setminus Y_0$. And $\langle P_X, [L] \rangle = 1$ for a standard space L , since P_X is the k -th power of the hyperplane class cut on X and L is a linear space of dimension k . So P_X is primitive and $V^\perp = Z \cdot P_X$. Now fix $J \in K$ and β with $\beta^m = -1$, and let $\Lambda_J := \{z_{k_i} = \beta z_{j_i} : i = 1, \dots, k\}$, of dimension $k+1$. On Λ_J one has $z_{j_i}^m + z_{k_i}^m = 0$ for $i \geq 1$, so $X \cap \Lambda_J = \{z_0^m + z_{k_0}^m = 0\} \cap \Lambda_J$, the reduced union of the m standard spaces $L_{J,(\beta^p, \beta, \dots, \beta)}$, $\beta_0^m = -1$. Hence $P_X \in L_K(X)$, and therefore

$$H_n(X)/L_K(X) \cong V^\vee/L'_K, \text{ rank } L_K(X) = \text{rank } L'_K + 1,$$

where $L'_K \subseteq V^\vee \subseteq R$ is the image of $L_K(X)$. Since the maps are R -linear and $L_K(X)$ is an R -submodule, L'_K is an ideal of R .

Step 4 (the image of L_J is generated by ψ_J). For $v \in (Z/m)^{n+1}$ write $\gamma^v := \prod \gamma_i^{v_i}$ and put $v_0 := 0$. Since γ^v multiplies z_i by ζ^{v_i} , one has $(\gamma^v)^{-1}(L_{J,\beta}) = L_{J,\beta'}$ with $\beta'_i = \zeta^{v_{j_i} - v_{k_i}} \beta_i$. For fixed J these translates exhaust all β (choose $v_{j_i} = 0$ and adjust v_{k_i}), so $L_{[J]}(X)$ is the R -module generated by $[L_{J,\eta}]$, $\eta := (\eta, \dots, \eta)$. By Step 2 and (T3), its image is $\psi'_J := \sum_v a_v t^v$ with

$$a_v = \langle L_{J,\eta}, \gamma^v[S] \rangle = \langle (\gamma^v)^{-1} L_{J,\eta}, S \rangle = \text{sgn}(\sigma_J) \cdot \prod_{i=0}^k e(v_{k_i} - v_{j_i}),$$

where $e(c) := s(\zeta^{-c}\eta)$, that is $e(0) = 1$, $e(1) = -1$ (because $\zeta^{-1}\eta = \eta^{-1}$), and $e(c) = 0$ otherwise. The indices k_0 and j_i , k_i ($i \geq 1$) are $1, \dots, n+1$, each once, so the sum factorises, and $\sum_c e(c)t^c = 1 - t$ and $\sum_{c,c'} e(c' - c)t_j^c t_k^{c'} = (1 - t_k)\sum_c (t_j t_k)^c = (1 - t_k)\varphi(t_j t_k)$ give

$$\psi'_J = \text{sgn}(\sigma_J) \cdot (1 - t_{k_0}) \cdot \prod_{i=1}^k (1 - t_{k_i}) \varphi(t_{j_i} t_{k_i}) = \pm \psi_J.$$

Hence $L'_K = (\psi_J : J \in K)$.

Step 5 (torsion transfer): Theorem 0(a). The exact sequence $0 \rightarrow V^\vee/L'_K \rightarrow R/L'_K \rightarrow R/V^\vee \rightarrow 0$ has a torsion-free cokernel, so every torsion element of R/L'_K lies in $V^\vee/L'_K$, and the torsion subgroups of $V^\vee/L'_K$ and $R/L'_K = A_K$ coincide. With Step 3, $\text{Tors}(H_n(X)/L_K(X)) \cong \text{Tors}(A_K)$.

Step 6: Theorem 0(b)–(d). (b) If $\text{char } K_p = p$ with $p = 0$ or $p \nmid m$, then $t^m - 1$ is separable over K_p , so $R \otimes K_p \cong K_p^{\mu_m^{n+1}}$ is the ring of functions on μ_m^{n+1} , and $A_K \otimes K_p$ is the ring of functions on the common zeros of the ψ_J , $J \in K$. For $b \in \mu_m$, $\varphi(b) = m \neq 0$ if $b = 1$ and $\varphi(b) = 0$ otherwise. So $\psi_J(a) \neq 0$ iff $a_{k_i} \neq 1$ for all i and $a_{j_i} a_{k_i} = 1$ for $i \geq 1$; since then $a_{j_i} = a_{k_i}^{-1} \neq 1$, this says exactly $a \in \Gamma_J$. Hence $\dim(A_K \otimes K_p) = m^{n+1} - |\Gamma_K|$. (c) By Step 3 and (b) with $p = 0$, $\text{rank } L_K(X) = \text{rank } L'_K + 1 = (m^{n+1} - \text{rank } A_K) + 1 = |\Gamma_K| + 1$. (d) For $p \nmid m$, (b) with $K_p = \bar{F}_p$ gives $\dim_{F_p}(A_K \otimes F_p) = \text{rank } A_K$, so A_K has no p -torsion, and by (a) neither has $H_n(X)/L_K(X)$. ■

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